

The Gastrointestinal Tract

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CHAPTER CONTENTS

Congenital Abnormalities, 685

Atresia, Fistulae, and Duplications, 685
Diaphragmatic Hernia, Omphalocele, and Gastroschisis, 686

Ectopia, 686

Meckel Diverticulum, 686

Pyloric Stenosis, 686

Hirschsprung Disease, 686

Esophagus, 688

Esophageal Obstruction, 688

Achalasia, 688

Esophagitis and Related Disorders, 689

Lacerations, 689

Chemical and Infectious Esophagitis, 689

Reflux Esophagitis, 690

Eosinophilic Esophagitis, 690

Esophageal Varices, 691

Barrett Esophagus, 691

Esophageal Tumors, 692

Adenocarcinoma, 693

Squamous Cell Carcinoma, 694

Stomach, 696

Gastropathy and Acute Gastritis, 696

Stress-Related Mucosal Disease, 697

Chronic Gastritis and Its Complications, 697

Helicobacter pylori Gastritis, 698

Autoimmune Atrophic Gastritis, 699

Uncommon Forms of Gastritis, 700

Eosinophilic Gastritis, 700

Lymphocytic Gastritis, 700

Granulomatous Gastritis, 701

Peptic Ulcer Disease, 701

Other Complications of Chronic Gastritis, 702

Mucosal Atrophy and Intestinal Metaplasia, 702

Dysplasia, 702

Gastritis Cystica, 702

Hypertrophic Gastropathies, 702

Ménétrier Disease, 702

Zollinger-Ellison Syndrome, 703

Gastric Polyps and Tumors, 704

Inflammatory and Hyperplastic Polyps, 704

Fundic Gland Polyps, 704

Gastric Adenoma, 705

Gastric Adenocarcinoma, 705

Lymphoma, 706

Neuroendocrine Tumors, 707

Gastrointestinal Stromal Tumor, 709

Small Intestine and Colon, 710

Intestinal Obstruction, 710

Hernias, 710

Adhesions, 710

Volvulus, 710

Intussusception, 711

Ischemic Bowel Disease, 711

Angiodysplasia, 712

Malabsorption and Diarrhea, 713

Cystic Fibrosis, 713

Celiac Disease, 713

Environmental Enteric Dysfunction, 715

Autoimmune Enteropathy, 715

Lactase (Disaccharidase) Deficiency, 716

Microvillus Inclusion Disease, 716

Abetalipoproteinemia, 716

Infectious Enterocolitis, 716

Bacterial Gastroenteritis, 717

Cholera, 717

Campylobacter Enterocolitis, 718

Shigellosis, 719

Salmonella, 720

Typhoid Fever, 720

Yersinia, 721

Escherichia coli, 721

Pseudomembranous Colitis, 722

Whipple Disease, 722

Viral Gastroenteritis, 723

Norovirus, 723

Rotavirus, 724

Adenovirus, 725

SARS-CoV-2, 725

Parasitic Enterocolitis, 725

Ascaris lumbricoides, 725

Strongyloides, 725

Necator duodenale and *Ancylostoma duodenale*, 725

Trichuris trichiura, 725

Enterobius vermicularis and *Enterobius gregorii*, 725

Schistosomiasis, 725

Intestinal cestodes, 725

Entamoeba histolytica, 726

Giardia lamblia, 726

Cryptosporidium, 726

Irritable Bowel Syndrome, 727

Inflammatory Bowel Disease, 727

- Epidemiology, 727
- Crohn Disease, 729
- Ulcerative Colitis, 730
- Indeterminate Colitis, 733
- Colitis-Associated Neoplasia, 733
- Other Causes of Colitis, 733
 - Diversion Colitis*, 733
 - Microscopic Colitis*, 733
 - Graft-Versus-Host Disease*, 734
 - Apoptotic Enterocolitis*, 734

Sigmoid Diverticular Disease, 734**Intestinal Polyps, 735**

- Hyperplastic Polyps, 735
- Inflammatory Polyps, 735
- Hamartomatous Polyps, 736
- Juvenile Polyps*, 736

Peutz-Jeghers Syndrome, 737**Premalignant and Malignant Lesions, 738**

- Adenomas, 738
- Familial Adenomatous Polyposis, 739
- Hereditary Nonpolyposis Colorectal Cancer, 741
- Adenocarcinoma, 741

Anal Canal, 745

- Tumors, 745
- Hemorrhoids, 745

Appendix, 745

- Acute Appendicitis, 745
- Tumors, 746

Peritoneal Cavity, 747**Peritonitis, 747**

- Sclerosing Retroperitonitis, 747

Tumors, 747

The gastrointestinal (GI) tract is a hollow tube extending from the oral cavity to the anus that consists of anatomically distinct segments, including the esophagus, stomach, small intestine, colon, rectum, and anus. Each segment has unique and complementary functions that are highly integrated and together serve to regulate the intake, processing, and absorption of ingested nutrients and the disposal of waste products. The regional variations in structure and function are reflected in diseases of the GI tract, which often mainly affect one or another segment. Accordingly, following consideration of several important congenital abnormalities, the discussion is organized anatomically. Disorders affecting more than one segment of the GI tract, such as Crohn disease, are discussed with the region that is involved most frequently.

CONGENITAL ABNORMALITIES

Depending on both the nature and the timing of the insult during gestation, a variety of developmental anomalies can affect the GI tract. Importantly, because many organs develop simultaneously during embryogenesis and are susceptible to the same insults, the presence of congenital GI anomalies should prompt evaluation of other organs.

ATRESIA, FISTULAE, AND DUPLICATIONS

Atresia, fistulae, and duplications may occur in any part of the GI tract. When present within the esophagus they are discovered shortly after birth, usually due to regurgitation during feeding. Most of the lesions are incompatible with survival without prompt surgical repair. Absence, or agenesis, of the esophagus is extremely rare, but atresia, in which development is incomplete, is present in 1 to 5 infants per 10,000 live births. In atresia, a segment of the esophagus does not develop, leaving only a thin, noncanalized cord and causing a mechanical obstruction (Fig. 17.1A). Atresia occurs most commonly at or near the tracheal bifurcation and is usually associated with a fistula connecting the upper or lower esophageal segments to a bronchus or the trachea (Fig. 17.1B). In other cases, a fistula may be present without atresia (Fig. 17.1C). Any of the three primary forms of tracheoesophageal fistula can lead to aspiration, suffocation, pneumonia, and severe fluid and electrolyte imbalances. Developmental abnormalities of the esophagus

are associated with a range of congenital abnormalities, including syndromes known as VACTERL (malformations including vertebral, anal, cardiac, tracheoesophageal, renal, and limb defects); TACRD (tracheal agenesis/atresia, complex congenital cardiac abnormalities, radial ray defects, and duodenal atresia), as well as other abnormalities involving the heart, lung, great vessels, genitourinary tract, and anus. The last-mentioned includes imperforate anus, which is due to a failure of the cloacal diaphragm to involute.

Stenosis is an incomplete form of atresia with markedly reduced luminal caliber due to fibrous thickening of the wall. This results in either partial or complete obstruction. In addition to congenital forms, stenosis can be acquired because of inflammatory scarring, such as that caused by chronic gastroesophageal reflux, irradiation, systemic sclerosis, or caustic injury. Although stenosis can involve any part of the GI tract, clinically significant stenosis most often affects the esophagus and small intestine due to their smaller luminal diameters.

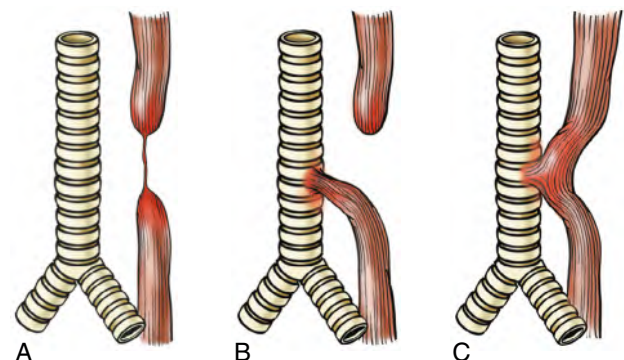


Fig. 17.1 Esophageal atresia and tracheoesophageal fistula. (A) Blind upper and lower esophagus with thin cord of connective tissue linking the two segments. (B) Blind upper segment with fistula between lower segment and trachea. (C) Fistula (without atresia) between patent esophagus and trachea. The developmental anomaly shown in (B) is the most common. (Modified from Morson BC, Dawson IMP, eds. *Gastrointestinal Pathology*. Oxford: Blackwell Scientific Publications; 1972:8.)

DIAPHRAGMATIC HERNIA, OMPHALOCELE, AND GASTROSCHISIS

Diaphragmatic hernia occurs when incomplete formation of the diaphragm allows the abdominal viscera to herniate into the thoracic cavity, most commonly on the left side. When severe, the space-filling effect of displaced viscera can lead to potentially fatal pulmonary hypoplasia.

Omphalocele occurs when the extraembryonic gut fails to return to the abdominal cavity and closure of the abdominal musculature is incomplete. As a result, abdominal viscera (sometimes including the liver) herniate ventrally into a membrane covered by amnion and peritoneum separated by Wharton jelly. Omphalocele may be associated with other birth defects as well as chromosomal abnormalities.

Gastroschisis is similar to omphalocele except that it is not covered by a membrane and involves all the layers of the abdominal wall, from the peritoneum to the skin. It is usually limited to the intestine but may be associated with other abnormalities of the GI tract. The incidence of gastroschisis is rising, possibly due to environmental factors. Surgical repair of omphalocele and gastroschisis is generally successful.

ECTOPIA

Ectopic tissues (developmental rests) are common in the GI tract. The most frequent site of ectopic gastric mucosa is the upper third of the esophagus, where it is referred to as an inlet patch. While generally asymptomatic, acid released by gastric mucosa within the esophagus can result in dysphagia, esophagitis, Barrett esophagus, or, rarely, adenocarcinoma. Less frequently, ectopic pancreatic tissue is found in the distal esophagus or stomach. These nodules are most often asymptomatic but, when present in the pylorus, can induce damage and inflammation that leads to scarring and pyloric outlet obstruction. Rests may be present within any layer of the gastric wall and, therefore, can mimic invasive cancer. Gastric heterotopia, small patches of ectopic gastric mucosa in the small bowel or colon, may cause occult blood loss due to peptic ulceration of adjacent mucosa.

MECKEL DIVERTICULUM

A true diverticulum is a blind outpouching of the alimentary tract that communicates with the lumen and includes all three layers of the bowel wall. The most common true diverticulum, and the most common congenital anomaly of the GI tract, is *Meckel diverticulum*, which occurs in the ileum. Meckel diverticulum occurs due to failed involution of the vitelline duct, which connects the lumen of the developing gut to the yolk sac. This solitary diverticulum extends from the antimesenteric side of the bowel (Fig. 17.2). The “rule of 2’s” is often used to help remember characteristics of Meckel diverticula, which:

- Occur in approximately 2% of the population
- Are generally present within 2 feet (60 cm) of the ileocecal valve
- Are approximately 2 inches (5 cm) long
- Have a male:female ratio of 2:1
- Are most often symptomatic by age 2 (only approximately 4% are ever symptomatic).

The mucosal lining of Meckel diverticula may resemble normal small intestine, but ectopic pancreatic or gastric tissue may also be present. The latter may secrete acid that causes peptic ulceration of adjacent small intestinal mucosa. Such cases may present with occult bleeding or abdominal pain that resembles acute appendicitis or obstruction.

Virtually all other diverticula are acquired and either lack muscularis entirely or have an attenuated muscularis propria. These are, technically, pseudodiverticula, but are generally also referred to as diverticula. They occur most often in the sigmoid colon (discussed later).



Fig. 17.2 Meckel diverticulum. The blind pouch is located on the antimesenteric side of the small bowel.

PYLORIC STENOSIS

Pyloric stenosis may be either congenital or acquired. The most common congenital form is congenital or infantile hypertrophic pyloric stenosis. It is three to five times more common in males and has an overall incidence of 1 per 300 to 900 live births. Monozygotic twins have a high rate of concordance; risk is increased to a lesser degree in dizygotic twins and siblings of affected individuals. Turner syndrome and trisomy 18 also confer an increased risk. While the underlying mechanisms are not understood, erythromycin or azithromycin exposure, either orally or via mother’s milk, in the first 2 weeks of life is associated with this disorder.

Congenital hypertrophic pyloric stenosis generally presents between the third and sixth weeks of life and is rare after 12 weeks. It is five times more common in males and is associated with being the first-born child, prematurity, and young maternal age. Presentation before 2 weeks is frequently associated with a family history of congenital hypertrophic pyloric stenosis. In the general population, congenital hypertrophic pyloric stenosis incidence is ~0.2%. This increases to ~6% in siblings of affected patients. Genome-wide association studies have linked several genes related to gastrointestinal development to congenital hypertrophic pyloric stenosis, but further study is needed to understand how these variants are linked to pathogenesis.

Congenital hypertrophic pyloric stenosis typically presents as forceful projectile nonbilious vomiting after feeding and frequent demands for refeeding. In some cases, abnormal left-to-right peristaltic waves are seen across the upper abdomen just prior to emesis. Physical examination reveals a firm, ovoid, 1- to 2-cm abdominal mass in up to 90% of cases. When present, this mass, or “olive,” is nearly pathognomonic for congenital hypertrophic pyloric stenosis. Ultrasonographic analysis of muscle thickness and pyloric canal length is a sensitive and specific way to establish the diagnosis. Anatomically, hyperplasia of the pyloric muscularis propria causes outflow tract obstruction and may be exacerbated by mucosal and submucosal edema and inflammation. Surgical splitting of the hypertrophic muscularis (pyloromyotomy) is generally curative.

Acquired pyloric stenosis occurs in adults because of antral gastritis or peptic ulcers close to the pylorus. Other causes include carcinomas of the distal stomach and pancreas that narrow the pyloric channel due to fibrosis or malignant infiltration.

HIRSCHSPRUNG DISEASE

Hirschsprung disease causes functional obstruction of the colon due to failure of ganglion cells to migrate to the wall of the colon.

Hirschsprung disease, also known as congenital aganglionic megacolon, occurs in approximately 1 in 5000 live births. It may be isolated or occur in combination with other developmental abnormalities; 10% of all cases occur in children with Down syndrome, and serious neurologic abnormalities are present in 5% of infants with Hirschsprung disease.

Pathogenesis

The enteric neuronal plexus develops from neural crest cells that migrate into the bowel wall during embryogenesis. Hirschsprung disease results when the migration of neural crest cells from cecum to rectum is arrested prematurely or when the ganglion cells undergo premature death. This produces a distal intestinal segment that lacks both the Meissner submucosal plexus and the Auerbach myenteric plexus, termed aganglionosis. Coordinated peristaltic contractions are absent, and functional obstruction occurs, resulting in dilation proximal to the affected segment.

Heterozygous loss-of-function mutations in the receptor tyrosine kinase *RET* account for the majority of familial and approximately 15% of sporadic Hirschsprung disease cases. In addition to *RET*, germline mutations in *EDNRB* or *EDN3*, which encode a receptor-ligand pair that regulates neural crest cell development, have also been implicated in Hirschsprung disease development (Fig. 17.3). Other genes encoding secreted proteins, transporters, and transcription factors have been

implicated in small numbers of cases. Importantly, incomplete penetrance in those with pathogenic mutations suggests that environmental factors or modifying genes are also important. The increased frequency of disease in males and association with Down syndrome further suggest that these disease modifying genes are sex-linked or encoded on chromosome 21, respectively.

MORPHOLOGY

Diagnosis of Hirschsprung disease requires histologic confirmation that ganglion cells are absent within the affected segment. In addition to their characteristic morphology in hematoxylin and eosin–stained sections, ganglion cells can be identified using immunohistochemical stains for acetylcholinesterase.

The rectum is always affected, but the length of the additional involved segments varies widely. In most cases the abnormality extends to the sigmoid colon, but in severe cases, the entire colon may be involved. Disease limited to the rectosigmoid is termed *short-segment Hirschsprung disease* (the more common type), while cases with more proximal extension are termed *long-segment Hirschsprung disease*.

The aganglionic region may have a grossly normal or contracted appearance. In contrast, the normally innervated proximal colon undergoes progressive dilation (Fig. 17.4) and may become massively distended (**megacolon**) due to the distal obstruction. The colonic wall may be stretched to the point of rupture, which occurs most frequently near the cecum. Mucosal inflammation or shallow ulcers may also be present in normally innervated segments, making gross identification of the extent of aganglionosis difficult. Hence, intraoperative histopathologic analysis (frozen section) is commonly used to confirm the presence of ganglion cells at the anastomotic site.

Clinical Features

Hirschsprung disease typically presents with a failure to pass meconium in the immediate postnatal period. Obstruction or constipation follows, often with visible but ineffective peristalsis. When only a few centimeters of rectum are involved, some stool may be passed, obscuring the diagnosis. More often, the obstruction is obvious and is marked by abdominal distention and bilious vomiting as luminal material backs up into the proximal colon resulting in dramatic luminal dilation referred to as *megacolon*. The major threats to life are enterocolitis, fluid and electrolyte disturbances, perforation, and peritonitis. Treatment consists of surgical resection of the aganglionic segment and anastomosis of the normal proximal colon to the rectum. Even after successful surgery, it may take years to attain normal bowel function and continence.

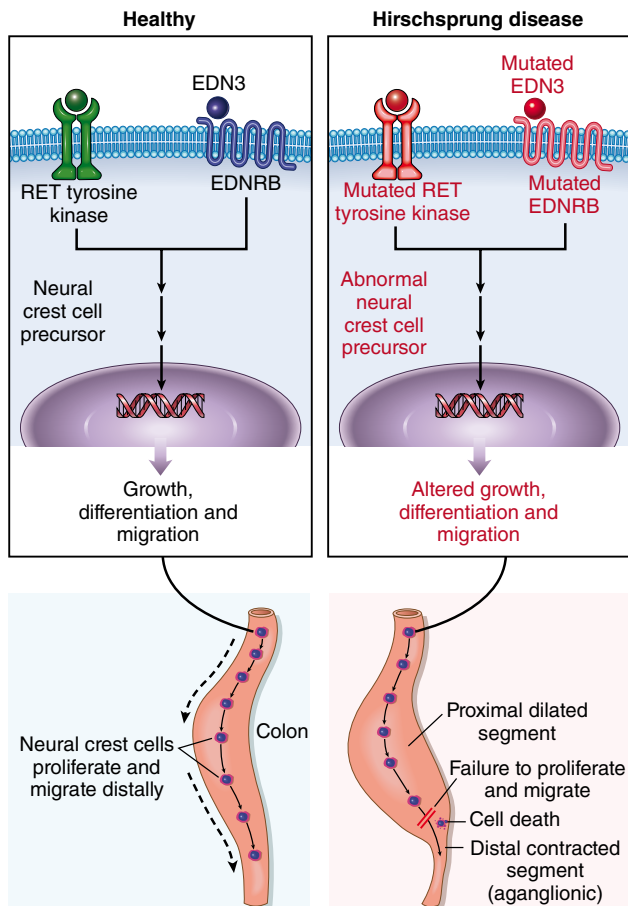


Fig. 17.3 Hirschsprung disease. Signaling via the RET tyrosine kinase and the EDNRB-EDN3 receptor-ligand pair direct proliferation and differentiation as neural crest cells move from proximal to distal within the submucosal (Meissner) and myenteric (Auerbach) plexuses. Mutation of either *RET*, *EDNRB*, or *EDN3* lead to failure of migration and, ultimately, cell death. The ganglion cells lost would normally signal to relax the muscularis propria. Thus, the distal aganglionic region is hypercontracted. The proximal segment is innervated and dilated secondarily due to distal obstruction.

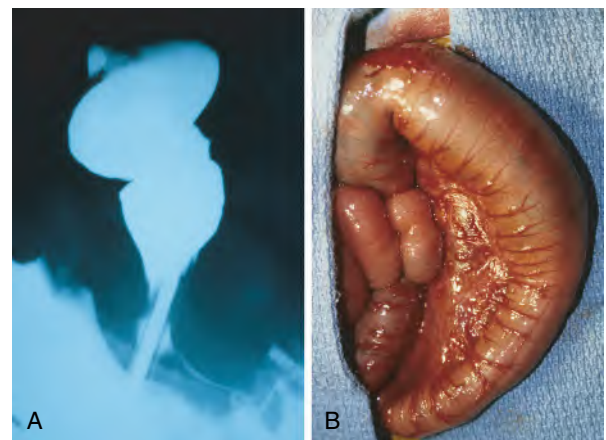


Fig. 17.4 Hirschsprung disease. (A) Preoperative barium enema study showing constricted rectum (bottom of the image) and dilated sigmoid colon. (B) Corresponding intraoperative photograph showing constricted rectum and dilation of the sigmoid colon. (Courtesy Dr. Aliya Husain, University of Chicago.)

Patients are also at risk for developing Hirschsprung disease-associated enterocolitis, which occurs in ~25% of cases either prior to or up to 20 months following surgical repair. Features of Hirschsprung disease-associated enterocolitis range from mild abdominal distention and ileus to obstipation, pneumatosis intestinalis, and pneumoperitoneum and can be associated with severe dehydration and electrolyte imbalances. Early diagnosis of Hirschsprung disease-associated enterocolitis is critical, as advances in treatment have reduced mortality from ~30% to less than 1% over the past 50 years. Although many theories have been proposed, the pathogenesis of Hirschsprung disease-associated enterocolitis is poorly understood.

In contrast to the congenital megacolon of Hirschsprung disease, acquired megacolon may occur at any age due to Chagas disease, obstruction by a neoplasm or inflammatory stricture, or a complication of ulcerative colitis. Of these, ganglion cell loss is only present in Chagas disease (discussed later), which shares pathophysiologic features with Hirschsprung disease.

KEY CONCEPTS

Congenital Malformations of the Gastrointestinal Tract

- The gastrointestinal (GI) tract is a common site of developmental abnormalities, often in association with congenital defects of other organ systems.
- Atresia and fistulae are developmental anomalies that typically present at birth. Imperforate anus is the most common form of congenital intestinal atresia, while the esophagus is the most common site of fistulization.
- Stenosis may be developmental or acquired. Both forms are characterized by a thickened wall and partial or complete luminal obstruction. Acquired forms are often due to inflammatory scarring.
- Diaphragmatic hernia is characterized by incomplete diaphragm development and herniation of abdominal organs into the thorax, often producing pulmonary hypoplasia. Omphalocele and gastroschisis refer to ventral herniation of abdominal organs.
- Ectopia refers to normally formed tissues at an abnormal site. This is common in the GI tract, with ectopic gastric mucosa in the upper third of the esophagus being the most common form.
- Meckel diverticulum is a true diverticulum, defined by the presence of all three layers of the bowel wall, that stems from failed involution of the vitelline duct. It is a frequent site of gastric ectopia, which may result in peptic injury and occult bleeding.
- Congenital hypertrophic pyloric stenosis presents as forceful projectile vomiting between the third and sixth weeks of life and is more common in males.
- Hirschsprung disease is caused by the failure of neural crest–derived ganglion cells to migrate into the distal colon. The defect, which always involves the rectum, extends proximally for variable distances.

ESOPHAGUS

The esophagus develops from the cranial portion of the foregut and is recognizable by the third week of gestation. It is a hollow, highly distensible muscular tube that extends from the epiglottis in the pharynx to the gastroesophageal junction. Acquired diseases of the esophagus run the gamut from highly lethal cancers to the heartburn of gastroesophageal reflux that ranges from chronic and incapacitating to occasional discomfort.

ESOPHAGEAL OBSTRUCTION

The primary function of the esophagus is to deliver ingested solid food and fluids to the stomach. This can be impeded by physical or functional obstruction. The latter results from disruption of coordinated

peristalsis after swallowing. Esophageal dysmotility falls into three principal patterns.

- *Nutcracker esophagus* describes functional obstruction of the esophagus by intense, high amplitude, uncoordinated contractions of inner circular and outer longitudinal smooth muscle. Barium swallow can be normal or have a nutcracker-like appearance, with narrow and distended regions. Manometry is, however, required for diagnosis.
- *Diffuse esophageal spasm*, also known as corkscrew esophagus due to the appearance on barium swallow, is characterized by repetitive, simultaneous contractions of the distal esophageal smooth muscle. Unlike nutcracker esophagus, these contractions are of normal amplitude.
- *Lower esophageal sphincter dysfunction*, including high resting pressure or incomplete relaxation, may be present as an isolated anomaly or accompany nutcracker esophagus or diffuse esophageal spasm.

Because wall stress is increased, esophageal dysmotility may result in development of small diverticula, particularly in the epiphrenic region immediately above the lower esophageal sphincter. Similarly, impaired relaxation and spasm of the cricopharyngeus muscle after swallowing can result in increased pressure within the distal pharynx and development of a *Zenker (pharyngoesophageal) diverticulum*, located immediately above the upper esophageal sphincter. Zenker diverticula are uncommon, typically develop after age 50, and may reach several centimeters in size. When small they may be asymptomatic, but larger Zenker diverticula may accumulate significant amounts of food, producing a mass and symptoms that include regurgitation and halitosis. In contrast, benign or malignant strictures that cause mechanical obstruction typically present as progressive dysphagia that begins with inability to swallow solids.

Benign esophageal stricture is generally caused by fibrous thickening of the submucosa and associated with atrophy of the muscularis propria and secondary epithelial damage. Although occasionally congenital, stenosis is most often due to inflammation and scarring that may be caused by chronic gastroesophageal reflux, irradiation, or caustic injury. In general, patients with functional obstruction or benign strictures maintain their appetite and weight. In contrast, malignant strictures are often associated with weight loss.

Esophageal mucosal webs are uncommon ledge-like mucosal protrusions. These usually occur in females older than age 40 and may be associated with gastroesophageal reflux, chronic graft-versus-host disease, or blistering skin diseases. In the upper esophagus, webs may be accompanied by iron deficiency anemia, glossitis, and cheilosis as part of *Plummer-Vinson syndrome*. Morphologically, esophageal webs are recognized as semi-circumferential lesions composed of a fibrovascular connective tissue and overlying epithelium. The main symptom of webs is partial obstruction, primarily of incompletely chewed food, and nonprogressive dysphagia.

Esophageal rings, or *Schatzki rings*, are similar to webs but are circumferential, thicker, and include mucosa, submucosa, and occasionally hypertrophic muscularis propria. When present in the distal esophagus, above the gastroesophageal junction, they are termed *A rings* and are covered by squamous mucosa; in contrast, those located at the squamocolumnar junction of the lower esophagus are designated *B rings* and may have gastric cardia-type mucosa on their undersurface.

ACHALASIA

Achalasia is characterized by the triad of incomplete lower esophageal sphincter relaxation, increased lower esophageal sphincter tone, and aperistalsis. Symptoms include dysphagia, difficulty in belching, and chest pain. Although there is some increased risk for esophageal cancer, it is not considered great enough to warrant surveillance endoscopy.

Primary achalasia is rare and results from degeneration of nitric oxide-producing neurons that normally induce lower esophageal sphincter relaxation. Degenerative changes in the extraesophageal vagus nerve or the dorsal motor nucleus of the vagus may also occur. The cause is unknown; rare familial cases have been described.

Secondary achalasia may arise in Chagas disease, in which *Trypanosoma cruzi* infection causes destruction of the myenteric plexus, failure of peristalsis, and esophageal dilation. Additional sites, including duodenal, colonic, and ureteric myenteric plexi can also be affected in Chagas disease. Other causes of achalasia-like disease include diabetic autonomic neuropathy; infiltrative disorders such as malignancy, amyloidosis, and sarcoidosis; systemic sclerosis; and lesions of dorsal motor nuclei (e.g., following polio). Lower esophageal sphincter dysfunction also occurs in association with Down syndrome. The association of some achalasia cases with remote herpes simplex virus 1 (HSV1) infection, linkage of immunoregulatory gene polymorphisms to achalasia, and occasional coexistence of Sjögren syndrome or autoimmune thyroid disease suggest that achalasia may also be driven by immune-mediated destruction of inhibitory esophageal neurons. Treatments for both primary and secondary achalasia aim to overcome the obstruction and include pneumatic balloon dilation, botulinum neurotoxin (Botox) injection to inhibit contraction-promoting cholinergic neurons, or surgically cutting the muscles of the esophagus.

ESOPHAGITIS AND RELATED DISORDERS

Lacerations

Longitudinal mucosal tears near the gastroesophageal junction, called Mallory-Weiss tears, are most often associated with severe retching or vomiting secondary to acute alcohol intoxication. Normally, a reflex relaxation of the gastroesophageal musculature precedes the antiperistaltic contractile wave associated with vomiting. It is speculated that this relaxation fails during prolonged vomiting, with the result that refluxing gastric contents overwhelm the gastric inlet and cause the esophageal wall to stretch and tear. The roughly linear lacerations of *Mallory-Weiss syndrome* are longitudinally oriented and range in length from millimeters to several centimeters. These tears usually cross the gastroesophageal junction and may also be located in the proximal gastric mucosa. Up to 10% of upper GI bleeding, which often presents as hematemesis (Table 17.1), is due to superficial esophageal lacerations such as those associated with Mallory-Weiss syndrome. These do not generally require surgical intervention, and healing tends to be rapid and complete.

Like Mallory-Weiss tears, *Boerhaave syndrome* is caused by a sudden increase in intraesophageal pressure combined with negative

intrathoracic pressure, as occurs with severe straining or retching. In Boerhaave syndrome, which is relatively uncommon, tears extend transmurally to cause esophageal perforation. Presenting symptoms of acute onset chest or abdominal pain, tachypnea, or shock may mimic those of myocardial infarction, pancreatitis, peptic ulcer perforation, aortic aneurysm dissection, pneumothorax, or pneumonia. Early diagnosis is critical, as untreated Boerhaave syndrome is fatal. Conservative management can be attempted for selected patients, but endoscopic or surgical therapy is required for severe cases and those that deteriorate during medical management.

Chemical and Infectious Esophagitis

The stratified squamous mucosa of the esophagus may be damaged by a variety of irritants including alcohol, corrosive acids or alkalis, excessively hot fluids, and heavy smoking. Symptoms range from self-limited pain, particularly on swallowing, called *odynophagia*, to hemorrhage, stricture, or perforation in severe cases.

Esophageal chemical injury is often secondary to accidental or intentional, as in suicide attempts, or ingestion of household cleaning products and other corrosive agents. Less severe mucosal injury can occur when medicinal pills lodge and dissolve in the esophagus, a condition termed *pill-induced esophagitis*. Cytotoxic chemotherapy, radiation therapy, or graft-versus-host disease may all cause iatrogenic injury to the esophagus, which may also be involved by desquamative skin diseases and, rarely, Crohn disease.

Esophageal infections in otherwise healthy individuals are most often due to herpes simplex virus (HSV). Infections are more common in patients who are debilitated or immunosuppressed and may be caused by HSV, cytomegalovirus (CMV), or fungal organisms. Among fungi, candidiasis is most common, but mucormycosis and aspergillosis also occur.

MORPHOLOGY

The morphology of chemical and infectious esophagitis varies with etiology. Dense infiltrates of neutrophils are usually present but may be absent in the setting of acute chemical esophagitis with necrosis induced by lye, acid, or detergent ingestion. Pill-induced esophagitis frequently occurs at sites of preexisting strictures.

Esophageal irradiation damage is marked by intimal proliferation and luminal narrowing of submucosal and mural blood vessels, similar to changes in other tissues. Mucosal damage is, in part, secondary to this radiation-induced vascular injury, as discussed in Chapter 9.

Infection by fungi or bacteria can either cause injury or complicate a preexisting ulcer. Nonpathogenic oral bacteria are frequently found in ulcer beds, while pathogenic bacteria and fungi, which account for about 10% of infectious esophagitis, may invade and cause mucosal necrosis. Candidiasis is characterized by adherent, gray-white **pseudomembranes** composed of densely matted fungal hyphae and inflammatory cells covering the esophageal mucosa.

The presence of discrete punched-out ulcers suggests a viral etiology. The punched-out ulcers (Fig. 17.5A) caused by herpes viruses can be recognized by nuclear inclusions within degenerating, often multinucleate, epithelial cells at ulcer margins (Fig. 17.5B). In contrast, CMV causes shallower ulcerations and is marked by nuclear and cytoplasmic inclusions within capillary endothelium and stromal cells (Fig. 17.5C). Although the histologic appearance is characteristic, immunohistochemical stains for virus-specific antigens are useful ancillary diagnostic tools.

Histologic features of esophageal **graft-versus-host disease** are similar to those in the skin and include basal epithelial cell apoptosis, mucosal atrophy, and submucosal fibrosis without significant acute inflammatory infiltrates. The microscopic appearances of esophageal involvement in bullous pemphigoid, epidermolysis bullosa, and Crohn disease are also similar to those seen in other sites (Chapter 25).

TABLE 17.1 Esophageal Causes of Hematemesis

Lacerations (Mallory-Weiss tears)
Esophageal perforation (cancer or Boerhaave syndrome)
Varices (cirrhosis)
Esophageal-aortic fistula (usually with cancer)
Chemical and pill esophagitis
Infectious esophagitis (<i>Candida</i> , herpes)
Benign strictures
Vasculitis (autoimmune, cytomegalovirus)
Reflux esophagitis (erosive)
Eosinophilic esophagitis
Esophageal ulcers (many etiologies)
Barrett esophagus
Adenocarcinoma
Squamous cell carcinoma
Hiatal hernia

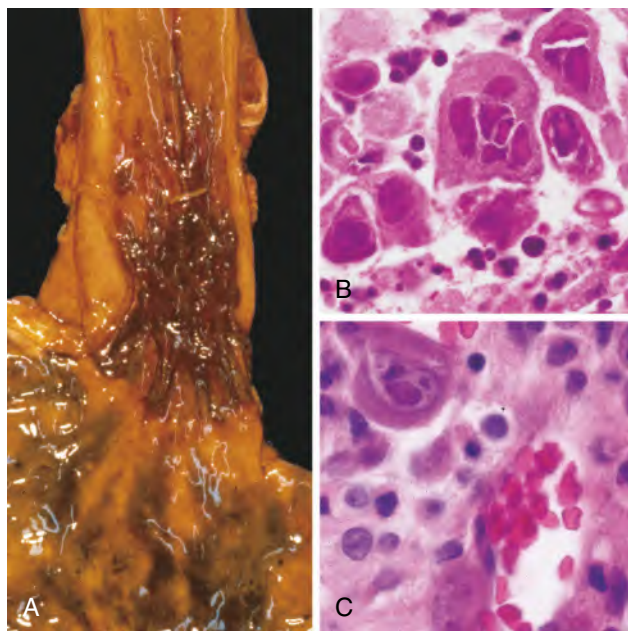


Fig. 17.5 Viral esophagitis. (A) Postmortem specimen with multiple, overlapping herpetic ulcers in the distal esophagus. (B) Multinucleate squamous cells containing herpesvirus nuclear inclusions. (C) Cytomegalovirus-infected endothelial cells with nuclear and cytoplasmic inclusions.

Reflux Esophagitis

Reflux of gastric contents into the lower esophagus is the most frequent cause of esophagitis and the most common outpatient GI diagnosis in the United States. The associated clinical condition, termed gastroesophageal reflux disease (GERD), reflects sensitivity of the abrasion-resistant esophageal epithelium to gastric acid.

Pathogenesis

Transient lower esophageal sphincter relaxation is thought to be a major cause of GERD. This relaxation is mediated via vagal pathways and can be triggered by gastric distention. Gastroesophageal reflux can also occur following abrupt increases in intraabdominal pressure, (e.g., after coughing, straining, or bending). Other conditions that are associated with GERD include alcohol and tobacco use, obesity, central nervous system depressants, pregnancy, hiatal hernia (discussed later), delayed gastric emptying, and increased gastric volume.

MORPHOLOGY

Simple hyperemia, evident to the endoscopist as redness, may be the only alteration in mild GERD; mucosal histology is often unremarkable. More significant gastric reflux is associated with erosions (Fig. 17.6A) and influx of eosinophils into the squamous mucosa (Fig. 17.7A). Basal zone hyperplasia and elongation of lamina propria papillae are also common. Neutrophil infiltration is less frequent and is generally associated with bacterial or fungal infection or chemical damage.

Clinical Features

The prevalence of GERD varies widely across geographies and approaches 30% in Europe, North America, South America, and the Middle East but is only 10% or less in East Asia and Australia. Typical symptoms, which may occasionally be mistaken for ischemic heart disease, include heartburn, dysphagia, and regurgitation of sour-tasting gastric contents, most often postprandially. The severity of symptoms does not correlate well with histologic damage, and neither endoscopy nor biopsy are generally required for GERD diagnosis. Treatment with

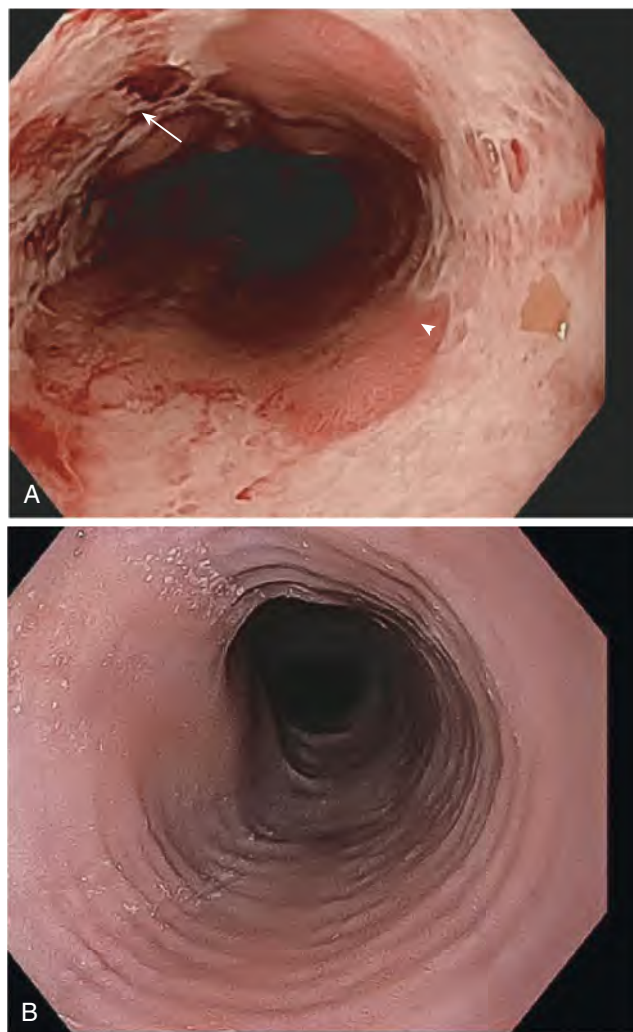


Fig. 17.6 Esophagitis. (A) Endoscopic view of reflux esophagitis with multiple erosions within the squamous-lined esophagus (arrow), metaplastic patches (in Barrett esophagus, discussed later), and distal gastric mucosa. Note the tan islands of metaplastic epithelium within the pale squamous mucosa (arrowhead). (B) The “feline” or “ringed” endoscopic appearance of the esophagus is typical of eosinophilic esophagitis. (Courtesy Dr. Linda S. Lee, Brigham and Women’s Hospital and Harvard Medical School, Boston.)

proton pump inhibitors to reduce gastric acidity provides symptomatic relief in up to 90% of patients.

Endoscopy is typically employed for evaluation of proton pump inhibitor-refractory patients and if eosinophilic esophagitis, Barrett esophagus, strictures, ulcerations, and other GERD complications are suspected.

Hiatal hernia is symptomatic in fewer than 10% of adults but may cause lower esophageal sphincter incompetence and symptoms that are similar to GERD. Most hiatal hernias are of the sliding type, in which the diaphragmatic crura are spread and the gastroesophageal junction herniates into the thorax. Less common paraesophageal hiatal hernias penetrate through a defect in the phrenoesophageal membrane, can involve the stomach and other organs, and generally require surgical repair.

Eosinophilic Esophagitis

Eosinophilic esophagitis is an immune-mediated disorder that is frequently associated with atopy. Consistent with an allergic etiology, many patients also have atopic dermatitis, allergic rhinitis, asthma, or modest peripheral eosinophilia. Mast cells have been implicated in pathogenesis

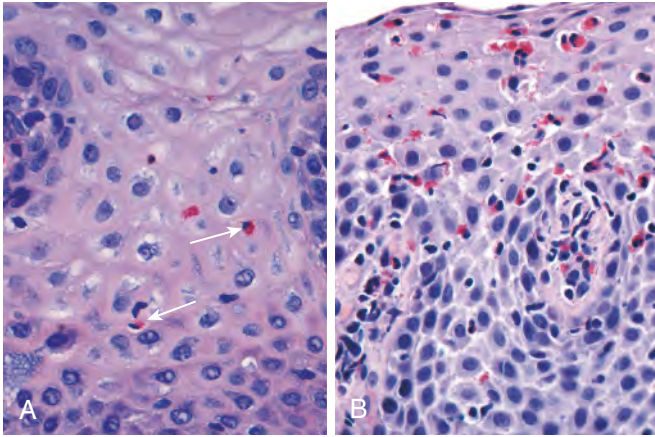


Fig. 17.7 Esophagitis. (A) Reflux esophagitis with scattered intraepithelial eosinophils (arrows) and mild basal zone expansion. (B) Eosinophilic esophagitis is characterized by numerous intraepithelial eosinophils.

and eosinophilic esophagitis-associated pain. In addition to GERD-like symptoms, patients may experience food impaction, dysphagia, and vomiting. Infants may also experience feeding intolerance. Endoscopically, the esophageal mucosa can resemble stacked circular rings (referred to as feline esophagus because of supposed endoscopic resemblance to a striped cat's tail) with strictures and linear furrows (Fig. 17.6B). The cardinal histologic feature is large numbers of intraepithelial eosinophils, particularly superficially, that form clusters and sheets, and intercellular edema (Fig. 17.7B). While characteristic, these histologic findings are not specific to eosinophilic esophagitis, and other diagnoses, including GERD, Crohn disease, autoimmune disorders, and infections must be excluded. There is no association with Barrett esophagus.

Diet and other environmental factors are thought to underlie eosinophilic esophagitis. Empiric elimination of cow's milk, cross-reacting mammalian milk, and milk products is recommended as initial therapy for patients who prefer a dietary approach to therapy. Addition of other dietary restrictions can be guided by skin patch or immunoglobulin E (IgE) testing. Pharmacological approaches, including proton pump inhibitors and topical corticosteroids, can be added to dietary interventions. The incidence of the disease has increased in the last 20 years, paralleling the increasing incidence of other Th2-mediated allergic disorders. Interleukin (IL)-4/IL-13 receptor antagonists can be beneficial. Endoscopic dilation of esophageal strictures can relieve dysphagia symptoms but does not reduce inflammation and must be performed in small increments over multiple sessions to limit risk of mucosal tears and esophageal perforation.

Esophageal Varices

Esophageal varices are dilated veins within the lower esophagus. Although most small varices never bleed, rupture of large varices can result in exsanguination.

Pathogenesis

Esophageal varices are caused by portal hypertension, which is due to impaired blood flow through the portal venous system and liver. Increased portal venous pressure results in the development of collateral channels at sites where the portal or caval systems communicate with the systemic venous circulation. These collateral veins allow some drainage to occur, but also result in congestion and dilation of subepithelial and submucosal venous plexuses within the distal esophagus and proximal stomach. These dilated vessels, termed *varices*, are common in patients with cirrhosis. They are most frequently seen in patients with alcoholic

liver disease, but can complicate cirrhosis of any cause, including nonalcoholic fatty liver disease. Noncirrhotic causes of portal hypertension include presinusoidal, (e.g., schistosomiasis or polycystic liver disease), sinusoidal, (e.g., amyloidosis or Gaucher disease), or postsinusoidal, (e.g., Budd-Chiari syndrome or venoocclusive disease), obstruction. Portal hypertension is discussed in more detail in Chapter 18.

MORPHOLOGY

Esophageal varices are tortuous dilated veins within the mucosa and submucosa of the distal esophagus and proximal stomach (Fig. 17.8). Variceal rupture can result in hemorrhage into the lumen or the esophageal wall and may be associated with mucosal ulceration and necrosis. If rupture has occurred in the past, venous thrombosis, inflammation, and evidence of prior sclerotherapy may also be present.

Clinical Features

Gastroesophageal varices are present in 30% of those with compensated cirrhosis and in 60% of those with decompensated cirrhosis. Variceal hemorrhage is an emergency that can be treated medically by inducing splanchnic vasoconstriction or endoscopically by sclerotherapy (injection of thrombotic agents), balloon tamponade, or variceal ligation. Despite these interventions, each episode of variceal hemorrhage has 15% to 20% risk of mortality, and over half of patients who survive a first variceal bleed have recurrent hemorrhage within 1 year, with mortality risk similar to the first episode. Risk factors for hemorrhage include large or tortuous varices, elevated hepatic venous pressure gradient, previous bleeding, and advanced liver disease. Treatments include beta-blockers to reduce portal blood flow and endoscopic variceal ligation.

Barrett Esophagus

Barrett esophagus is a complication of chronic GERD that is characterized by intestinal metaplasia of the esophageal squamous mucosa and is associated with an increased risk of cancer. The incidence of Barrett esophagus is rising; it is estimated to occur in as many as 10% of those with symptomatic GERD and up to 2% of the general population. Barrett esophagus is most common in males of European descent between 40 and 60 years of age. The greatest clinical concern is that it confers an increased risk of esophageal adenocarcinoma. Genomic sequencing of biopsies involved by Barrett esophagus has revealed the presence of driver mutations in cancer genes that are also found in esophageal adenocarcinoma. Potentially oncogenic mutations are more numerous when biopsies demonstrate dysplasia, a preinvasive neoplastic change that is associated with prolonged symptoms, longer segment length, increased patient age, and European ethnicity. While most esophageal adenocarcinomas occur in association with Barrett esophagus, most individuals with Barrett esophagus do not develop esophageal tumors.

MORPHOLOGY

Barrett esophagus appears as tongues of red, velvety mucosa extending upward from the gastroesophageal junction. This metaplastic mucosa alternates with residual smooth, pale squamous (esophageal) mucosa and interfaces with light-brown columnar (gastric) mucosa distally (Fig. 17.9A–B). Barrett esophagus can be subclassified as long segment (≥ 3 cm), or short segment (< 3 cm). Patients with short-segment disease may not experience GERD symptoms and are at lower risk of developing dysplasia or carcinoma relative to those with long-segment disease.

Diagnosis of Barrett esophagus requires endoscopic evidence of metaplastic columnar mucosa above the gastroesophageal junction. Microscopically, **intestinal-type metaplasia** is marked by replacement of the squamous epithelium with columnar epithelial cells. When present, goblet cells are diagnostic of Barrett esophagus; they have distinct mucus vacuoles that stain

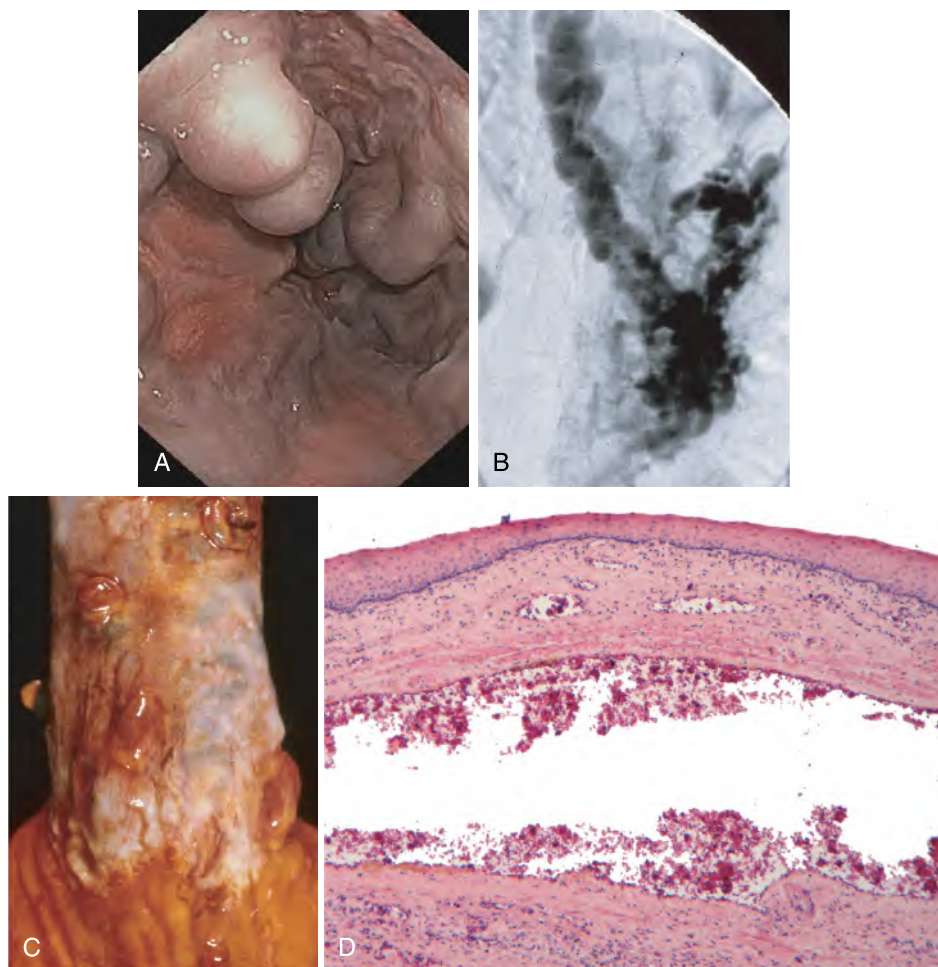


Fig. 17.8 Esophageal varices. (A) Endoscopically, distended varices form submucosal nodules. (B) Angiogram demonstrating several tortuous esophageal varices. (C) Collapsed varices are present in this postmortem specimen corresponding to the angiogram in (B). The polypoid areas represent previous sites of variceal hemorrhage that have been ligated with bands. (D) Dilated varices beneath intact squamous mucosa.

pale blue and impart the shape of a wine goblet to the remaining cytoplasm (Fig. 17.9C). Non-goblet columnar cells, such as gastric-type foveolar cells, are also present, but there is not universal agreement as to whether they are sufficient for the diagnosis. The risk of developing cancer is greatest for those with goblet cells

Dysplasia is classified as low grade or high grade. Atypical mitoses, nuclear hyperchromasia, irregularly clumped chromatin, increased nuclear-to-cytoplasmic ratio, and failure of epithelial cells to mature as they migrate to the esophageal surface are present in both grades (Fig. 17.10A). Dysplastic glands display budding, irregular shapes, and cellular crowding. High-grade dysplasia (Fig. 17.10B) exhibits more severe cytologic and architectural changes than low-grade dysplasia. With progression, dysplastic cells may invade the lamina propria, a feature that defines progression to intramucosal carcinoma.

Clinical Features

Barrett esophagus is identified by endoscopy and biopsy. Management after diagnosis is controversial. Many experts support surveillance using a standardized biopsy protocol. Advanced imaging techniques (i.e., chromoendoscopy or narrow band imaging) have been shown to increase the diagnostic yield of surveillance biopsies and are becoming common. However, the effect of surveillance on patient outcomes, including survival, is unsettled. The debate is also fueled by uncertainty regarding the

potential of low-grade dysplasia to regress spontaneously or progress to cancer. Low-grade dysplasia can be managed conservatively, with more aggressive and frequent surveillance. Low- and high-grade dysplasia can also be treated by radiofrequency ablation or endoscopic mucosal resection. Multifocal high-grade dysplasia, which carries a significant risk of progression, or intramucosal carcinoma can be treated by ablation, endoscopic mucosal resection, or surgical esophagectomy. In the past, invasive carcinoma was generally resected surgically, but advances in endoscopic mucosectomy have made it a reasonable alternative, especially for superficial lesions and in patients who are not candidates for esophagectomy because of age, life expectancy, or comorbidities.

ESOPHAGEAL TUMORS

Nearly all esophageal cancers are either adenocarcinoma or squamous cell carcinoma. Squamous cell carcinoma is more common worldwide, but adenocarcinoma is on the rise in the United States and other Western countries. Other malignancies of the esophagus are far less common and include unusual forms of adenocarcinoma, undifferentiated carcinoma, neuroendocrine carcinoma, melanoma, lymphoma, and sarcoma; these are not discussed here. Benign tumors of the esophagus are generally mesenchymal and arise within the esophageal wall, with leiomyoma being most common.

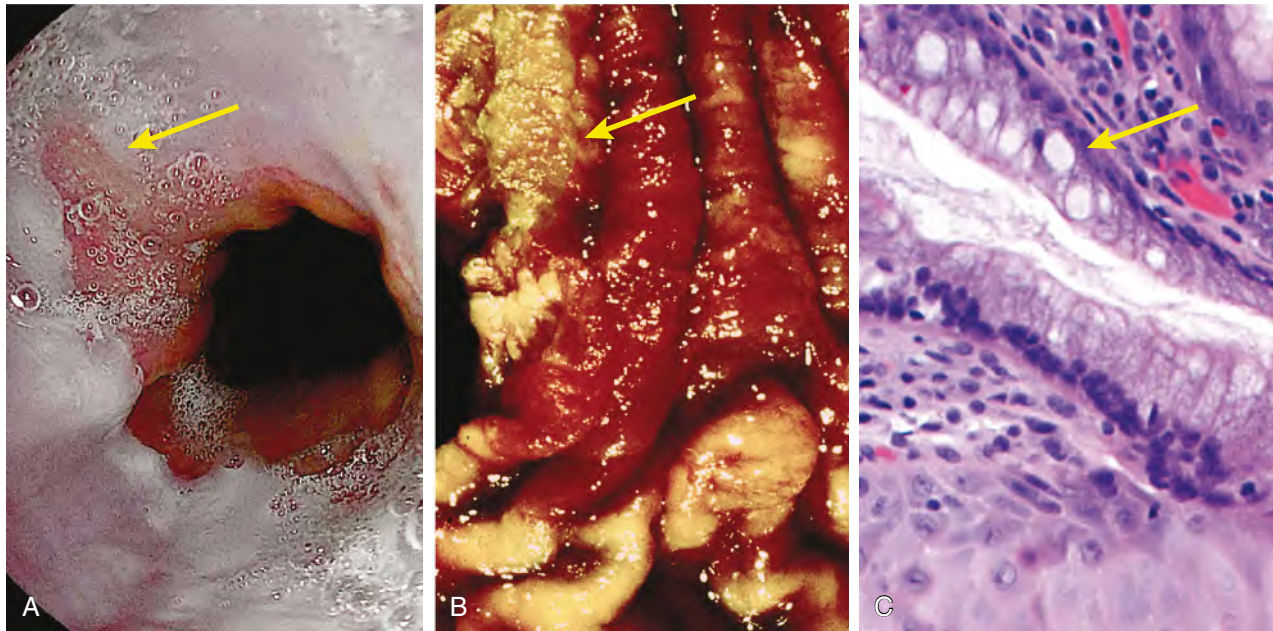


Fig. 17.9 Barrett esophagus. (A) This endoscopic view shows an irregular z-line between gastric and squamous mucosa. A tongue of metaplastic salmon-colored columnar (metaplastic) mucosa extends up and to the left (*arrow*). (B) Grossly, regions of residual pale squamous mucosa (*arrow*) are present within the reddish metaplastic mucosa. The tan gastric mucosa is below the irregular z-line. (C) Stratified squamous epithelium (*bottom*) and intestinal Barrett metaplasia (*top*) in a biopsy taken within the tubular esophagus (above the z-line). Compare the distended vacuole filled with pale mucin (*arrow*) to the small, pink granules within foveolar mucus cells that are adjacent to the squamous mucosa. (A, Courtesy Dr. Jajoo Kunal, Brigham and Women's Hospital and Harvard Medical School, Boston.)

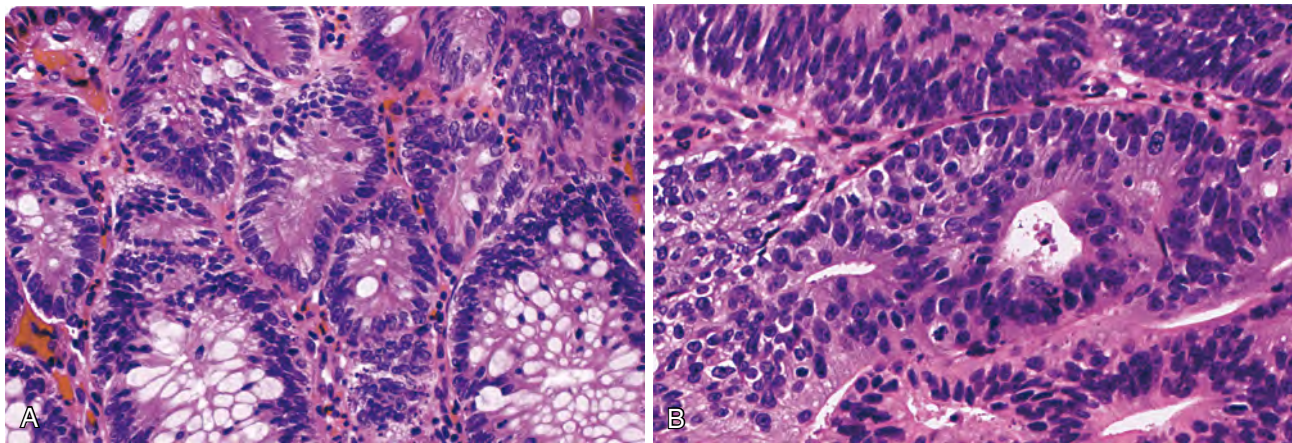


Fig. 17.10 Dysplasia in Barrett esophagus. (A) Abrupt transition from metaplasia (*bottom*) to low-grade dysplasia (*top*). Note nuclear elongation, hyperchromasia, and stratification. (B) High-grade dysplasia is characterized by nuclear rounding, loss of polarity, and gland-within-gland (cribriform) architecture.

Adenocarcinoma

Esophageal adenocarcinoma typically arises in the background of Barrett esophagus and long-standing GERD. Thus, increased rates of esophageal adenocarcinoma may be partly due to the increased incidence of obesity-related and other causes of GERD and Barrett esophagus. Other causes of reduced lower esophageal sphincter pressure and tobacco use, dietary nitrosamines, and radiation exposure are also associated with increased esophageal adenocarcinoma risk. Conversely, reduced risk is associated with diets rich in fresh fruits, vegetables, whole grains, fiber, vitamins C and B6, and folate. Some

serotypes of *Helicobacter pylori* are associated with decreased risk of esophageal adenocarcinoma, perhaps because they cause gastric atrophy, which decreases acid secretion and reduces reflux-induced esophageal damage. Unlike esophageal squamous cell carcinoma, alcohol is not associated with increased adenocarcinoma risk.

Esophageal adenocarcinoma occurs most frequently in individuals of Northern European descent and shows a strong sex bias, being seven-fold more common in males. The incidence varies widely worldwide, with rates being highest in the United States, the United Kingdom, Canada, Australia, the Netherlands, and Brazil and lowest

in Korea, Thailand, Japan, and Ecuador. In countries where esophageal adenocarcinoma is more common, the incidence has increased markedly since 1970, more rapidly than almost any other cancer. For unknown reasons, in the United States these increases have been restricted to males and females of Northern European descent and males of Hispanic descent. As a result, esophageal adenocarcinoma, which represented less than 5% of esophageal cancers in the United States before 1970, now accounts for more than 60% of such tumors. Squamous cell carcinoma, discussed below, remains more common elsewhere in the world.

Pathogenesis

Molecular studies indicate that progression of Barrett esophagus to adenocarcinoma occurs over an extended period through the step-wise accumulation of genetic and epigenetic changes. In molecular terms, esophageal adenocarcinomas are characterized by chromosomal instability, with chromosomal abnormalities and mutations of the tumor suppressor genes *TP53* and *CDKN2A* detected at early stages. *CDKN2A* encodes two tumor suppressor proteins, p16 and p19-ARF, and both allelic loss and hypermethylation-induced epigenetic silencing of these have been described. Neoplastic progression may be accompanied by amplification of several oncogenes, including *ERBB2*, *VEGFA*, *EGFR*, *KRAS*, *CCND1*, and *CDK6*.

MORPHOLOGY

Esophageal adenocarcinoma usually occurs in the distal third of the esophagus and may invade the adjacent gastric cardia (Fig. 17.11A). It initially appears as flat or raised patches in otherwise intact mucosa and may grow into large masses of 5 cm or more in diameter. Alternatively, tumors may infiltrate diffusely or ulcerate and invade deeply. Advanced lesions may be impossible to differentiate from adenocarcinoma of the gastric cardia. There is also significant molecular overlap between these and, as a result, distal esophageal and gastric cardia tumors are often grouped as adenocarcinoma of the esophago-gastric junction.

Microscopically, tumors typically consist of mucin-producing glands (Fig. 17.12A), often with intestinal-type morphology; less frequently, tumors are composed of diffusely infiltrative signet-ring cells (similar to those seen in diffuse gastric cancers) or, in rare cases, small poorly differentiated cells (similar to small cell carcinoma of the lung). Residual Barrett esophagus is frequently adjacent to the tumor and when present suggests an origin within the esophagus rather than the gastric cardia.

Clinical Features

Although esophageal adenocarcinomas are occasionally discovered incidentally during the evaluation of GERD or surveillance of Barrett esophagus, they more commonly present with pain or difficulty in swallowing, progressive weight loss, hematemesis, chest pain, or vomiting. By the time symptoms appear, the tumor has often spread to submucosal lymphatic vessels. As a result, 5-year survival is typically less than 25%. In contrast, 5-year survival is approximately 80% in cases in which adenocarcinoma is limited to the mucosa or submucosa.

Squamous Cell Carcinoma

Esophageal squamous cell carcinoma occurs in adults older than age 45 years. The incidence varies up to 180-fold between and within countries, being more common in lower resource communities. The regions with highest incidence are Iran, central China, Hong Kong, Brazil, and South Africa. A pocket of extremely high incidence in western Kenya has been linked to consumption of a traditional fermented milk, termed *mursik*, which contains the carcinogen acetaldehyde (Chapter 9). In other geographic areas with high esophageal

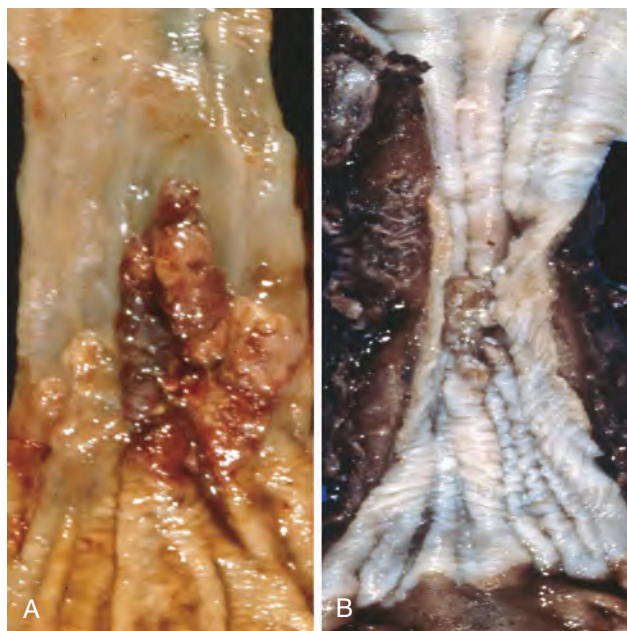


Fig. 17.11 Esophageal cancer. (A) Adenocarcinoma usually occurs distally and, as in this case, often extends into the gastric cardia. (B) Squamous cell carcinoma is most frequently found in the mid-esophagus, where it commonly causes strictures.

squamous cell carcinoma rates, regular ingestion of very hot tea is common. Its incidence is higher in urban areas of the United States, whereas esophageal squamous cell carcinoma is more common in rural areas in other countries.

In low-risk regions, esophageal squamous cell carcinoma is more common in males. There is, however, no sex preference in high-risk areas. In the United States, it is nearly eight-fold more frequent in Americans of African descent than Northern European descent, a striking risk disparity that can only partially be explained by differences in exposure to known risk factors.

Esophageal squamous cell carcinoma risk factors include alcohol and tobacco use, caustic esophageal injury, achalasia, Plummer-Vinson syndrome, diets deficient in fruits or vegetables, and frequent consumption of very hot beverages. Previous radiation to the mediastinum also predisposes individuals to esophageal carcinoma, with most cases occurring 5 or more years after exposure. Tylosis, a rare autosomal dominant disorder marked by hyperkeratosis of the palms and soles, is also strongly associated with esophageal squamous cell carcinoma.

Pathogenesis

Most esophageal squamous cell carcinomas in Europe and the United States are linked to alcohol and tobacco use, which synergize to increase risk. However, esophageal squamous cell carcinoma is also common in some regions where alcohol and tobacco use is uncommon. Thus, nutritional deficiencies, as well as exposure to polycyclic hydrocarbons, nitrosamines, other mutagenic compounds, such as those found in fungus-contaminated foods, and recurrent injury (e.g., from extremely hot tea) are also suspected to contribute to carcinogenesis. Human papillomavirus (HPV) infection might contribute to esophageal squamous cell carcinoma in high-risk regions.

There are several molecular genetic subtypes of esophageal squamous cell carcinoma. The first subtype is characterized by mutations that overlap with those found in head and neck squamous cell

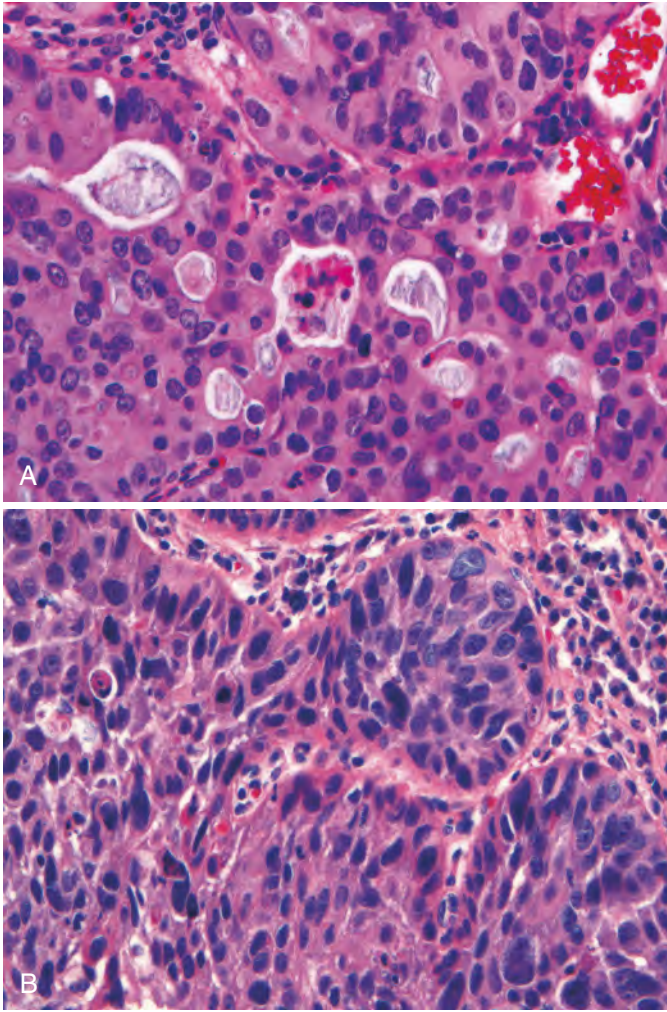


Fig. 17.12 Esophageal cancer. (A) Sheets of esophageal adenocarcinoma cells form irregular glandular lumens. (B) Squamous cell carcinoma is composed of nests of malignant cells that partially recapitulate the organization of benign squamous epithelium.

carcinoma, including genes associated with the Nrf2 pathway, which regulates responses to oxidative stress. The second subtype lacks Nrf2 pathway gene mutations but displays pronounced immune cell infiltration and increased executioner caspase-7 cleavage. These two subtypes, which are chromosomally unstable, share mutations or amplifications in genes that affect cell differentiation and cell cycle regulation. Esophageal squamous cell carcinomas of an uncommon third molecular subtype uniformly carry missense mutations of *SMARCA4*, which regulates chromatin structure, and either *PIK3CA* or *PTEN* but, in contrast to the first two subtypes, is chromosomally stable.

MORPHOLOGY

Squamous cell carcinomas primarily occur in the proximal esophagus, with half developing in the middle third (Fig. 17.11B). Squamous cell carcinoma begins as an in situ lesion termed **squamous dysplasia** (this histopathology is referred to as intraepithelial neoplasia or carcinoma in situ at sites outside the GI tract, e.g., uterine cervix).

Early squamous cell carcinomas are small, gray-white, plaque-like thickenings that grow into tumor masses over months to years. Tumors may be

polypoid or exophytic, and protrude into and obstruct the lumen, or ulcerated or diffusely infiltrative lesions that spread within the wall to cause thickening, rigidity, and luminal narrowing. Squamous cell cancers may also invade surrounding structures including the respiratory tree (causing pneumonia), the aorta (causing catastrophic exsanguination), or the mediastinum and pericardium.

Most squamous cell carcinomas are moderately to well differentiated (Fig. 17.12B). Regardless of histology, symptomatic tumors are generally large and invasive at diagnosis. The rich lymphatic network of the esophagus promotes metastases as well as circumferential and longitudinal spread; intramural satellite tumor nodules may be present several centimeters away from the principal mass. The sites of lymph node metastases vary with tumor location: cancers in the upper third of the esophagus favor cervical lymph nodes; those in the middle third favor mediastinal, paratracheal, and tracheobronchial nodes; and those in the lower third spread to gastric and celiac nodes.

Clinical Features

The onset of esophageal squamous cell carcinoma is insidious. Patients typically have dysphagia, odynophagia (pain on swallowing), or obstruction at presentation and may have unknowingly adjusted to progressive esophageal obstruction by altering their diet from solid to liquid foods. Weight loss and debilitation ensue due to impaired nutrition and tumor cachexia. Hemorrhage and sepsis may accompany tumor ulceration, and symptoms of iron deficiency are often present. Occasionally, the first symptoms are caused by aspiration of food via a tracheoesophageal fistula (caused by extension of the tumor into the trachea).

Increased prevalence of endoscopic screening has led to earlier detection of esophageal squamous cell carcinoma. This is significant because 5-year survival rates are 75% in individuals with superficial lesions but much lower in patients with more advanced tumors. Lymph node metastases are associated with poor prognosis. The overall 5-year survival rate in the United States remains less than 20% and varies by tumor stage and patient age, ethnicity, and sex.

KEY CONCEPTS

Esophageal Diseases

- Abnormalities of esophageal motility include nutcracker esophagus and diffuse esophageal spasm.
- Achalasia, characterized by incomplete lower esophageal sphincter relaxation, increased lower esophageal sphincter tone, and esophageal aperistalsis, is a common form of functional esophageal obstruction. It can be primary or secondary, with the latter form most commonly due to *Trypanosoma cruzi* infection.
- Mallory-Weiss tears of mucosa at the gastroesophageal junction develop as a result of severe retching or vomiting.
- Esophagitis can result from chemical or infectious mucosal injury. Infection is most common in immunocompromised individuals.
- The most prevalent cause of esophagitis is reflux of gastric acid into the esophagus (GERD).
- Eosinophilic esophagitis is an immune-mediated disorder that is strongly associated with food allergy, allergic rhinitis, or asthma. Its prevalence is increasing.
- Gastroesophageal varices are a consequence of portal hypertension and are present in half of cirrhosis patients.

- Barrett esophagus develops in patients with chronic GERD and represents columnar metaplasia of the esophageal squamous mucosa.
- Barrett esophagus is a risk factor for development of esophageal adenocarcinoma.
- Esophageal squamous cell carcinoma is associated with alcohol and tobacco use, caustic esophageal injury, achalasia, tylosis, and Plummer-Vinson syndrome.

STOMACH

Disorders of the stomach are a frequent cause of clinical disease, with inflammatory and neoplastic lesions being most common. In the United States, disorders with gastric symptoms account for nearly one-third of all health care spending on GI disease. Gastric cancer remains a leading cause of death worldwide despite decreasing incidence in certain locales such as the United States.

The stomach is divided into four major anatomic regions: cardia, fundus, body, and antrum, each with somewhat distinctive cells that influence the location and pathogenesis of specific disorders. The cardia and antrum are lined mainly with mucin-secreting foveolar cells that form small glands. The antral glands are similar but also contain endocrine cells such as G cells that release gastrin to stimulate luminal acid secretion by parietal cells within the gastric fundus and body. The well-developed glands of the body and fundus contain H^+ -secreting parietal cells and chief cells, which produce and secrete digestive enzymes.

GASTROPATHY AND ACUTE GASTRITIS

Inflammation of the gastric mucosa occurs in many conditions and is referred to as gastritis when neutrophils are present and

gastropathy when inflammatory cells are rare or absent. Irritants including nonsteroidal antiinflammatory drugs (NSAIDs), alcohol, and bile are the most common causes of gastropathy. Both gastropathy and gastritis may be asymptomatic or associated with epigastric pain, nausea, and vomiting. In more severe cases there may be mucosal erosion, ulceration, hemorrhage, hematemesis, melena, or, rarely, massive blood loss.

Pathogenesis

The gastric lumen has a pH close to 1, more than a million times more acidic than the blood. This harsh environment contributes to digestion but also has the potential to cause damage. Multiple mechanisms have evolved to protect the gastric mucosa (Fig. 17.13). Foveolar cell secretions form a thin layer of mucus and phospholipids that prevents large food particles from directly touching the epithelium. Mucus also contributes to the formation of an “unstirred” layer of fluid with a neutral pH due to bicarbonate ion secretion by surface epithelial cells. Beneath the mucus, gastric epithelial cells form a physical barrier that limits back diffusion of acid and leakage of other luminal materials, including digestive enzymes, into the lamina propria. Surface foveolar cells are replaced every 3 to 7 days, while parietal and chief cells are more long lived. The rich mucosal vasculature delivers oxygen and nutrients and removes whatever gastric acid may have diffused into the lamina propria.

Gastropathy and gastritis occur when the damaging forces overwhelm the protective factors (see Fig. 17.13). Disruption of protective mechanisms is illustrated by the following examples:

- **Inhibition of cyclooxygenase (COX) by NSAIDs.** NSAIDs inhibit COX-dependent synthesis of prostaglandins E_2 and I_2 , which contribute to nearly all of the above defense mechanisms, including: mucus, bicarbonate, and phospholipid secretion; mucosal

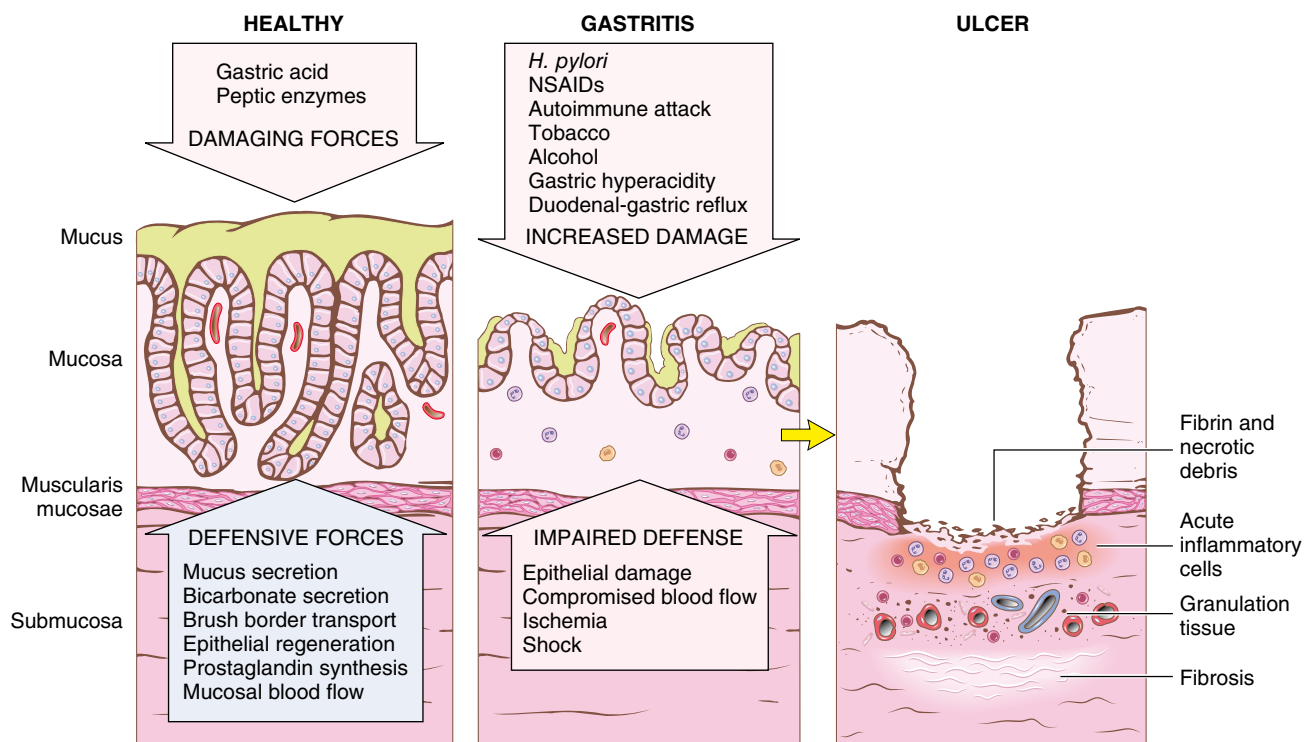


Fig. 17.13 Mechanisms of gastric injury and protection. Defensive forces are normally sufficient to prevent mucosal damage (*left*). When damage increases or defense wanes, the imbalance leads to gastritis with accelerated turnover, mucus disruption, and inflammatory cell infiltration (*middle*). Unchecked, this can progress to form an ulcer (*right*).

blood flow; and epithelial restitution. They also reduce acid secretion. Although COX-1 plays a larger role than COX-2, both isoenzymes contribute to mucosal protection. Thus, while the risk of NSAID-induced gastric injury is greatest with nonselective inhibitors such as aspirin, ibuprofen, and naproxen, selective COX-2 inhibitors such as celecoxib and etoricoxib can also cause gastropathy or gastritis.

- *Inhibition of gastric bicarbonate transporters* by ammonium ions can result in gastric injury in uremic patients and those infected with urease-secreting *H. pylori*.
- *Reduced mucin and bicarbonate secretion* have been suggested as factors that explain the increased susceptibility of older adults to gastritis.
- *Decreased oxygen delivery* may account for an increased incidence of acute gastritis at high altitudes.

Ingestion of harsh chemicals, particularly acids or bases, either accidentally or as a suicide attempt, causes severe gastric injury. Direct cellular damage also contributes to gastritis induced by alcohol consumption, NSAIDs, radiation therapy, and chemotherapy. DNA synthesis or mitotic inhibitors, including those used in cancer chemotherapy, may also cause mucosal damage by inhibiting epithelial renewal.

MORPHOLOGY

Histologically, gastropathy and mild acute gastritis may be difficult to recognize since the lamina propria shows only moderate edema and slight vascular congestion. The surface epithelium is intact, but foveolar cell hyperplasia, with characteristic corkscrew profiles and epithelial proliferation, is often present. In gastritis, neutrophils will be interspersed among epithelial cells or within mucosal gland lumina. The lamina propria contains only a few lymphocytes and plasma cells in acute gastritis.

The presence of neutrophils that have traversed the basement membrane to contact epithelial cells directly is abnormal in all parts of the GI tract and signifies active inflammation, or, in this case, gastritis (rather than gastropathy). Throughout the GI tract, the term *active inflammation* is preferred over acute inflammation since neutrophils may be present in both acute and chronic disease states. Thus, unlike other sites, the term *acute* refers to disease duration rather than the inflammatory pattern.

As injury progresses, erosions (i.e., superficial mucosal defects) are accompanied by neutrophilic infiltrates and fibrin-containing exudate. Hemorrhage may cause dark puncta within hyperemic mucosa. Concurrent erosion and hemorrhage are termed **acute erosive hemorrhagic gastritis**. Although large areas of the gastric surface may be denuded, the involvement is typically superficial.

STRESS-RELATED MUCOSAL DISEASE

Stress-related mucosal disease occurs in patients with severe trauma, extensive burns, intracranial disease, major surgery, serious medical disease, and other forms of severe physiologic stress. In some cases, stress-related ulcers are given specific names based on location and clinical associations. For example:

- *Stress ulcers* are most common in the setting of shock, sepsis, or severe trauma.
- *Curling ulcers* refer to ulcers occurring in the proximal duodenum and stomach in patients with severe burns or trauma.
- *Cushing ulcers* refer to gastric, duodenal, and esophageal ulcers arising in those with intracranial disease. They have a greater risk of perforation.

Pathogenesis

Stress-related gastric mucosal injury is most often related to local ischemia. Underlying causes include systemic hypotension or reduced blood flow caused by stress-induced splanchnic vasoconstriction. Upregulation and increased release of the vasoconstrictor endothelin-1 can also contribute to ischemic gastric mucosal injury.

Ulcers associated with intracranial injury are thought to be caused by direct stimulation of vagal nuclei, which causes hypersecretion of gastric acid. Systemic acidosis, a frequent finding in these settings, may also contribute to mucosal injury by reducing the pH gradient that promotes washout of acid that has diffused back into the lamina propria.

MORPHOLOGY

Stress-related gastric mucosal injury ranges from shallow erosions to lesions that penetrate the mucosa. These benign ulcers are typically rounded, less than 1 cm in diameter, and have a base that is frequently stained brown to black by acid digestion of extravasated blood. Unlike peptic ulcers, which are usually solitary and located in the antrum, acute stress ulcers are found throughout the stomach and are often multiple. Microscopically, acute stress ulcers are sharply demarcated, with relatively normal adjacent mucosa. Serositis may be present, but the scarring and blood vessel thickening that characterize chronic peptic ulcers are conspicuously absent. Healing with complete re-epithelialization occurs within days to weeks after successful treatment of the underlying condition or removal of the offending agent, such as NSAIDs.

Clinical Features

Most critically ill patients admitted to hospital intensive care units have histologic evidence of gastric mucosal damage. Fortunately, bleeding is limited, and only 1% to 4% of patients require transfusion. Other complications, including perforation, are even less common. Proton pump inhibitors are frequently used prophylactically to blunt the impact of stress ulceration, but correction of the underlying condition is, ultimately, the most effective approach to therapy.

Non-stress-related causes of gastric bleeding include the following conditions:

- *Dieulafoy lesion* consists of an abnormal submucosal arteriole that is usually found within the lesser curvature of the stomach near the gastroesophageal junction. Erosion of the overlying epithelium can cause recurrent gastric bleeding that, while usually self-limited, can be copious. NSAID use may enhance bleeding risk.
- *Gastric antral vascular ectasia* (GAVE) is responsible for 4% of nonvariceal upper GI bleeding. It can be recognized endoscopically as longitudinal stripes of edematous, erythematous mucosa that alternate with less severely injured, paler mucosa imparting an appearance sometimes referred to as watermelon stomach. The erythematous stripes are created by ectatic mucosal vessels. Histologically, the antral mucosa shows reactive gastropathy with dilated capillaries containing fibrin thrombi. While most often idiopathic, GAVE can be associated with cirrhosis and systemic sclerosis. Recurrent bleeding may produce a positive test for fecal blood and lead to iron deficiency anemia.
- *Systemic coagulopathy* is an additional cause of gastric damage that may produce lesions resembling stress ulcers.

CHRONIC GASTRITIS AND ITS COMPLICATIONS

***H. pylori* infection is a common cause of chronic gastritis.** Other causes include chronic NSAID use, autoimmune gastritis; radiation

injury; chronic bile reflux; mechanical injury (e.g., an indwelling naso-gastric tube); and involvement by systemic diseases such as Crohn disease, amyloidosis, or graft-versus-host disease.

Relative to acute gastritis, the symptoms associated with chronic gastritis are typically less severe but more persistent. Nausea and upper abdominal pain are typical, sometimes with vomiting, but hematemesis is uncommon.

Helicobacter pylori Gastritis

H. pylori are spiral-shaped or curved bacilli initially discovered by two Australians, pathologist Robin Warren and internal medicine resident Barry Marshall. Although Warren and Marshall ultimately shared the 2005 Nobel Prize, their initial report was met with skepticism. After failing to develop an animal model, Marshall had a baseline endoscopy and biopsy and then drank a broth containing cultured *H. pylori*. He developed mild nausea, and repeat endoscopy and biopsy documented *H. pylori* colonization and gastritis. Marshall recovered after taking antibiotics. Most often, symptoms, if any, of acute *H. pylori* infection are mild and do not typically come to medical attention. Patients ultimately present with chronic gastritis, gastric ulcer, or duodenal ulcer and *H. pylori* is detected in gastric biopsies (Fig. 17.14).

In the United States, *H. pylori* infection is associated with low income, household crowding, limited education, residence in rural areas,

birth outside of the United States, and age over 60 years. The prevalence of infection is also greater in those of African American or Mexican Hispanic descent as well as some immigrant populations. Humans are the primary carriers, suggesting that transmission is mainly by the fecal-oral route. Infection is typically acquired in childhood and persists for life without treatment. Improved sanitation explains reduced *H. pylori* infection rates among younger people in high-resource countries, where *H. pylori* infection is currently uncommon before 10 years of age, is found in only 10% of those between the ages of 18 and 30 years but is present in up to 50% of those older than 65. It follows that childhood environment is a critical risk factor for *H. pylori* colonization.

Pathogenesis

***H. pylori* infection most often presents as a predominantly antral gastritis with normal or increased acid production.** Organisms are most readily detected in the antrum, where inflammation enhances local gastrin production. The resulting gastric body parietal cell expansion and increased acid secretion lead to greater risk of gastric or duodenal peptic ulcer disease (see later). Alternatively, longstanding *H. pylori* gastritis may progress to involve the gastric body and fundus. This may result in atrophic gastritis with reduced parietal cell mass and intestinal metaplasia. In contrast to autoimmune gastritis (see later), atrophy induced by *H. pylori* is not associated with autoantibodies and is typically patchy. Parietal cell loss in *H. pylori* gastritis can result in reduced acid secretion that, in turn, stimulates gastrin production. However, because some parietal cells survive, limited acid secretion continues and gastrin is not increased to the extent seen in autoimmune gastritis. Nevertheless, decreased acid secretion in *H. pylori* gastritis with atrophy reduces the risk of gastric and duodenal ulcers. This results in an inverse relationship between gastric adenocarcinoma, which is associated with atrophy and intestinal metaplasia, and duodenal ulcers, which are associated with increased acid secretion.

H. pylori has adapted to the ecologic niche provided by gastric mucus. Its virulence is linked to:

- **Flagella**, which allow the bacteria to be motile in viscous mucus
- **Urease**, which generates ammonia from endogenous urea and thereby elevates local gastric pH to enhance bacterial survival
- **Adhesins** that enhance bacterial adherence to surface foveolar cells
- **Toxins**, such as cytotoxin-associated gene A (*CagA*).

Urease and adhesins are both highly conserved and essential for colonization (see Fig. 17.14). Other bacterial virulence factors, including the *CagA* pathogenicity island, which encodes *CagA* and the vacuolating cytotoxin *VacA*, have been studied in experimental systems and shown to affect epithelial functions. Specifically, *CagA*, which is injected into the epithelial cell cytoplasm, activates signaling and structural remodeling. However, suspected contributions of these factors remain correlative. For example, the *CagA* pathogenicity island is present in 50% of *H. pylori* isolates overall but in 90% of *H. pylori* isolates in populations with an increased prevalence of gastric cancer. This may, in part, be because, in addition to the antrum, *CagA*-expressing strains colonize the gastric body and induce proinflammatory cytokine secretion, atrophy, and intestinal metaplasia more effectively than *CagA*-negative *H. pylori*. *CagA* is also a dominant antigen targeted by the immune response to *H. pylori*.

Host factors also play an important role in the outcome of *H. pylori* infection (see Fig. 17.14). Genetic polymorphisms that lead to increased expression of the proinflammatory cytokines tumor necrosis factor (TNF) and IL-1 or decreased expression of the antiinflammatory

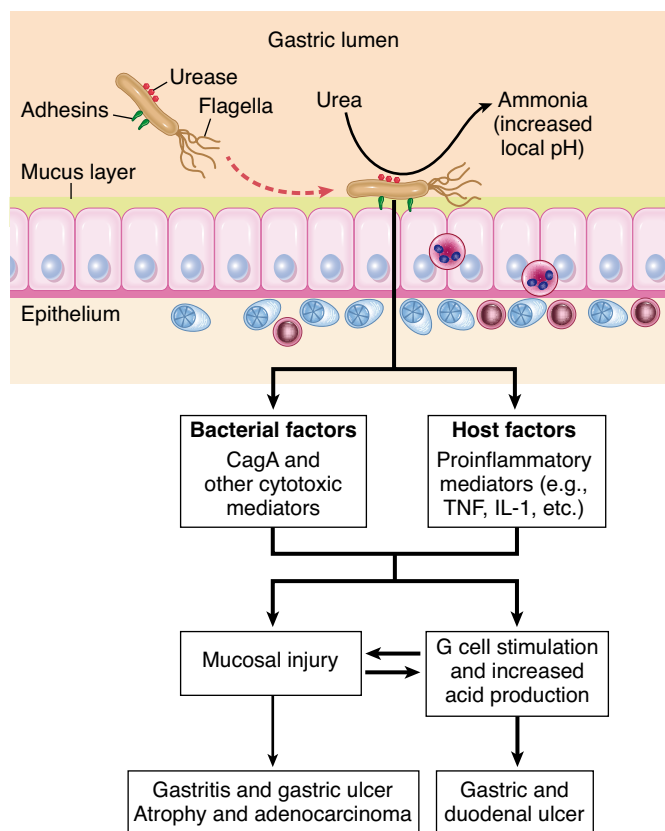


Fig. 17.14 Pathogenesis of *Helicobacter pylori* gastritis. Bacteria bind to gastric foveolar epithelial cells via adhesion proteins and colonize the mucin layer. Urease generates ammonia to buffer local acidity and protects *H. pylori* from gastric acid. *H. pylori* also sense local pH and use flagella to avoid acidic microenvironments and remain close to the epithelial surface. Toxins are thought to directly affect epithelial cells and to modify disease by interacting with host factors, such as cytokine production. *CagA*, Cytotoxin-associated gene A; *G* cell, gastrin-producing cell; *IL*, interleukin; *TNF*, tumor necrosis factor.

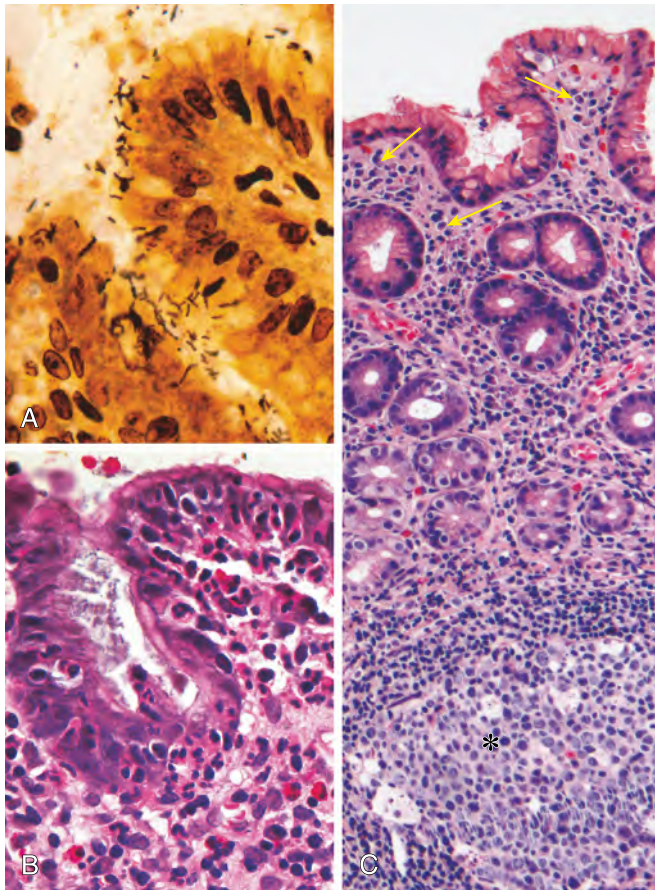


Fig. 17.15 *Helicobacter pylori* gastritis. (A) Spiral-shaped *H. pylori* are highlighted in this Warthin-Starry silver stain. Organisms are abundant within surface mucus. (B) Intraepithelial and lamina propria neutrophils are prominent. (C) Lymphoid aggregates with germinal centers (asterisk) and abundant subepithelial plasma cells (arrows) within the superficial lamina propria are characteristic of *H. pylori* gastritis.

cytokine IL-10 may be associated with development of pangastritis, atrophy, intestinal metaplasia, and gastric cancer. The course of *H. pylori* gastritis is, therefore, the result of interplay between gastroduodenal mucosal defenses, host genetics and immune responses, and bacterial virulence factors.

MORPHOLOGY

Gastric biopsy specimens generally demonstrate *H. pylori* in infected individuals. The organism is concentrated within the superficial mucus overlying epithelial cells in the surface and neck regions. The distribution can be irregular, with areas of heavy colonization adjacent to those with few organisms. Organisms are visible on routine H&E-stained sections but are most readily identified with immunostains or histochemical stains (see Fig. 17.15A).

The antrum is the preferred biopsy site for evaluation of *H. pylori* gastritis because this area is the most commonly infected. In dense colonization, organisms may also be found in oxyntic (acid-producing) mucosa of the fundus and body. *H. pylori*-infected antral mucosa is usually erythematous and has a coarse or even nodular appearance. The inflammatory infiltrate includes large numbers of plasma cells, often in clusters or sheets, within the superficial lamina propria. This is accompanied by increased numbers of lymphocytes,

macrophages, and neutrophils within the lamina propria. Neutrophils infiltrate across the basement membrane (see Fig. 17.15B) and accumulate in the lumens of gastric glands, or pits, to create pit abscesses. When intense, inflammatory infiltrates may create thickened rugal folds, mimicking the endoscopic appearance of early cancers. Lymphoid aggregates, some with germinal centers, are frequently present (see Fig. 17.15C) and represent induced **mucosa-associated lymphoid tissue (MALT)** that has the potential to transform into lymphoma. Thus, risk of lymphoma, as well as adenocarcinoma, is increased in chronic *H. pylori* gastritis.

Clinical Features

The changes seen in atrophic *H. pylori* gastritis are sometimes referred to as environmental metaplastic atrophic gastritis, to distinguish it from autoimmune metaplastic atrophic gastritis (described next). In addition to histologic identification of the organism, several noninvasive diagnostic tests have been developed, including serology for anti-*H. pylori* antibodies, fecal bacterial detection, and the urea breath test, which detects ammonia produced by the bacterial urease. Other diagnostic tests performed on biopsies include the rapid urease test, bacterial culture, or polymerase chain reaction (PCR)-based detection of *H. pylori* DNA.

Effective treatments for *H. pylori* infection include combinations of antibiotics and proton pump inhibitors. Individuals with *H. pylori* gastritis usually improve after treatment, although relapses can occur after incomplete eradication or reinfection, particularly in regions with high endemic colonization rates.

Autoimmune Atrophic Gastritis

Autoimmune atrophic gastritis, in contrast to *H. pylori*-associated gastritis, typically spares the antrum and is often associated with

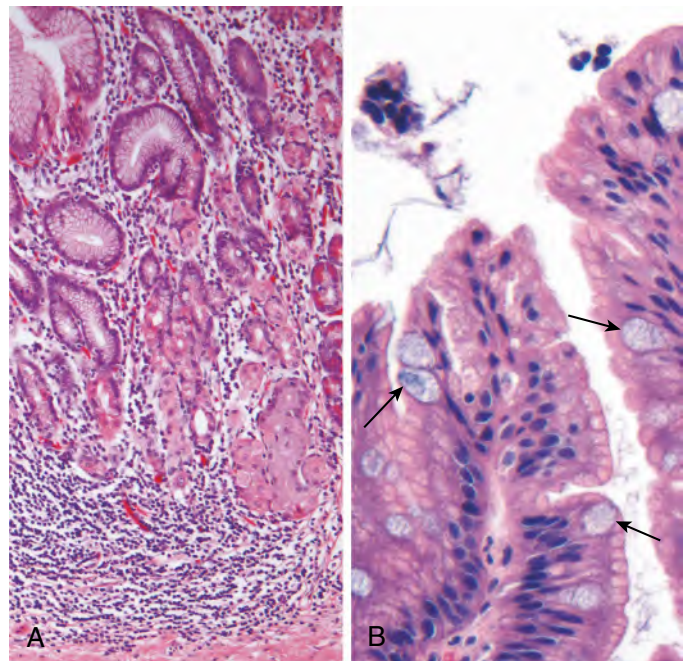


Fig. 17.16 Autoimmune gastritis. (A) Low-magnification image of gastric body demonstrating deep inflammatory infiltrates, primarily composed of lymphocytes, and glandular atrophy. (B) Intestinal metaplasia, recognizable as the presence of goblet cells (arrows) within gastric foveolar epithelium.

TABLE 17.2 Characteristics of *Helicobacter pylori*-Associated and Autoimmune Gastritis

	<i>H. pylori</i> -Associated	Autoimmune
Location	Antrum	Body
Inflammatory infiltrate	Neutrophils, subepithelial plasma cells	Lymphocytes, macrophages
Acid production	Increased to slightly decreased	Decreased
Gastrin secretion	Normal to increased	Increased to markedly increased
Other lesions	Hyperplastic/inflammatory polyps	Neuroendocrine hyperplasia
Serology	Antibodies to <i>H. pylori</i>	Antibodies to parietal cells (H^+ , K^+ -ATPase, intrinsic factor)
Sequelae	Peptic ulcer, adenocarcinoma, MALToma	Atrophy, pernicious anemia, adenocarcinoma, carcinoid tumor
Associations	Low socioeconomic status, poverty, residence in rural areas	Autoimmune disease: thyroiditis, diabetes mellitus, Graves disease

H^+ , K^+ ATPase, Hydrogen potassium adenosine triphosphatase; MALToma, mucosa-associated lymphoid tissue lymphoma.

marked hypergastrinemia (Table 17.2). It accounts for less than 10% of cases of chronic gastritis and has an estimated prevalence of 2% in those over 60 years of age. Autoimmune atrophic gastritis is characterized by:

- Antibodies to parietal cells and intrinsic factor that can be detected in serum and gastric secretions
- Reduced serum pepsinogen I concentration
- Endocrine cell hyperplasia
- Vitamin B₁₂ deficiency
- Defective gastric acid secretion (achlorhydria)

Pathogenesis

Autoimmune atrophic gastritis is associated with loss of parietal cells, which are responsible for secretion of gastric acid, pepsin, and intrinsic factor. The absence of acid production stimulates gastrin release, resulting in hypergastrinemia and hyperplasia of antral gastrin-producing G cells. Intrinsic factor loss results in defective ileal vitamin B₁₂ absorption, which ultimately leads to vitamin B₁₂ deficiency and *pernicious anemia* (a form of megaloblastic anemia described in Chapter 14).

CD4⁺ T cells and antibodies directed against parietal cell components, including H^+ , K^+ -ATPase (proton pump), are considered to be the principal agents of injury in autoimmune atrophic gastritis. Antibodies against H^+ , K^+ -ATPase and intrinsic factor are present in up to 80% of patients and are useful in establishing the diagnosis.

MORPHOLOGY

Autoimmune atrophic gastritis is characterized by **diffuse damage to the oxyntic (acid-producing) mucosa within the body and fundus.** The

antrum and cardia are typically spared. With diffuse atrophy, oxyntic mucosa of the body and fundus appears markedly thinned, and rugal folds are lost. If vitamin B₁₂ deficiency is severe, nuclear enlargement (megaloblastic change) may be seen in epithelial cells. The **inflammatory infiltrate** is composed of lymphocytes, macrophages, and plasma cells, often in association with lymphoid aggregates and follicles. In contrast to the superficial lamina propria inflammation that is typical of *H. pylori* gastritis, inflammation in autoimmune atrophic gastritis is deeper and centered on the gastric glands (see Fig. 17.16A), and neutrophils are not prominent. Parietal and chief cell loss can be extensive. When atrophy is incomplete, residual islands of oxyntic mucosa may give the endoscopic appearance of multiple small polyps or nodules. In other areas, small surface elevations may represent sites of intestinal metaplasia, characterized histologically by the presence of intestinal-type goblet cells, absorptive cells, and, in some cases, Paneth cells (see Fig. 17.16B). Endocrine and enterochromaffin-like cell hyperplasia is commonly present. Rarely, endocrine cell hyperplasia may progress to form small, multicentric, low-grade neuroendocrine (previously referred to as carcinoid) tumors within the gastric body.

Clinical Features

Antibodies to parietal cell components, including H^+ , K^+ -ATPase and intrinsic factor, are present 2 to 3 decades prior to atrophy and complications such as pernicious anemia; hence, progression to clinically evident disease is slow. The median age at diagnosis is 60 years. Females are affected slightly more often than males. Pernicious anemia and autoimmune atrophic gastritis are often associated with other autoimmune diseases including Hashimoto thyroiditis, type 1 diabetes mellitus, Addison disease, and others. These associations, along with concordance in some monozygotic twins and clustering of disease in families, support a genetic predisposition, but the responsible genetic variants are not known.

The clinical features are mainly related to vitamin B₁₂ deficiency, including megaloblastic anemia; atrophic glossitis, in which the tongue becomes smooth and beefy red; epithelial megaloblastosis; malabsorptive diarrhea; and many others. Clinical features and pathologic changes related to the effects of vitamin B₁₂ deficiency in the marrow and the CNS (e.g., subacute combined degeneration of the spinal cord) are described in detail in Chapters 14 and 28, respectively.

Uncommon Forms of Gastritis

Eosinophilic Gastritis

Eosinophilic gastritis is characterized by tissue damage associated with dense infiltrates of eosinophils in the mucosa and muscularis, usually in the antral or pyloric region. Other regions of the GI tract may also be involved, and there are typically peripheral eosinophilia and increased serum IgE levels. Allergic reactions are one cause of eosinophilic gastritis, with cow's milk and soy protein being the most common allergens in children. Eosinophilic gastritis can also occur in association with immune disorders such as systemic sclerosis and inflammatory myopathy, parasitic infections, and even *H. pylori* infection.

Lymphocytic Gastritis

This disease preferentially affects adult females and produces nonspecific abdominal symptoms. It is idiopathic, but approximately 40% of cases are associated with celiac disease, suggesting an immune-mediated pathogenesis. Lymphocytic gastritis typically affects the entire stomach and is often referred to as *varioliform gastritis* based on the distinctive endoscopic appearance (characterized by thickened folds covered by small nodules with central aphthous ulceration). Histologically there is a marked increase in the number of intraepithelial T lymphocytes.

TABLE 17.3 Risk Factors for Peptic Ulcer Disease

<i>Helicobacter pylori</i> infection
Cigarette use (synergizes with <i>H. pylori</i> for gastric PUD)
Chronic obstructive pulmonary disease
Illicit drugs, e.g., cocaine, that reduce mucosal blood flow
NSAIDs (potentiated by corticosteroids)
Alcoholic cirrhosis (primarily duodenal PUD)
Psychological stress (can increase gastric acid secretion)
Endocrine cell hyperplasia (can stimulate parietal cell growth and gastric acid secretion)
Zollinger-Ellison syndrome (PUD of stomach, duodenum, and jejunum)
Viral infection (cytomegalovirus, herpes simplex virus)

NSAIDs, Nonsteroidal antiinflammatory drugs; PUD, peptic ulcer disease.

Granulomatous Gastritis

The descriptive term granulomatous gastritis is applied to any gastritis that contains granulomas. It encompasses a diverse group of diseases with widely varying clinical and pathologic features. Many cases are idiopathic. In Western populations, gastric involvement by Crohn disease is the most common specific cause of granulomatous gastritis followed by sarcoidosis and infections that include mycobacteria, fungi, CMV, and *H. pylori*. Transmural granulomatous inflammation can lead to narrowing and rigidity of the gastric antrum.

Peptic Ulcer Disease

Peptic ulcer disease (PUD) refers to chronic mucosal ulceration affecting the duodenum or stomach and is almost always associated with *H. pylori* infection, NSAIDs, or cigarette smoking. The most common form of PUD occurs within the gastric antrum or duodenum due to chronic *H. pylori*-induced antral gastritis, which is associated with increased gastric acid secretion and decreased duodenal bicarbonate secretion. Because of reduced acid secretion, individuals with gastric mucosal atrophy are generally protected from antral and duodenal ulcers. Acid secreted by ectopic gastric mucosa within the duodenum, an ileal Meckel diverticulum, or esophageal ectopic gastric mucosa (an inlet patch) may also cause peptic injury.

The incidence of PUD is falling as the prevalence of *H. pylori* infection decreases. However, PUD in patients older than 60 years has increased due to growing NSAID use. This effect is amplified by *H. pylori* infection, which interacts with the doses of aspirin that many take for cardiovascular benefits to increase gastric injury. PUD is also associated with cigarette use and cardiovascular disease, likely due to reductions in mucosal blood flow, oxygenation, and healing. Other risk factors for PUD are listed in Table 17.3.

Pathogenesis

PUD results from imbalances between defense mechanisms and damaging factors that cause chronic gastritis (discussed earlier). Thus, PUD generally develops on a background of chronic gastritis. The reasons why some people develop only chronic gastritis while others develop PUD are poorly understood. However, as with *H. pylori* gastritis, it is likely that host factors as well as variation in bacterial pathogenicity are involved.

MORPHOLOGY

Peptic ulcers occur in the context of chronic gastritis but are most frequently found in the proximal duodenum, within a few centimeters of the pyloric valve. Foveolar metaplasia, in which gastric-type mucus cells are present, is common

in chronic duodenal peptic disease and may be a protective response, as gastric epithelia are less sensitive to acid than intestinal epithelia. Gastric peptic ulcers are predominantly located along the lesser curvature near the interface of the body and antrum.

Peptic ulcers are solitary in more than 80% of patients and form a round to oval, **sharply punched-out defect** (Fig. 17.17A). The mucosal margin is usually level with the surrounding mucosa but may overhang the base, particularly on the proximal side. In contrast, **heaped-up margins are more characteristic of cancers**. **Perforation** into the peritoneal cavity is a surgical emergency that may be identified as free air under the diaphragm on upright abdominal radiographs.

The base of peptic ulcers is smooth and clean because of peptic digestion of exudate. Active ulcers may be lined by a thin layer of fibrinoid debris with a predominantly neutrophilic inflammatory infiltrate. Beneath this, granulation tissue with immature vessels, neutrophils, mononuclear leukocytes, and a fibrous or collagenous scar forms the ulcer base (Fig. 17.17B). Larger vessels within the scarred area are typically thickened and are occasionally thrombosed. Bleeding from these vessels may cause life-threatening hemorrhage. Scarring may involve the entire thickness of the wall and draw the surrounding mucosa into folds that radiate outward.

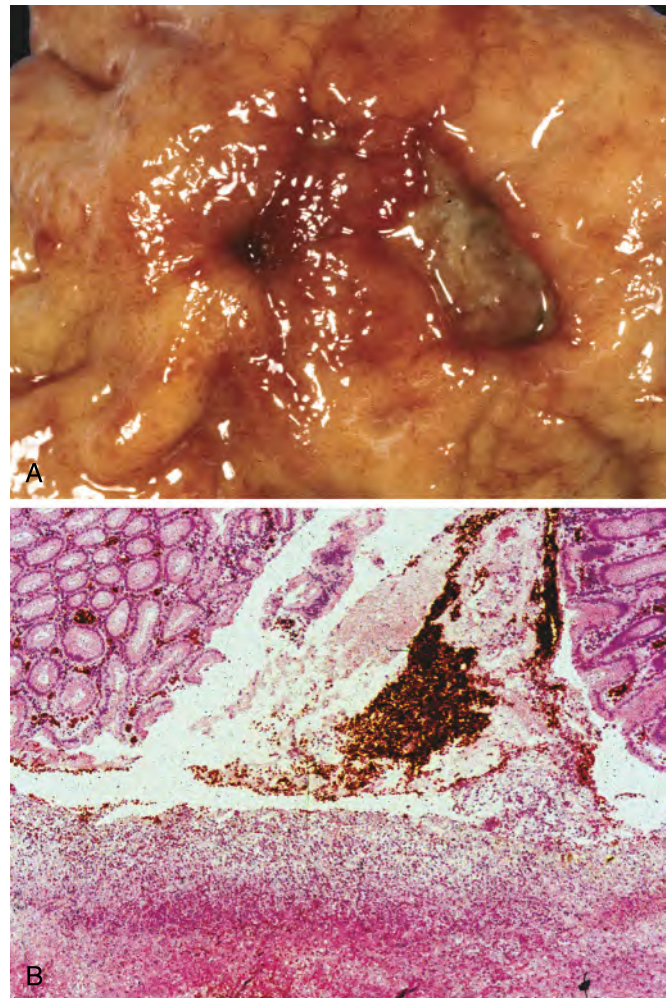


Fig. 17.17 Peptic ulcer. (A) Mucosal defect with clean edges. (B) The ulcer base is composed of granulation tissue.

TABLE 17.4 Complications of Peptic Ulcer Disease

Bleeding
Occurs in 15%–20% of patients
Most frequent complication
May be life threatening
Accounts for 25% of ulcer-related deaths
May be the first indication of an ulcer
Perforation
Occurs in up to 5% of patients
Accounts for two-thirds of ulcer-related deaths
Is rarely first indication of an ulcer
Obstruction
Mostly in chronic ulcers
Secondary to edema or scarring
Occurs in about 2% of patients
Most often associated with pyloric channel ulcers
May occur with duodenal ulcers
Causes incapacitating, crampy abdominal pain
Can rarely cause total obstruction and intractable vomiting

Clinical Features

Peptic ulcers typically come to clinical attention because of epigastric burning or aching pain. Others present with iron deficiency anemia, hem-orrhage, or perforation (Table 17.4). The pain tends to occur 1 to 3 hours after meals during the day, is worse at night (usually between 11 p.m. and 2 a.m.), and is relieved by alkali or food. Nausea, vomiting, bloating, belching, and significant weight loss are additional manifestations. With penetrating ulcers, the pain is occasionally referred to the back, left upper quadrant, or chest, where it may be misinterpreted as cardiac in origin.

Current therapies for PUD are aimed at *H. pylori* eradication and neu-tralization of gastric acid, primarily with proton pump inhibitors. It is also important to withdraw other offending agents such as NSAIDs, including selective COX-2 inhibitors, that may interfere with mucosal healing. While peptic ulcers were previously notoriously difficult to eradicate, the recur-rence rate is now less than 20% following successful clearance of *H. pylori*.

Other Complications of Chronic Gastritis

Mucosal Atrophy and Intestinal Metaplasia

As noted above, chronic gastritis may lead to significant loss of parietal cell mass. This oxyntic atrophy is frequently associated with intestinal metaplasia, recognized by the presence of goblet cells, and is strongly associated with increased risk of gastric adenocarcinoma. The risk of adenocarcinoma is greatest in autoimmune metaplastic atrophic gas-tritis. This may be because achlorhydria of gastric mucosal atrophy permits overgrowth of bacteria that produce carcinogenic nitrosa-mines. Intestinal metaplasia may regress after *H. pylori* eradication.

Dysplasia

Chronic gastritis exposes the epithelium to inflammation-related free radi-cal damage and proliferative stimuli that include regenerative responses to injury and increased production of gastrin, a gastric epithelial mitogen. Over time this combination of stressors can lead to the accumulation of genetic alterations that result in carcinoma. Preinvasive in situ lesions can be recognized histologically as dysplasia. The morphologic hallmarks of dysplasia are variations in epithelial size, shape, and orientation along with coarse chromatin texture, hyperchromasia, and nuclear enlargement. The distinction between dysplasia and regenerative epithelial changes induced by active inflammation can be a challenge for the pathologist, since in-creased epithelial proliferation and mitotic figures may be prominent in both. One clue is that reactive epithelial cells mature as they reach the mu-cosal surface, while dysplastic lesions remain cytologically immature.

Gastritis Cystica

Gastritis cystica is an exuberant reactive epithelial proliferation associ-ated with mural entrapment of epithelial-lined cysts. These may be found within the submucosa (gastritis cystica polyposia) or deeper layers of the gastric wall (gastritis cystica profunda). Because of the association with chronic gastritis and partial gastrectomy, it is pre-sumed that gastritis cystica is trauma-induced, but the reasons for the development of epithelial cysts within deeper portions of the gastric wall are not clear. Regenerative epithelial changes can be prominent in the entrapped epithelium, and gastritis cystica can therefore be mistaken for invasive adenocarcinoma.

HYPERTROPHIC GASTROPATHIES

Hypertrophic gastropathies are uncommon diseases characterized by giant “cerebriform” enlargement of the rugal folds due to epithe-lial hyperplasia without inflammation. As might be expected, the hypertrophic gastropathies are linked to excessive growth factor release. Two well-defined examples are Ménétrier disease and Zollinger-Ellison syndrome, the morphologic features of which are compared with other gastric proliferations in Table 17.5.

Ménétrier Disease

Ménétrier disease is a rare protein-losing disorder associated with excessive secretion of transforming growth factor (TGF)- α , which is an EGF receptor (EGFR) ligand, and hyperactivation of the epider-mal growth factor receptor on gastric epithelial cells. Some cases occur in association with viral or *H. pylori* infection. Diffuse hyper-plasia of foveolar epithelium within the body and fundus as well as hypoproteinemia due to albumin loss (which can approach 10 g/day) across the gastric mucosa characterize Ménétrier disease. Secondary symptoms such as weight loss, diarrhea, and peripheral edema are common. Systemic effects elevated levels of TGF- α , including hyper-plasia of the liver, pancreas, and GI tract (beyond the stomach), muscle and adipose tissue loss, and psoriasis, may also be present. Risk of gastric adenocarcinoma is also increased in adults with Mé-nétrier disease.

MORPHOLOGY

Ménétrier disease is characterized by **irregular enlargement of the gastric rugae**. Some areas may appear polypoid. Enlarged rugae are present in the body and fundus (Fig. 17.18A), but the antrum is generally spared. Histologically, the most characteristic feature is hyperplasia of foveolar mucous cells. The glands are elongated with a corkscrew-like appearance, and cystic dilation is common (Fig. 17.18B). Inflammation is usually modest, although some cases show marked intraepithelial lymphocytosis. Diffuse or patchy glandular atrophy, evident as hypoplasia of parietal and chief cells, is typically intermixed with epithelial hyperplasia.

Clinical Features

Treatment of Ménétrier disease is supportive, with intravenous albu-min and parenteral nutritional supplementation. In cases associated with herpesvirus, CMV, or *H. pylori*, treatment of the infection may be helpful. Antibodies that block epidermal growth factor receptor activa-tion can be effective. In severe cases gastrectomy remains a therapeutic option. In children, Ménétrier disease is often self-limited and fre-quently occurs after CMV or other infections, potentially due to CMV-induced EGF receptor activation. Regardless, viral clearance is associ-ated with normalization of epithelial proliferation and restoration of gastric anatomy.

TABLE 17.5 Hypertrophic Gastropathies and Gastric Polyps

	Location	Predominant Cell Type	Inflammatory Infiltrate	Symptoms	Risk Factors	Association With Dysplasia or Adenocarcinoma
Ménétrier Disease	Body and fundus	Mucous	Limited, lymphocytes	Hypoproteinemia, weight loss, diarrhea	None (adults) CMV (children)	Yes
Zollinger-Ellison Syndrome	Fundus	Parietal > mucous, endocrine	Neutrophils	Peptic ulcers	Multiple endocrine neoplasia	No
Inflammatory and Hyperplastic Polyps	Antrum > body	Mucous	Neutrophils and lymphocytes	Similar to chronic gastritis	Chronic gastritis, <i>H. pylori</i>	Yes
Gastritis Cystica	Body	Mucous, cyst-lining	Neutrophils and lymphocytes	Similar to chronic gastritis	Trauma, prior surgery	No
Fundic Gland Polyps	Body and fundus	Parietal and chief	None	None, nausea	PPIs, FAP	Syndromic (FAP) only
Gastric Adenomas	Antrum > body	Dysplastic, intestinal	Variable	Similar to chronic gastritis	Chronic gastritis, atrophy, intestinal metaplasia	Frequent

CMV, Cytomegalovirus; FAP, familial adenomatous polyposis; PPIs, proton pump inhibitors.

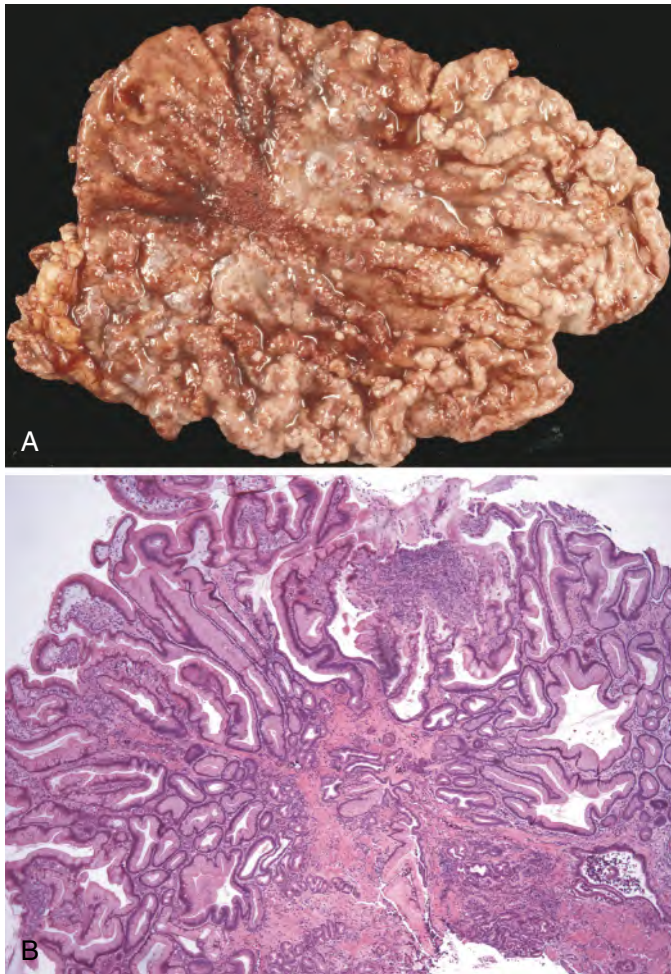


Fig. 17.18 Ménétrier disease. (A) Marked hypertrophy of rugal folds. (B) Foveolar hyperplasia with elongated and focally dilated glands. (Courtesy Dr. M. Kay Washington, Vanderbilt University, Nashville, TN.)

Zollinger-Ellison Syndrome

Zollinger-Ellison syndrome is caused by gastrin-secreting neuroendocrine tumors. These gastrinomas are almost always found in the small intestine or pancreas. Patients often present with duodenal ulcers or chronic diarrhea. Within the stomach, the most remarkable feature is a marked increase in oxyntic mucosal thickness due to massive parietal cell hyperplasia. Gastrin also induces hyperplasia of mucous neck and endocrine cells within oxyntic mucosa.

Treatment of individuals with Zollinger-Ellison syndrome includes blockade of acid hypersecretion, typically with proton pump inhibitors. Acid suppression allows peptic ulcers to heal and prevents gastric perforation, allowing treatment of the underlying gastrinoma, which is the main determinant of long-term survival.

Gastrinomas are sporadic, solitary lesions in 75% of patients. These tumors are typically >2 cm in diameter and located in the pancreas. The remaining 25% of patients with gastrinomas have multiple endocrine neoplasia type 1 (MEN1). These patients are younger and generally have multiple, <2 cm duodenal tumors that commonly metastasize. Metastasis is also more common with sporadic gastrinomas with *MEN1* mutations. Somatostatin receptor scintigraphy or endoscopic ultrasonography can locate tumors. Although proton pump inhibitors are first-line therapy, somatostatin analogues can provide some benefit to patients with metastatic disease.

KEY CONCEPTS

Gastritis

- Gastritis is a mucosal inflammatory process. When inflammatory cells are absent or rare, the term gastropathy can be applied.
- The clinical spectrum of acute gastritis ranges from asymptomatic disease to mild epigastric pain, nausea, and vomiting. Causative factors include any agent or disease that interferes with gastric mucosal protective mechanisms.
- The most common cause of chronic gastritis is *H. pylori* infection. Other injurious agents include NSAIDs and alcohol.
- *H. pylori* gastritis typically affects the antrum and is associated with increased gastric acid production. Later in the course of disease, the body can

be involved, and resulting glandular atrophy can lead to mildly reduced acid production. Host immune responses and bacterial characteristics determine whether the infection remains antral or progresses to pangastritis and atrophy.

- *H. pylori* gastritis induces MALT that can give rise to B-cell lymphomas (MALTomas).
- After *H. pylori* and NSAIDs, autoimmune atrophic gastritis is the most frequent cause of chronic gastritis. It results in atrophy of the gastric body oxyntic glands, which leads to decreased gastric acid production, antral G-cell hyperplasia, and vitamin B₁₂ deficiency. Anti-parietal cell and anti-intrinsic factor antibodies are typically present.
- Intestinal metaplasia develops in all forms of chronic gastritis and is a risk factor for gastric adenocarcinoma.
- Peptic ulcer disease is usually secondary to chronic *H. pylori*-induced gastritis and the resulting hyperchlorhydria. Ulcers can develop in the stomach or duodenum and usually heal after suppression of gastric acid production and *H. pylori* eradication.
- Ménétrier disease is a rare disorder caused by excessive secretion of TGF- α that is characterized by diffuse foveolar hyperplasia and protein-losing enteropathy.
- Zollinger-Ellison syndrome is caused by gastrin-secreting tumors that cause parietal cell hyperplasia and acid hypersecretion; 60% to 90% of these gastrinomas are malignant.

GASTRIC POLYPS AND TUMORS

Polyps, nodules or masses that project above the level of the surrounding mucosa are identified in up to 5% of upper GI endoscopies. Polyps may develop because of epithelial or stromal cell hyperplasia, inflammation, ectopia, or neoplasia. Only the most common types of gastric polyps will be discussed here (Peutz-Jeghers and juvenile polyps are discussed with intestinal polyps).

Inflammatory and Hyperplastic Polyps

Up to 75% of all gastric polyps are inflammatory or hyperplastic polyps, which share a common etiology and are differentiated based on histology. Polyp incidence is correlated with the regional prevalence of *H. pylori* infection. Inflammatory or hyperplastic polyps are most common in individuals between 50 and 60 years of age and usually

develop in association with chronic gastritis, which initiates the injury that leads to reactive hyperplasia and polyp growth. The risk of dysplasia in inflammatory polyps ranges from 1% to 20% and increases sharply in pedunculated polyps greater than 1 cm in diameter.

MORPHOLOGY

The majority of inflammatory or hyperplastic polyps are smaller than 1 cm in diameter. Polyps are frequently multiple, particularly in individuals with atrophic gastritis. Hyperplastic polyps are ovoid in shape and have a smooth surface. In contrast, superficial erosions are common in inflammatory polyps. Microscopically, polyps have irregular, cystically dilated, and elongated foveolar glands (Fig. 17.19A). The lamina propria is typically edematous with variable degrees of acute and chronic inflammation (Fig. 17.19B).

Fundic Gland Polyps

Fundic gland polyps occur sporadically and in individuals with germline mutations in the *APC* gene (the cause of familial adenomatous polyposis) or the DNA repair gene *MUTYH* (both discussed later). The prevalence of sporadic fundic gland polyps has increased markedly in recent years due to increased use of proton pump inhibitors. These drugs inhibit acid production, which leads to increased gastrin secretion and, in turn, oxyntic gland growth. Fundic gland polyps may be asymptomatic or associated with nausea, vomiting, or epigastric pain.

MORPHOLOGY

Fundic gland polyps occur in the gastric body and fundus and are well-circumscribed lesions with a smooth surface. They may be single or multiple and are composed of cystically dilated, irregular glands lined by parietal, chief, and foveolar mucus cells. Inflammation is typically absent or minimal (Fig. 17.19C). Dysplasia may occur in familial adenomatous polyposis (FAP)-associated fundic gland polyps, and rare cancers have been reported. In contrast, sporadic fundic gland polyps, including those arising in association with proton pump inhibitor therapy, do not carry a significant risk of neoplasia.

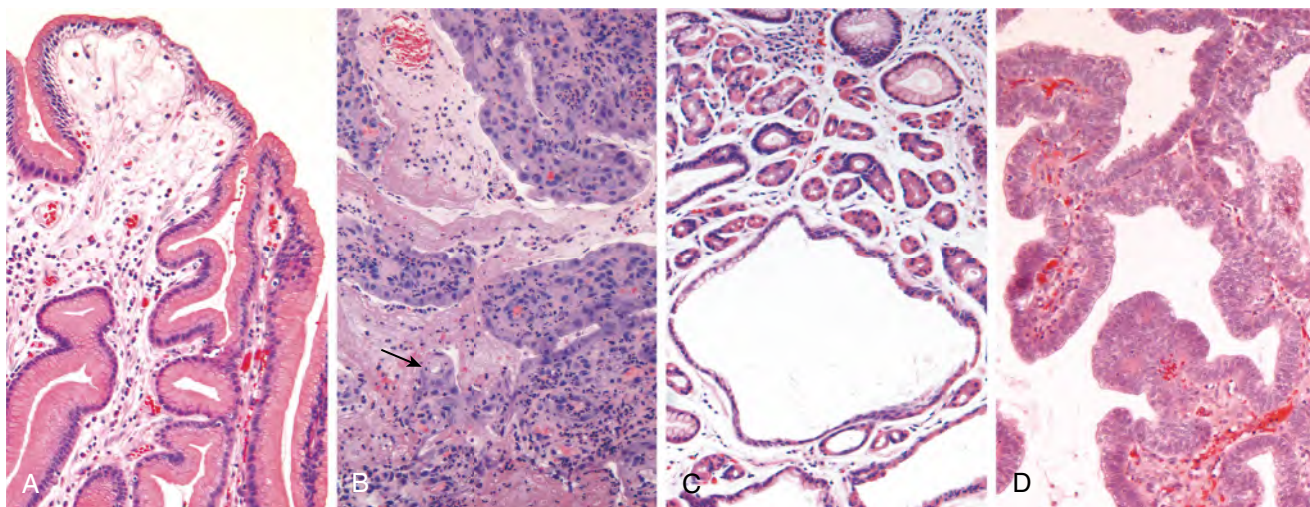


Fig. 17.19 Gastric polyps. (A) Hyperplastic polyp containing corkscrew-shaped foveolar glands. (B) Hyperplastic polyp with inflammation (arrow). (C) Fundic gland polyp composed of cystically dilated glands lined by parietal, chief, and foveolar cells. (D) Gastric adenoma with epithelial nuclear hyperchromasia and pseudostratification.

Gastric Adenoma

Most gastric adenomas develop in a background of chronic gastritis with atrophy and intestinal metaplasia, except for those associated with germline mutations in *APC* or *MUTYH*. They represent up to 10% of all gastric polyps (see Table 17.5). Their frequency increases progressively with age, and there is marked variation in prevalence among different populations that parallels the incidence of gastric adenocarcinoma. Patients are usually between 50 and 60 years of age, and males are affected three times more often than females. The risk of adenocarcinoma within gastric adenomas correlates with size and is particularly elevated in lesions greater than 2 cm in diameter. Gastric adenomas have a greater risk of cancer (up to 30% when large) and must be managed more aggressively than colonic adenomas.

MORPHOLOGY

Gastric adenomas are usually solitary antral lesions. All GI adenomas exhibit **dysplasia**, which is classified as low- or high-grade. Both grades of dysplasia are characterized by nuclear enlargement, elongation, pseudostratification, and hyperchromasia (Fig. 17.19D). High-grade dysplasia is marked by more severe cytologic atypia and architectural abnormalities, including glandular budding and gland-within-gland, or cribriform, structures.

Gastric Adenocarcinoma

Adenocarcinoma is the most common malignancy of the stomach, comprising more than 90% of all gastric cancers. As discussed in more detail later, gastric adenocarcinoma is classified morphologically into intestinal-type, which tends to form bulky masses, and diffuse-type, which infiltrates and thickens the gastric wall. Early symptoms of both subtypes resemble those of chronic gastritis and peptic ulcer disease, including dyspepsia, dysphagia, and nausea. As a result, these tumors are often discovered at advanced stages, when symptoms such as weight loss, anorexia, early satiety (primarily in diffuse cancers), anemia, and hemorrhage appear. Gastric cancer incidence varies markedly with geography. In Japan, Chile, Costa Rica, and Eastern Europe, the incidence is up to 20-fold higher than in North America, northern Europe, Africa, and Southeast Asia. Mass endoscopic screening programs have been successful in regions where the incidence is high, such as Japan, where 35% of newly detected cases are early gastric cancers limited to the mucosa and submucosa. Unfortunately, mass screening programs are not cost-effective in low-incidence regions, and fewer than 20% of cases are detected at an early stage in North America and northern Europe. Metastases are often present at diagnosis. Sites most commonly involved include left supraclavicular sentinel lymph nodes (*Virchow node*), periumbilical lymph nodes (*Sister Mary Joseph nodule*), ovaries (*Krukenberg tumor*), left axillary lymph nodes, and the pouch of Douglas.

Gastric cancer is more common in low-resource settings and in individuals with multifocal mucosal atrophy and intestinal metaplasia. As noted earlier, peptic ulcer disease is associated with reduced risk of gastric cancer, but patients who have had partial gastrectomies for peptic ulcer disease are at risk of developing cancer in the residual gastric stump, possibly due to hypochlorhydria and bile reflux.

Adenocarcinoma of the stomach was the most common cause of cancer death in the United States in 1930. However, since that time the incidence has fallen by 85%, and gastric adenocarcinoma now accounts for only 2.5% of cancer deaths in the United States. Decreased gastric cancer incidence is largely attributed to reduced rates of *H. pylori* infection and primarily relates to intestinal-type cancers.

Other environmental and dietary factors, including decreased consumption of dietary carcinogens such as N-nitroso compounds and benzo[a]pyrene (because of the reduced use of salt and smoking for food preservation) and the widespread availability of refrigeration may have also contributed to the decreased incidence. Consistent with an environmental, rather than genetic, cause, migrants from high- to low-risk regions maintain the risk of their original country, but their children have rates of intestinal-type gastric cancer similar to those of the new country of residence.

Although the overall incidence of gastric adenocarcinoma is falling, cancer of the gastric cardia is on the rise. This is probably related to Barrett esophagus and may reflect the increasing incidence of chronic GERD and obesity. Gastric cardia and distal esophageal adenocarcinomas, which typically have *TP53* and *KRAS* mutations as well as intestinal-type morphology, display similar clinical behaviors and therapeutic responses.

Pathogenesis

While most gastric cancers are not hereditary, mutations identified in hereditary gastric cancer have provided important insights into mechanisms of carcinogenesis in sporadic cases. Most prominent among these are loss-of-function mutations in the tumor suppressor gene *CDH1* (discussed in Chapter 7) which are responsible for up to ~30% of hereditary diffuse gastric cancers (see later). *CDH1* encodes the adherens junction protein E-cadherin. *CDH1* loss-of-function mutations are also present in about 50% of sporadic diffuse gastric tumors, and E-cadherin expression is drastically decreased in the remaining tumors, often by hypermethylation and silencing of the *CDH1* promoter. Similarly, hypermethylation frequently inhibits expression of the wild type *CDH1* allele in tumors occurring in patients with germline mutations. Thus, **E-cadherin loss is a key step in the development of hereditary and sporadic diffuse gastric cancer.** *CDH1* mutations are also common in sporadic and familial lobular carcinoma of the breast, which, like diffuse gastric cancer, tends to infiltrate as single cells. Hereditary diffuse gastric cancer is also associated with germline mutations in α -catenin (*CTNNA1*), another adherens junction component that binds to the E-cadherin/ β -catenin complex and regulates actin organization. Sporadic diffuse gastric cancers are additionally associated with *RHOA* mutations or *CLDN18::ARHGAP26* fusions, which are mutually exclusive, and other mutations that affect cell adhesion. Overall, however, diffuse tumors are genomically stable and carry fewer mutations than intestinal-type gastric cancers.

Intestinal-type gastric cancers can be divided into molecular subtypes characterized by chromosomal instability (CIN), microsatellite instability (MSI), and involvement of Epstein-Barr virus (EBV). CIN gastric cancers are characterized by aneuploidy and enriched for mutations that activate receptor tyrosine kinase-RAS signaling, including gain of function mutations in *ERBB2* (HER2), *EGFR*, *MET*, *KRAS*, and *VEGFA*. Mutations of genes encoding components of the Wnt pathway (*APC* and *CTNNB1*) or the TGF- β pathway (*SMAD4* and *SMAD2*) and *TP53* are also prevalent in CIN gastric cancers, which are most common in the antrum and pyloric channel.

MSI cancers favor the gastric body and typically display *MLH1* silencing as part of the CpG island hypermethylation phenotype (CIMP), which is also seen in microsatellite unstable colon cancers without DNA mismatch repair protein mutations (see later). Mutations of *BRAF* and genes that regulate mitosis are also common in MSI gastric cancers, but *TP53* and *KRAS* are only rarely affected.

Most gastric cancers are associated with *H. pylori* infection which, as noted above, interacts with host genes. However, ~10% of gastric cancers are associated with EBV, rather than *H. pylori*, infection. These tumors, which are often in the fundus and proximal body of the

stomach, show much more extensive promoter hypermethylation than MSI tumors and silencing of *CDKN2A*, but not *MLH1*. EBV-positive gastric cancers also have a high rate of *PIK3CA* mutations and increased PD-L1/2 expression, which may explain the lymphocytic infiltrates that are common in these tumors and their responsiveness to immune checkpoint inhibitor therapy.

MORPHOLOGY

Gastric adenocarcinomas are classified according to their location and gross and histologic morphology. Most distal gastric adenocarcinomas occur in the gastric antrum; the lesser curvature is involved more often than the greater curvature. Gastric tumors with an **intestinal morphology form bulky tumors** (Fig. 17.20A) and are composed of glandular structures (Fig. 17.21A), while cancers with a **diffuse infiltrative growth pattern** (Fig. 17.20B) are typically composed of **signet-ring cells** (Fig. 17.21B). Although they may penetrate the gastric wall, intestinal-type adenocarcinomas frequently grow along broad cohesive fronts to form either an exophytic mass or an ulcerated infiltrative tumor. The neoplastic cells often contain apical mucin vacuoles, and abundant mucin may be present in gland lumina. In contrast, diffuse cancers permeate the gastric wall as small clusters and individual discohesive cells due to the loss of E-cadherin. These cells do not form glands but instead have large mucin vacuoles that expand the cytoplasm and push the nucleus to the periphery, creating a signet-ring cell morphology. They may be mistaken for inflammatory cells at low magnification. Release of extracellular mucin in either type of gastric cancer can result in formation of large mucin lakes that dissect tissue planes.

A mass may be difficult to appreciate in diffuse gastric cancer, but the infiltrative tumors often evoke a **desmoplastic reaction** that stiffens the gastric wall. When there are large areas of infiltration, diffuse rugal flattening and a rigid, thickened wall may impart a leather bottle appearance termed **linitis plastica** (see Fig. 17.20B).

Clinical Features

Intestinal-type gastric cancer predominates in high-risk areas and develops from precursor lesions including flat dysplasia and adenomas. The mean age of presentation is 55 years, and the male-to-female ratio is 2:1. In contrast, the incidence of diffuse gastric cancer is relatively uniform across countries, and there are no identified precursor lesions. Although sporadic diffuse gastric cancer prevalence is similar in males and females, there is a slightly greater lifetime risk in males, relative to females, that carry germline *CDH1* mutations. Tumors also develop at younger ages, with a mean of 40 years, in patients with germline *CDH1* mutations. This risk and the absence of precursor lesions has led most specialists to recommend prophylactic gastrectomy to prevent cancer development. Although only a minority are hereditary, the frequency of diffuse gastric cancer has not changed in recent years. In contrast, rates of intestinal-type gastric cancer have decreased, particularly in high-risk areas. As a result, the incidences of intestinal-type and diffuse gastric cancers are now similar in many regions.

The depth of invasion and the extent of nodal and distant metastases are the most powerful prognostic indicators in gastric cancer. Local invasion into the duodenum, pancreas, and retroperitoneum is common. In such cases efforts are usually focused on chemotherapy or radiation therapy and palliative care. When possible, surgery remains the preferred treatment approach. With surgical resection, the 5-year survival rate of early gastric cancer exceeds 90%, even if lymph node metastases are present. In contrast, the 5-year survival rate for advanced

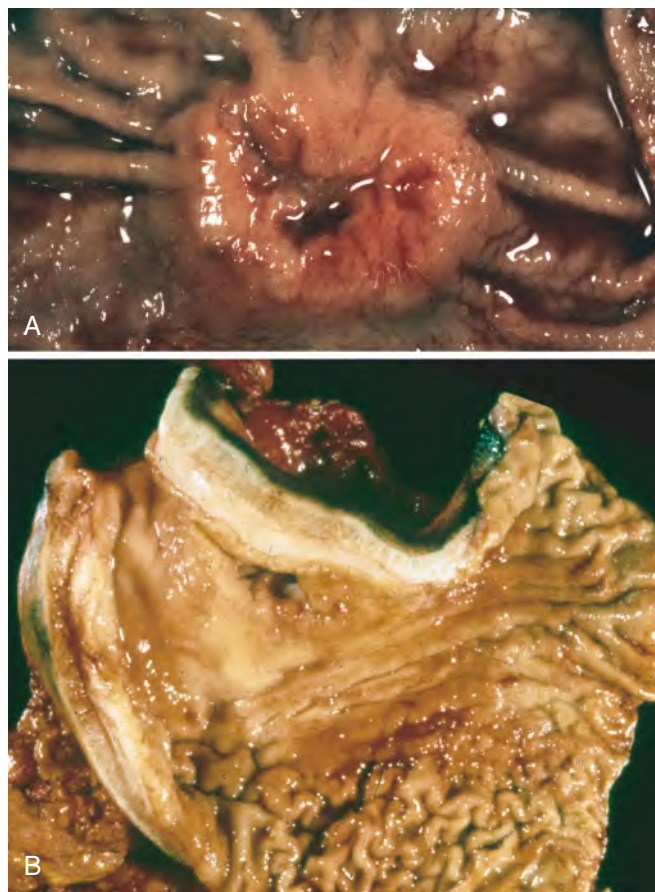


Fig. 17.20 Gastric adenocarcinoma. (A) Intestinal-type adenocarcinoma consisting of an elevated mass with heaped-up borders and central ulceration. Compare to the peptic ulcer in Fig. 17.17A. (B) Infiltrative type (linitis plastica) gastric cancer. The gastric wall is markedly thickened, and rugal folds are partially lost, but there is no dominant mass.

gastric cancer remains less than 20%. Because of the advanced stage at which most gastric cancers are discovered in the United States, the overall 5-year survival is less than 30%.

Lymphoma

Although extranodal lymphomas can arise in virtually any tissue, they do so most commonly in the GI tract, particularly the stomach. In allogeneic hematopoietic stem cell and organ transplant recipients, the bowel is also the most frequent site for Epstein-Barr virus-positive B-cell lymphoproliferations. This preferential location is most likely because the deficits in T-cell function caused by oral immunosuppressive agents (e.g., cyclosporine) are greatest at intestinal sites of drug absorption. Nearly 5% of all gastric malignancies are primary lymphomas, the most common of which are indolent extranodal marginal zone B-cell lymphomas. In the gut these tumors are often referred to as lymphomas of mucosa-associated lymphoid tissue (MALT), or MALTomas. These and other lymphomas of the gut are discussed in Chapter 13.

Pathogenesis

Extranodal marginal zone B-cell lymphomas usually arise in the setting of chronic inflammation. They can originate in the GI tract at sites of preexisting MALT such as the Peyer patches of the small intestine, but more commonly arise within tissues that are normally devoid of organized lymphoid tissue. MALT is not present in the normal stomach

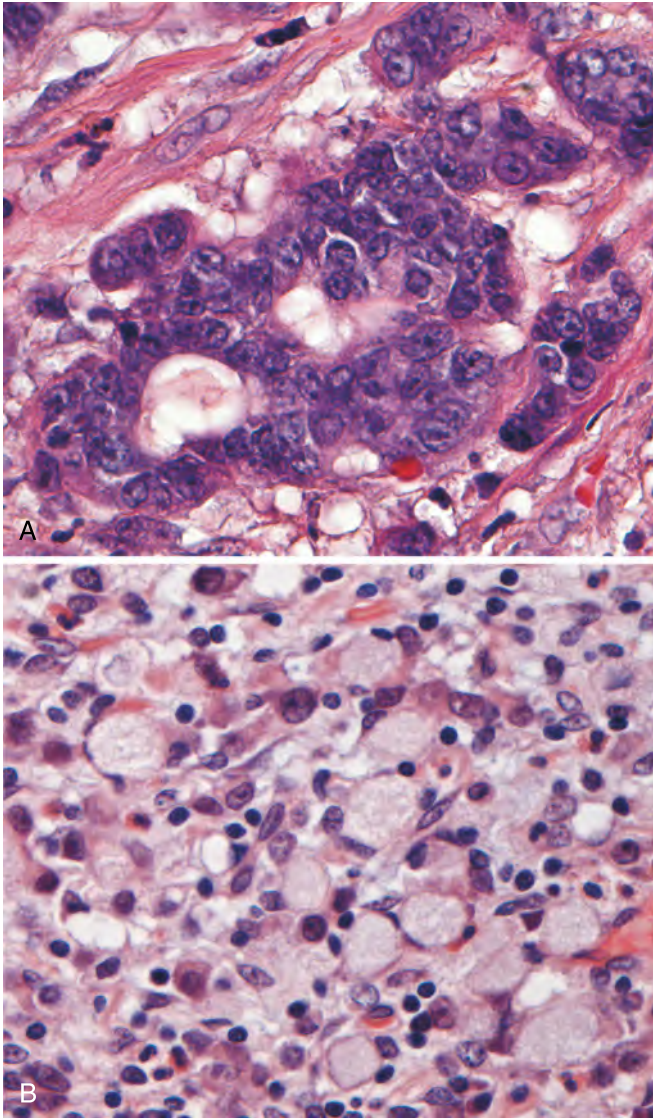


Fig. 17.21 Gastric adenocarcinoma. (A) Intestinal-type adenocarcinoma composed of columnar, gland-forming cells infiltrating through desmoplastic stroma. (B) In diffuse type or infiltrative tumors, signet-ring cells can be recognized by their large cytoplasmic mucin vacuoles and peripherally displaced, crescent-shaped, thin nuclei. Note the absence of gland formation.

but can be induced, typically secondary to chronic gastritis. *H. pylori* infection is the most common inducer of gastric MALT and therefore is found in association with most gastric MALTomas.

Four translocations are associated with gastric MALToma: $t(11;18)(q21;q21)$ and, less commonly, $t(1;14)(p22;q32)$, $t(14;18)(q32;q21)$, or $t(3;14)(p13;q32)$. The $t(11;18)(q21;q21)$ translocation brings together the apoptosis inhibitor 2 (*API2*) gene on chromosome 11 with the “mutated in MALT lymphoma” (*MALT1*) gene on chromosome 18. This creates a chimeric *API2::MALT1* fusion gene that encodes an *API2*-*MALT1* fusion protein. The $t(14;18)(q32;q21)$ and $t(1;14)(p22;q32)$ translocations cause increased expression of intact *MALT1* and *BCL-10* proteins, respectively. Finally, $t(3;14)(p13;q32)$ fuses *FOXP1* and *IGH* to increase *FOXP1* transcription factor expression.

The first three translocations cause constitutive activation of NF- κ B, a transcription factor that promotes B-cell growth and survival.

Antigen-dependent activation of NF- κ B in normal B and T cells requires both *BCL-10* and *MALT1*, which work together in a signal transduction pathway downstream of lymphocyte antigen receptors. Thus, *H. pylori*-induced inflammation may trigger NF- κ B activation through the *MALT1*/*BCL-10* pathway in MALTomas that lack these translocations. Removal of this stimulus may explain why these tumors often respond to *H. pylori* eradication. In contrast, tumors bearing translocations involving *MALT1* or *BCL10* are generally insensitive to *H. pylori* clearance. Other tumor characteristics such as invasion into or through the muscularis propria and lymph node involvement also correlate with resistance to *H. pylori* eradication.

As with other low-grade lymphomas, MALTomas can transform into more aggressive tumors that are histologically identical to diffuse large B-cell lymphomas. This is often associated with additional genetic changes such as inactivation of the tumor suppressor genes that encode p53 and p16.

MORPHOLOGY

Histologically, gastric MALToma takes the form of a dense lymphocytic infiltrate in the lamina propria (Fig. 17.22A). The neoplastic lymphocytes infiltrate gastric glands to create **lymphoepithelial lesions** (Fig. 17.22A, inset). Reactive-appearing B-cell follicles may be present, and in about 40% of tumors plasmacytic differentiation is observed. At other sites GI lymphomas may disseminate as discrete small nodules (Fig. 17.22B) or infiltrate the wall diffusely (Fig. 17.22C).

Like other tumors of mature B cells, MALTomas express the B-cell markers CD19 and CD20. They do not express CD5 or CD10 but are positive for CD43 in about 25% of cases, an unusual feature that can be diagnostically helpful. In cases lacking lymphoepithelial lesions, monoclonality may be demonstrated by restricted expression of either κ or λ immunoglobulin light chains or by molecular detection of clonal IgH rearrangements.

Clinical Features

The most common presenting symptoms are dyspepsia and epigastric pain. Hematemesis, melena, and constitutional symptoms such as weight loss can also be present. Because gastric MALTomas and *H. pylori* gastritis often coexist and have overlapping clinical symptoms and endoscopic appearances, diagnostic difficulties may arise, particularly in small biopsy specimens.

Neuroendocrine Tumors

These tumors arise from the diffuse components of the endocrine system. Those previously termed carcinoid tumors are now properly referred to as well-differentiated neuroendocrine tumors, although the term carcinoid continues to be used in the GI tract and other sites. Most are found in the GI tract, and more than 40% occur in the small intestine. The tracheobronchial tree and lungs are the next most commonly involved sites. Gastric neuroendocrine tumors may be associated with endocrine cell hyperplasia, autoimmune chronic atrophic gastritis, MEN1, and Zollinger-Ellison syndrome. Gastric endocrine and enterochromaffin-like cell hyperplasia has been linked to proton pump inhibitor therapy, but the risk of progression to a neuroendocrine neoplasm in this circumstance is negligible.

MORPHOLOGY

Grossly, neuroendocrine tumors are intramural or submucosal masses that create polypoid lesions (Fig. 17.23A). In the stomach they typically arise within oxyntic mucosa. The overlying mucosa may be intact or ulcerated, and in the

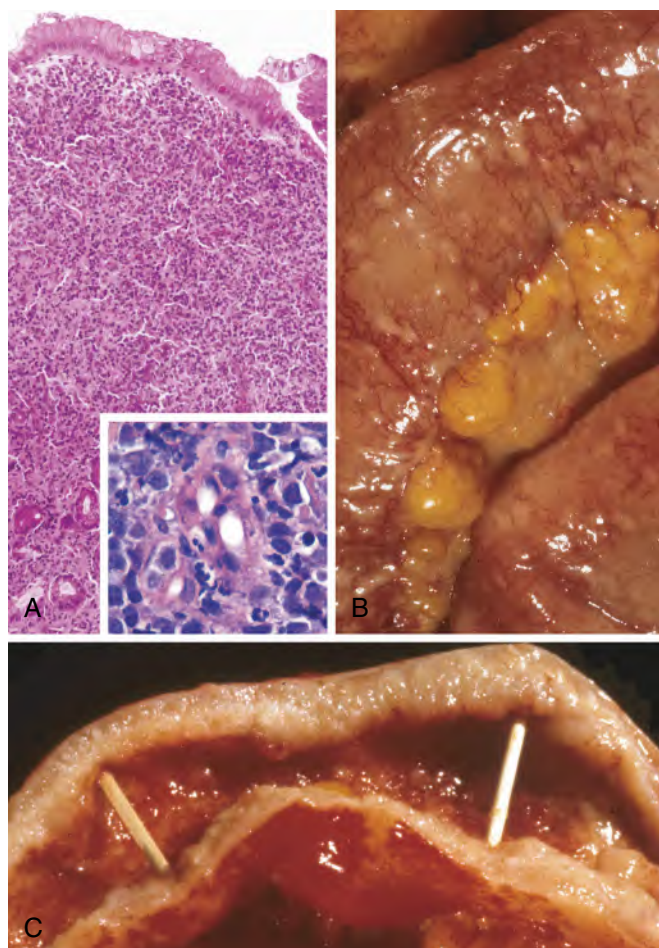


Fig. 17.22 Lymphoma. (A) Gastric mucosa-associated lymphoid tissue lymphoma replacing much of the gastric epithelium. *Inset* shows lympho-epithelial lesions with neoplastic lymphocytes surrounding and infiltrating gastric glands. (B) Disseminated lymphoma within the small intestine with numerous small serosal nodules. (C) Lymphoma infiltrating the small intestinal wall, producing diffuse thickening.

intestines the tumors may invade deeply to involve the mesentery. Neuroendocrine tumors tend to be yellow or tan in color and are very firm because of intense desmoplasia, which may cause kinking and obstruction of the small bowel. Histologically, well-differentiated neuroendocrine tumors are composed of islands, trabeculae, strands, glands, and sheets of uniform cells with scant, pink granular cytoplasm and a round to oval nucleus with a “salt and pepper” chromatin pattern (Fig. 17.23). Immunohistochemical stains are positive for endocrine granule markers such as synaptophysin and chromogranin A. Poorly differentiated neoplasms with high mitotic rates and Ki-67 proliferative indices are termed **neuroendocrine carcinomas** and subclassified as small cell or large cell types. Like neuroendocrine carcinomas at extraintestinal sites, mutations of *TP53* and *RB* are common.

Clinical Features

The peak incidence of GI neuroendocrine tumors is in the sixth decade, but they may appear at any age. Symptoms reflect the actions of hormones released by the tumor cells. For example, those that produce gastrin cause Zollinger-Ellison syndrome. Carcinoid syndrome, which develops in fewer than 10% of patients, is caused by vasoactive substances secreted by the tumor into the systemic circulation. This can lead to cutaneous flushing, sweating, bronchospasm, colicky abdominal pain, diarrhea, and right-sided cardiac valvular fibrosis. When neuroendocrine tumors are confined to the intestine, vasoactive substances released are metabolized to inactive forms by the liver, a “first-pass” effect similar to that exerted on oral drugs. Although this can be overcome by a large tumor burden or when tumors secrete hormones into a nonportal venous circulation, the carcinoid syndrome remains strongly associated with metastatic disease to sizes that allow tumor products to bypass the portal circulation and liver.

The most important prognostic factors for GI neuroendocrine tumors are degree of histological differentiation, mitotic rate, and Ki-67 proliferative index. Behavior is also affected by size and location, and staging is region specific.

- *Neuroendocrine tumors within the stomach and duodenum rarely metastasize and are generally cured by resection. This is particularly true for tumors that arise in association with atrophic gastritis-*

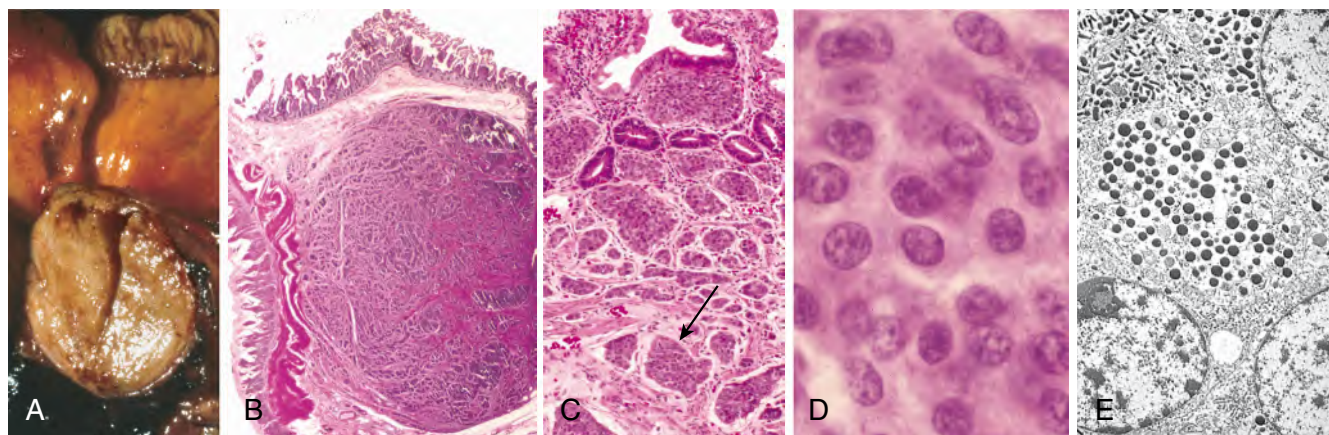


Fig. 17.23 Neuroendocrine tumor (carcinoid tumor). (A) Gross cross section of a submucosal tumor nodule. (B) Microscopically the nodule is composed of tumor cells embedded in dense fibrous tissue. (C) In other areas the tumor has spread extensively within mucosal lymphatic channels (*arrow*). (D) High magnification shows the characteristically bland cytology. The chromatin texture, with fine and coarse clumps, is frequently described as a “salt and pepper” pattern. (E) Electron microscopy reveals cytoplasmic dense core neurosecretory granules.

tis. Gastric neuroendocrine tumors without predisposing factors are often more aggressive.

- *Neuroendocrine tumors that arise in the jejunum and ileum* are often multiple and tend to be aggressive.
- *Neuroendocrine tumors arising in the appendix and colorectum* are typically discovered incidentally. Those in the appendix occur at any age and are generally located at the tip. These tumors are rarely more than 2 cm in diameter and are almost always benign. Rectal neuroendocrine tumors tend to produce polypeptide hormones and, when symptomatic, present with abdominal pain and weight loss. Because they are usually discovered when small, metastasis of rectal neuroendocrine tumors is uncommon.

Gastrointestinal Stromal Tumor

Gastrointestinal stromal tumor (GIST) is the most common mesenchymal tumor of the abdomen, with an annual incidence between 11 and 20 per 1 million people. More than half of these tumors occur in the stomach. The term *stromal* reflects historical confusion about the origin of this tumor, which is now recognized to arise from the *interstitial cells of Cajal*, or pacemaker cells. A wide variety of other mesenchymal neoplasms may arise in the stomach. Many are named according to the cell type they most resemble; for example, smooth muscle tumors are called leiomyomas or leiomyosarcomas, nerve sheath tumors are termed schwannomas, and tumors resembling glomus bodies (vascular structures found in the nail beds and other sites) are referred to as glomus tumors. These are all uncommon in the gut and are discussed in Chapter 26.

Clinically silent, microscopic proliferations that may represent precursors to GIST are common. These foci have low mitotic rates and extremely low risk of neoplastic transformation. Changes associated with progression to overt GIST are not well defined, but loss or partial deletion of chromosomes 9, 14, or 22 is common.

GISTs are diagnosed at a peak age of 60 years, with fewer than 10% occurring before 40 years of age. Of the uncommon GISTs in children, some are related to the Carney triad, a syndrome of unknown cause defined by the co-occurrence of GIST, paraganglioma, and pulmonary chondroma, usually occurring in females. There is also an increased incidence of GIST in individuals with neurofibromatosis type 1.

Pathogenesis

Approximately 75% of all GISTs have oncogenic, gain-of-function mutations in the receptor tyrosine kinase KIT, and another 8% have mutations that activate a related receptor tyrosine kinase, platelet-derived growth factor receptor α (PDGFRA). Constitutively active KIT and PDGFRA trigger the same downstream signaling pathways; hence, mutations in *KIT* and *PDGFRA* are mutually exclusive in GIST. Rare individuals with germline mutations of these genes have diffuse hyperplasia of Cajal cells and develop multiple GISTs. GISTs without mutated *KIT* or *PDGFRA* often have mutations in genes encoding components of the mitochondrial succinate dehydrogenase complex (*SDHA*, *SDHB*, *SDHC*, *SDHD*), an interesting (and still poorly understood) example of oncogenic mutations affecting components of a metabolic pathway, specifically the Krebs cycle. These mutations, which cause loss of succinate dehydrogenase function, are often inherited in the germline and confer an increased risk for GIST and paraganglioma (Carney-Stratakis syndrome), with the second copy of the affected gene being either mutated or lost in the tumor.

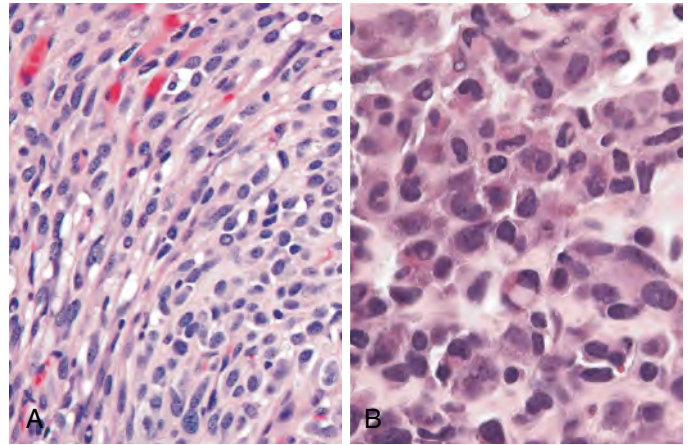


Fig. 17.24 Gastrointestinal stromal tumor. (A) Histologically, typical spindle cell gastrointestinal stromal tumors are composed of bundles, or fascicles, of spindle-shaped tumor cells. (B) Epithelioid cell gastrointestinal stromal tumors are composed of plump, epithelial-appearing cells without well-defined fascicles. These tumors can mimic adenocarcinoma.

MORPHOLOGY

Primary gastric GISTs can reach 30 cm in diameter. They usually form a solitary, well-circumscribed, fleshy mass. The overlying mucosa is frequently ulcerated, and the cut tumor surface has a whorled appearance. Metastases can form multiple serosal nodules throughout the peritoneal cavity or within the liver but spread outside of the abdomen is uncommon. GISTs composed of thin elongated cells are classified as **spindle cell type** (Fig. 17.24A), while GISTs dominated by plumper, epithelial-appearing cells are termed **epithelioid type** (Fig. 17.24B); mixtures of the two patterns are common. Nuclear pleomorphism is uncommon. The most useful diagnostic marker is KIT (CD117), which is immunohistochemically detectable in normal Cajal cells and 95% of gastric GISTs. GISTs with epithelioid types are usually KIT negative. Another marker *DOG1* (discovered on GIST 1) is positive in 95% of GISTs, including all epithelioid types.

Clinical Features

Symptoms of GISTs at presentation are typically related to mass effects. Mucosal ulceration can cause blood loss, and approximately half of individuals with GISTs present with anemia or related symptoms. GISTs may also be discovered incidentally during radiologic imaging, endoscopy, or abdominal surgery performed for other reasons. Complete surgical resection is the primary treatment for localized gastric GIST. The prognosis correlates with tumor size, mitotic index, and location, with gastric GISTs being less aggressive than those arising in the small intestine. Recurrence or metastasis is rare for gastric GISTs smaller than 5 cm but common for mitotically active tumors larger than 10 cm. Many tumors fall into an intermediate category where the malignant potential of the lesion cannot be predicted.

The molecular phenotype is an important consideration in the treatment of patients with unresectable, recurrent, or metastatic GISTs. Tumors with mutations in *KIT* or *PDGFRA* often respond to tyrosine kinase inhibitors such as imatinib, whereas tumors without these mutations are generally resistant. In treated patients, imatinib resistance often develops as a result of secondary *KIT* or *PDGFRA* mutations; such tumors may respond to other tyrosine kinase inhibitors that bypass the effects of the mutations that convey resistance to imatinib.

KEY CONCEPTS

Neoplastic and Nonneoplastic Proliferations of the Stomach

- Most gastric polyps are inflammatory or hyperplastic polyps. These reactive lesions are associated with chronic gastritis.
- Fundic gland polyps occur sporadically, most often as a consequence of proton pump inhibitor therapy and in patients with familial adenomatous polyposis and *MUTYH*-associated polyposis.
- Gastric adenomas develop in a background of chronic gastritis and are strongly linked to intestinal metaplasia and mucosal (glandular) atrophy. Adenocarcinoma is frequent in gastric adenomas, which therefore require more aggressive therapy than adenomas of the colon.
- Gastric intestinal-type adenocarcinoma incidence varies markedly with geography and has fallen in recent years. In contrast, diffuse gastric cancer incidence is more uniform and has not changed.
- Gastric tumors with an intestinal-type histology tend to form bulky tumors and may be ulcerated, while those composed of signet-ring cells typically display a diffuse infiltrative growth pattern that may thicken the gastric wall without forming a discrete mass. Gastric adenocarcinomas are linked to chronic gastritis.
- Primary gastric lymphomas are most often indolent B-cell tumors of mucosa-associated lymphoid tissue (MALTomas) that arise in the setting of chronic *H. pylori* gastritis.
- Neuroendocrine tumors arise from diffuse components of the endocrine system and are most common in the GI tract, particularly the small intestine. They can be subdivided into neuroendocrine tumors (previously termed carcinoid tumors) and neuroendocrine carcinomas based on histology and immunohistochemistry for the proliferative marker Ki-67. Prognosis is also affected by location; tumors of the small intestine tend to be most aggressive, while those of the appendix are typically benign.
- GIST is the most common mesenchymal tumor of the abdomen, occurs most often in the stomach, and develops from Cajal (pacemaker) cells. Most tumors have activating mutations in either the *KIT* or the *PDGFRA* tyrosine kinases gene and respond to targeted tyrosine kinase inhibitors.

SMALL INTESTINE AND COLON

The small intestine and colon make up most of the GI tract and are the sites of a broad array of diseases. Some of these relate to nutrient and water transport, perturbation of which results in malabsorption and diarrhea. The intestines are also the principal site where the immune system interfaces with a diverse array of antigens present in food and gut microbes. Thus, it is not surprising that the small intestine and colon are frequently affected by infectious and inflammatory disorders. Finally, the colon is the most common site of GI neoplasia in Western populations.

INTESTINAL OBSTRUCTION

Obstruction of the GI tract may occur at any level, but the small intestine is most often involved because of its relatively narrow lumen. Collectively, hernias, intestinal adhesions, intussusception, and volvulus account for 80% of mechanical obstructions (Fig. 17.25). Tumors, infarctions, and other causes of strictures, for example, Crohn disease, make up the remainder. The clinical manifestations of intestinal obstruction include abdominal pain and distention, vomiting, and constipation. Surgical intervention is usually required in cases where the obstruction has a mechanical basis or is associated with bowel infarction.

Hernias

Any weakness or defect in the abdominal wall may permit protrusion of a serosa-lined pouch of peritoneum called a hernia sac. Acquired

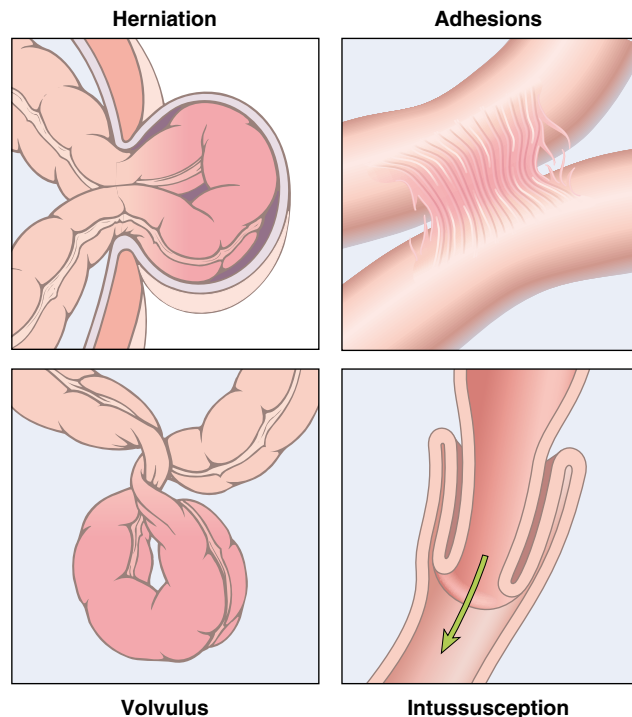


Fig. 17.25 Intestinal obstruction. The four major causes of intestinal obstruction are (1) herniation of a segment in the umbilical or inguinal regions, (2) adhesion between loops of intestine, (3) volvulus, and (4) intussusception.

hernias typically occur anteriorly, via the inguinal and femoral canals, via the umbilicus, or at sites of surgical defects and are found in up to 5% of the population. **Hernias are the most frequent cause of intestinal obstruction worldwide** and the third most common cause of obstruction in the United States. Obstruction usually occurs because of visceral protrusion (external herniation) and is most frequently associated with inguinal hernias, which tend to have narrow orifices and large sacs. Small bowel loops are typically involved, but omentum or large bowel may also protrude, and any of these may become entrapped. Pressure at the neck of the pouch may impair venous drainage of the entrapped viscus. The resultant stasis and edema cause swelling of the herniated loop, leading to permanent entrapment (incarceration) and, over time, arterial and venous compromise (strangulation) and infarction (Fig. 17.26).

Adhesions

Surgical procedures, infection, or other causes of peritoneal inflammation, such as endometriosis, may lead to development of adhesions between bowel segments, the abdominal wall, or operative sites. These fibrous bridges can create closed loops through which other viscera may slide and become entrapped, resulting in internal herniation. Sequelae, including obstruction and strangulation, are similar to external hernias.

Volvulus

Volvulus occurs when a loop of bowel twists about its mesenteric point of attachment, resulting in luminal obstruction and vascular compromise. It occurs most often in large redundant loops of sigmoid colon, followed in frequency by the cecum, small bowel, stomach, or, rarely, transverse colon. Because it is rare, volvulus can be overlooked clinically, often with catastrophic results.



Fig. 17.26 Intestinal obstruction. Portion of bowel incarcerated within an inguinal hernia. Note dusky areas of serosa and associated hemorrhage that indicate ischemic damage.

Intussusception

Intussusception occurs when a segment of the intestine (intussusceptum), constricted by a wave of peristalsis, telescopes into the immediately distal segment (intussusciens). Once trapped, the invaginated segment is propelled by peristalsis and pulls the mesentery along. Untreated, intussusception may progress to intestinal obstruction, compression of mesenteric vessels, and infarction.

Intussusception is the most common cause of intestinal obstruction in children younger than 2 years of age. In most cases there is no underlying anatomic defect, and the patient is otherwise healthy. Other cases are associated with viral infection and rotavirus vaccines, perhaps due to reactive hyperplasia of Peyer patches and other mucosa-associated lymphoid tissues that act as the leading edge of the telescoping bowel segment. Contrast enemas are used both diagnostically and therapeutically for idiopathic intussusception in infants and young children, in whom air enemas may also be used to reduce the intussusception. In older children and adults, intussusception is generally caused by an intraluminal mass or tumor that serves as the initiating point of traction. Surgical intervention is necessary when a mass is present. In cystic fibrosis, up to 1% of patients develop intussusception, possibly due to inadequate fecal hydration that allows adherent inspissated stool to act as a mass-like traction point.

ISCHEMIC BOWEL DISEASE

Most of the GI tract is supplied by the celiac, superior mesenteric, and inferior mesenteric arteries. As they approach the intestinal wall, the superior and inferior mesenteric arteries ramify into the mesenteric arcades. Interconnections between arcades, as well as collateral vessels from the proximal celiac and distal pudendal and iliac circulations, make it possible for the small intestine and colon to tolerate slowly progressive loss of blood supply from one artery. In contrast to chronic, progressive hypoperfusion, acute compromise of any major vessel can lead to infarction of several meters of intestine. Damage can range from mucosal infarction extending no deeper than the muscularis mucosae to mural infarction of mucosa and submucosa to transmural infarction involving all three wall layers. While mucosal or mural infarctions can follow acute or chronic hypoperfusion, transmural infarction is typically due to acute vascular obstruction. In a large majority of cases, acute obstruction is caused by thrombosis or embolism. The most important risk factor for thrombosis is severe atherosclerosis (which is often prominent at the origin of mesenteric vessels). Less common

causes of thrombosis include systemic vasculitides (Chapter 11). Obstructive emboli most commonly originate from aortic atheromas or cardiac mural thrombi. Mesenteric venous thrombosis, which can also lead to ischemic disease, is uncommon. It may result from inherited or acquired hypercoagulable states, invasive neoplasms, cirrhosis, trauma, or abdominal masses that compress the portal drainage. Intestinal hypoperfusion may also occur in the absence of vascular obstruction in the setting of cardiac failure, shock, dehydration, or use of vasoconstrictive drugs.

Pathogenesis

Intestinal responses to ischemia occur in two phases. The initial hypoxic injury (first phase) occurs at the onset of vascular compromise, but the greatest damage occurs during the second phase, reperfusion, when the blood supply is restored. While the mechanisms of reperfusion injury are incompletely understood, they likely include leakage of gut lumen bacterial products (e.g., lipopolysaccharide) into the systemic circulation, free radical production, neutrophil infiltration, and release of additional inflammatory mediators (Chapter 2).

The severity of vascular compromise, the time frame during which it develops, and the vessels affected are the major variables that determine the severity of ischemic bowel disease. Intestinal vascular anatomy also contributes to the distribution of ischemic damage in two ways:

- *Intestinal segments at the end of their respective arterial supplies are particularly susceptible to ischemia.* These watershed zones include the splenic flexure, where the superior and inferior mesenteric arterial circulations terminate, and, to a lesser extent, the sigmoid colon and rectum, where the inferior mesenteric, pudendal, and iliac arterial circulations end. Generalized hypotension or hypoxemia can cause localized injury in watershed zones, and ischemic disease should be considered in the differential diagnosis of focal colitis of the splenic flexure or rectosigmoid colon. Ischemic injury in watershed zones may also be triggered by ingestion of vasoconstrictive drugs such as cocaine.
- *The surface epithelium is more susceptible to ischemic injury than crypts* because intestinal capillaries run alongside the glands, from crypt to surface (villus), before making a hairpin turn and descending to the postcapillary venules. This pattern of circulation protects the epithelial stem cells, which are located within the crypts and are necessary for recovery from epithelial injury, but it makes the surface epithelium particularly vulnerable to ischemic injury. Surface epithelial atrophy with normal or hyperproliferative crypts is therefore a morphologic signature of ischemic intestinal disease.

MORPHOLOGY

Although the colon is the most common site of GI ischemia, mucosal and mural infarction may involve any level of the gut from stomach to anus. The lesions can be continuous but are most often segmental and patchy (Fig. 17.27A). The mucosa is hemorrhagic and often ulcerated (Fig. 17.27B). Edema may thicken the bowel wall, particularly when damage extends into the submucosa and muscularis propria.

Substantial portions of the bowel are generally involved in **transmural infarction caused by acute arterial obstruction**. The demarcation between normal and ischemic bowel is sharply defined, and initially the infarcted bowel is intensely congested and dusky to purple-red. Later, blood-tinged mucus or blood accumulates in the lumen, and the wall becomes edematous, thickened, and rubbery. There is coagulative necrosis of the muscularis propria within 1 to 4 days, and perforation may occur. Serositis, with purulent exudates and fibrin deposition, may be prominent.

In **mesenteric venous thrombosis**, arterial blood continues to flow for a time, resulting in a less abrupt transition from affected to normal bowel.

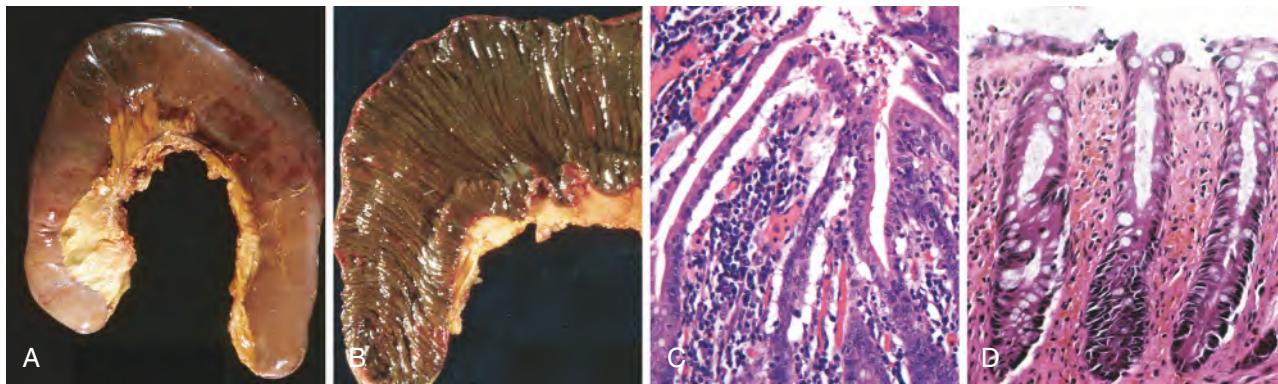


Fig. 17.27 Ischemic bowel disease. (A) Resected jejunum with dusky serosa due to acute ischemia (mesenteric thrombosis). (B) The mucosa is darkened by hemorrhage. (C) Characteristic attenuated villous epithelium in a case of acute mesenteric thrombosis. (D) Chronic colonic ischemia with atrophic surface epithelium and fibrotic lamina propria.

Propagation of the thrombus may lead to secondary involvement of the splanchnic bed. The ultimate result is similar to acute arterial obstruction because impaired venous drainage eventually prevents oxygenated arterial blood from entering the capillaries.

Microscopic examination of ischemic intestine demonstrates the characteristic atrophy or sloughing of surface epithelium (Fig. 17.27C), often with hyperproliferative crypts. Inflammatory infiltrates are initially absent, but neutrophils are recruited within hours of reperfusion. Chronic ischemia is accompanied by fibrous scarring of the lamina propria (Fig. 17.27D) and, uncommonly, stricture formation. In both acute and chronic ischemia, bacterial superinfection and enterotoxin release may induce **pseudomembrane formation** that resembles *Clostridioides* (formerly *Clostridium*) *difficile*-associated pseudomembranous colitis (discussed later).

Clinical Features

Ischemic disease of the colon is most common in patients older than 70 years of age and occurs slightly more often in females than in males. **Acute colonic ischemia typically presents with sudden onset of cramping, left lower abdominal pain, a desire to defecate, and passage of blood or bloody diarrhea.** In severe cases, patients may progress to shock and vascular collapse within hours. Surgical intervention is necessary in approximately 10% of cases and should be considered if evidence of infarction (e.g., diminished peristaltic sounds, guarding, or rebound tenderness) develops. Because these physical signs overlap with those of other abdominal emergencies (e.g., acute appendicitis, perforated ulcer, and acute cholecystitis), the diagnosis of intestinal infarction may be delayed or missed, with disastrous consequences.

With appropriate management, mortality in the first 30 days is approximately 10%. Mortality is doubled in patients with right-sided colonic disease, who have a more severe course in general. This may be because the right side of the colon is supplied by the superior mesenteric artery, which also supplies much of the small intestine. Thus, right-sided colonic ischemia may be the initial presentation of more widespread compromise of intestinal perfusion. Other poor prognostic indicators include coexisting chronic obstructive pulmonary disease and persistence of symptoms for more than 2 weeks. Ischemic bowel injury can take a variety of subacute or chronic forms.

- *Limited zones of mucosal and mural infarctions may progress to more extensive and transmural infarction* if the vascular supply is not restored by correction of the insult or, in chronic disease, by development of collateral supplies. The diagnosis of nonocclusive ischemic enteritis and colitis can be particularly difficult, as

common symptoms such as intermittent bloody diarrhea and intestinal obstruction are nonspecific.

- *Chronic ischemia* may masquerade as inflammatory bowel disease, with episodes of bloody diarrhea interspersed with periods of healing.
- *CMV infection* causes ischemic GI disease due to viral infection of endothelial cells and resulting localized vascular insufficiency. CMV infection, which can be a complication of immunosuppressive therapy, is discussed in Chapter 8.
- *Radiation enterocolitis* occurs when the GI tract is irradiated and is due to a combination of epithelial and endothelial injury. Beyond clinical history, the presence of highly atypical “radiation fibroblasts” within the stroma may provide an important clue to the etiology. Acute radiation enteritis manifests as anorexia, abdominal cramps, and malabsorptive diarrhea, while chronic radiation enteritis or colitis is often more indolent and may present as an inflammatory enterocolitis.
- *Necrotizing enterocolitis* is an acute disorder of the small and large intestines that can result in transmural necrosis. It is the most common acquired GI emergency of neonates, particularly those who are premature or of low birth weight, and frequently presents when oral feeding is initiated. It is discussed in more detail in Chapter 10 but is noted here because ischemic injury is thought to contribute to the pathogenesis.

ANGIODYSPLASIA

Angiodysplasia is characterized by malformed submucosal and mucosal blood vessels that are dilated and thin walled. It is most common in the cecum or right colon and usually comes to clinical attention after the sixth decade of life. Although the prevalence is less than 1% in adults, angiodysplasia accounts for 20% of major episodes of lower intestinal bleeding in older populations. The resultant bleeding ranges from chronic and intermittent to acute and massive.

The pathogenesis of angiodysplasia remains undefined but has been attributed to mechanical and congenital factors. Normal distention and contraction of the gut may intermittently occlude the submucosal veins that penetrate through the muscularis propria, leading to focal dilation and tortuosity of overlying submucosal and mucosal vessels. Because the cecum has the largest diameter of any colonic segment, it develops the greatest wall tension. This may explain the preferential distribution of angiodysplasia in the cecum and right colon.

Morphologically, angiodysplastic lesions are characterized by ectatic nests of tortuous veins, venules, and capillaries. The vascular channels are often separated from the intestinal lumen by only the thin

vessel wall and a layer of attenuated epithelial cells, making the vessels vulnerable to disruption by even minor injury.

MALABSORPTION AND DIARRHEA

Malabsorption is characterized by defective absorption of fats, fat- and water-soluble vitamins, proteins, carbohydrates, electrolytes and minerals, and water. A hallmark of malabsorption is steatorrhea, characterized by excessive fecal fat and bulky, frothy, greasy, yellow or clay-colored stools. The chronic malabsorptive disorders most commonly encountered in the United States are pancreatic insufficiency, celiac disease, and Crohn disease (Table 17.6). Intestinal graft-versus-host disease is an important cause of malabsorption and diarrhea after allogeneic hematopoietic stem cell transplantation.

Pathogenesis

Malabsorption results from disturbance of at least one of the four phases of nutrient absorption:

- *Intraluminal digestion*, in which proteins, carbohydrates, and fats are broken down into forms suitable for absorption
- *Terminal digestion*, involving hydrolysis of carbohydrates and peptides by disaccharidases and peptidases in the brush border of the small intestinal mucosa
- *Transepithelial transport*, in which nutrients, fluid, and electrolytes are transported across and processed within the small intestinal epithelium
- *Lymphatic transport* of absorbed lipids

In many malabsorptive disorders a defect in one of these processes predominates, but several usually contribute. As a result, the symptoms and consequences of malabsorption syndromes resemble each other more than they differ. Chronic malabsorption can lead to diarrhea, flatus, abdominal pain, weight loss, and muscle wasting, and is often associated with anorexia, abdominal distention, and borborrygmi. Inadequate absorption of vitamins and minerals can result in anemia and mucositis due to pyridoxine, folate, or vitamin B₁₂ deficiency; bleeding due to vitamin K deficiency; osteopenia and tetany due to calcium, magnesium, or vitamin D deficiencies; or peripheral neuropathy due to vitamin A or B₁₂ deficiencies. A variety of endocrine and skin disturbances (e.g., goiter and vitiligo) may also develop because of nutrient, micronutrient, and vitamin deficiencies.

Diarrhea is defined as an increase in stool mass, frequency, or fluidity, typically greater than 200 g per day. Both nutrient malabsorption and increased secretion of fluid by the intestine may contribute to diarrhea. In severe cases stool volume can exceed 14 L per day and, without fluid resuscitation, result in death. Painful, bloody, small-volume diarrhea is known as dysentery. Diarrhea can be classified into four major categories:

- *Secretory diarrhea* is characterized by isotonic stool and persists during fasting.
- *Osmotic diarrhea*, such as occurs with lactase deficiency, is due to the excessive osmotic force exerted by unabsorbed luminal solutes. The diarrhea fluid is more than 50 mOsm more concentrated than plasma, and diarrhea abates with fasting.
- *Malabsorptive diarrhea* follows generalized failure of nutrient absorption, is associated with steatorrhea, and is relieved by fasting.
- *Exudative diarrhea* due to inflammatory disease is characterized by purulent, often bloody stools that continue during fasting.

Cystic Fibrosis

Cystic fibrosis affects many organ systems, primarily the lungs, and is discussed in greater detail in Chapter 10. Only the malabsorption associated with cystic fibrosis is considered here. Individuals with cystic fibrosis have defects in chloride and, in certain tissues, bicarbonate ion transport across epithelia due to the absence or dysfunction of the epithelial cystic fibrosis transmembrane conductance regulator (CFTR). This interferes with bicarbonate, sodium, and water secretion, ultimately resulting in decreased hydration of luminal contents. The thick viscous stool may occasionally lead to intestinal obstruction, but more commonly results in formation of pancreatic intraductal concretions. The latter can begin in utero and result in duct obstruction, low-grade chronic pancreatitis and autodigestion of the pancreas, and eventually exocrine pancreatic insufficiency in more than 80% of patients. This in turn impairs the intraluminal phase of nutrient absorption, a defect that can be treated with oral enzyme supplementation. Novel treatments based on specific molecular defects have been developed (Chapter 10).

Celiac Disease

Celiac disease, also known as celiac sprue or gluten-sensitive enteropathy, is an immune-mediated disorder triggered by the ingestion of gluten-containing foods such as wheat, rye, or barley in genetically

TABLE 17.6 Defects in Malabsorptive and Diarrheal Diseases

Disease	Intraluminal Digestion	Terminal Digestion	Transepithelial Transport	Lymphatic Transport
Celiac disease		+	+	
Environmental enteropathy		+	+	
Chronic pancreatitis	+			
Cystic fibrosis	+			
Primary bile acid malabsorption	+		+	
Carcinoid syndrome			+	
Autoimmune enteropathy		+	+	
Disaccharidase deficiency		+		
Whipple disease				+
Abetalipoproteinemia			+	
Viral gastroenteritis		+	+	
Bacterial gastroenteritis		+	+	
Parasitic gastroenteritis		+	+	
Inflammatory bowel disease	+	+	+	

“+” indicates that the process is abnormal in the disease indicated.

predisposed individuals. Celiac disease has an overall worldwide incidence of 0.6% to 1%, but its prevalence varies widely among countries and regions. Some of these differences correlate with variation in wheat consumption. While previously uncommon in many countries, the incidence of celiac disease and gluten sensitivity is growing for unknown reasons.

Pathogenesis

Celiac disease is an immune enteropathy induced by an environmental stimulus (gluten), a genetic predisposition (HLA DQ2/8), and tissue transglutaminase 2 (TG2). Gluten, the major storage protein of wheat and similar grains, is digested by luminal and brush-border enzymes into amino acids and peptides that contain proline- and glutamine-rich sequences. These sequences are targeted by TG2, which deamidates glutamine residues to create glutamic acid. These negatively charged peptides bind to DQ2 on antigen-presenting cells, which is present in up to 90% of cases, or DQ8, which accounts for most remaining cases. This leads to activation of gluten-specific CD4⁺ T cells and production of cytokines, including IFN γ , IL-21, and IL-2. In addition, gluten peptide-TG2 complexes can elicit development of anti-TG2 IgA-producing plasma cells, which account for up to 20% of intestinal plasma cells during active disease. Tests that detect antibodies against gliadin and TG2 are extremely useful in diagnosing celiac disease. The CD4⁺ T cells and IL-15 produced by epithelial cells promote expansion and activation of intraepithelial cytotoxic CD8⁺ T cells. These T cells produce IFN γ and granzyme B and upregulate expression of NKG2D and NKG2C, which target epithelial cells by interacting with the unconventional MHC class I molecules HLA-E and MICA (Fig. 17.28).

The incidence of concordance for celiac disease in identical (monozygotic) twins is ~75%, about 5-fold greater than concordance between nonidentical (dizygotic) twins. However, only ~35% of celiac disease risk is determined by HLA DQ2/8. An additional 20% of risk is attributed to non-DQ HLA genes. Other genetic risk factors, such as polymorphisms within proinflammatory and immunoregulatory genes, also contribute to risk and may account for associations between celiac disease and other immune diseases including type 1 diabetes, thyroiditis, Sjögren syndrome, and IgA nephropathy.

MORPHOLOGY

Biopsy specimens from the second portion of the duodenum or proximal jejunum, which are exposed to the highest concentrations of dietary gluten, are generally diagnostic in celiac disease. The histopathology is characterized by **increased numbers of intraepithelial CD8⁺ T lymphocytes (intraepithelial lymphocytosis), crypt hyperplasia, and villous atrophy** (Fig. 17.29). This loss of mucosal and brush-border surface area contributes to malabsorption. Increased rates of epithelial turnover, reflected in increased crypt mitotic activity, may also limit the ability of absorptive enterocytes to fully differentiate and express proteins necessary for terminal digestion and transepithelial transport. Other features of fully developed celiac disease include increased numbers of plasma cells, mast cells, and eosinophils, especially within the upper part of the lamina propria. With increased serologic screening and early detection of disease-associated antibodies, it is now appreciated that increased numbers of intraepithelial lymphocytes, particularly within the villus, is a sensitive

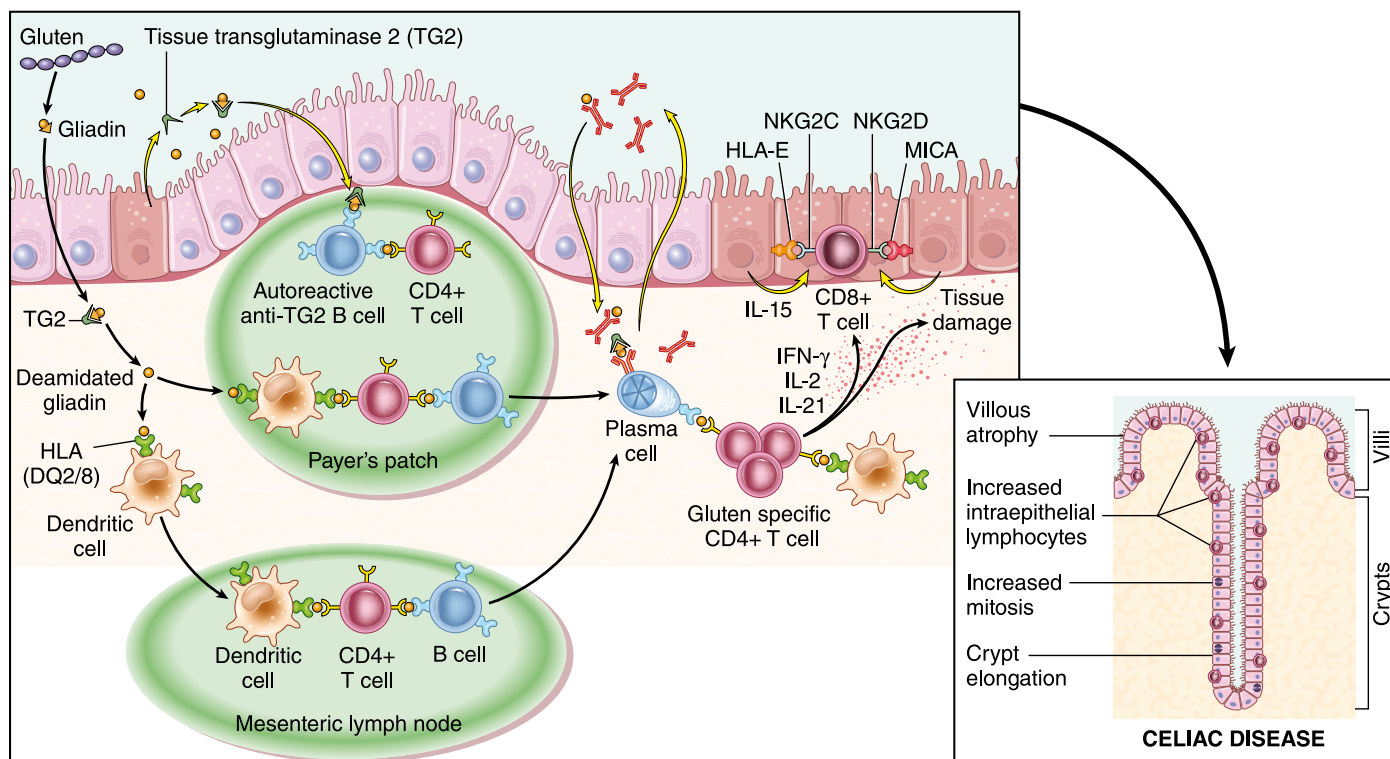


Fig. 17.28 Gliadin, a component of gluten, is deamidated by tissue transglutaminase (TG2). This triggers immune responses within the mucosa and mesenteric lymph nodes that induce gluten-specific CD4⁺ T cell expansion, CD8⁺ T cell development, cytokine release, and epithelial damage. As a result, epithelial turnover is accelerated, villi are shortened, and a compensatory proliferative response is activated within the crypts.

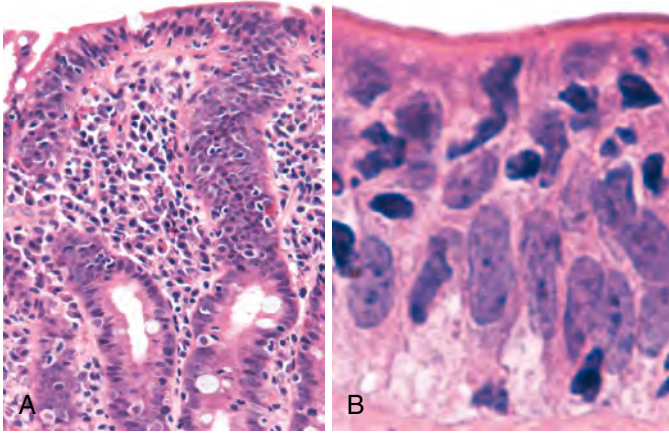


Fig. 17.29 Celiac disease. (A) Advanced cases of celiac disease show complete loss of villi, or total villous atrophy. Note the dense plasma cell infiltrates in the lamina propria. (B) Infiltration of the surface epithelium by T lymphocytes, which can be recognized by their small, densely stained nuclei relative to larger, pale-staining epithelial nuclei.

marker of celiac disease, even in the absence of epithelial damage and villous atrophy. However, intraepithelial lymphocytosis and villous atrophy are not specific for celiac disease and can be present in other diseases, including viral enteritis. The combination of histology and serology is therefore required for diagnosis.

Clinical Features

In adults, celiac disease presents most commonly between the ages of 30 and 60. Many cases escape clinical attention for extended periods because of atypical presentations. Other patients may have silent disease, defined as positive serology and villous atrophy without symptoms, or latent disease, in which positive serology is not accompanied by villous atrophy. Celiac disease may be associated with chronic diarrhea, bloating, or chronic fatigue, but can also be asymptomatic. These cases may present with anemia due to chronic iron and vitamin malabsorption. In adults, celiac disease is detected twice as frequently in females, perhaps because menstrual bleeding accentuates the effects of impaired iron absorption. A characteristic itchy, blistering skin lesion, *dermatitis herpetiformis* (Chapter 25), is present in as many as 10% of patients.

Pediatric celiac disease, which affects males and females equally, may present with malabsorption or atypical symptoms affecting almost any organ. In those with classic symptoms, disease typically begins after introduction of gluten to the diet, between 6 and 24 months of age, and manifests as irritability, abdominal distention, anorexia, chronic diarrhea, failure to thrive, weight loss, or muscle wasting. Children with nonclassical symptoms tend to present at older ages with complaints of abdominal pain, nausea, vomiting, bloating, or constipation. Common extraintestinal complaints include arthritis or joint pain, aphthous stomatitis, iron deficiency anemia, delayed puberty, and short stature.

Unfortunately, the only treatment currently available for celiac disease is a gluten-free diet. Exclusion of dietary gluten typically results in resolution of symptoms, reduced titers of anti-TG2 and other celiac disease-associated antibodies, and restoration of normal or near-normal mucosal histology within 6 to 24 months. A gluten-free diet may also reduce the risk of long-term complications including anemia, female infertility, osteoporosis, and cancer.

Noninvasive serologic tests are generally performed prior to biopsy. The most sensitive test is the measurement of IgA antibodies against TG2. IgA antiendomysial antibodies can also be present. IgG anti-tissue transglutaminase antibodies may be detected in patients with IgA deficiency. The absence of HLA-DQ2 and HLA-DQ8 is useful for its negative predictive value, but the presence of these alleles is not helpful in confirming the diagnosis.

Individuals with celiac disease have an increased risk of malignancy. The most common cancer is enteropathy-associated T-cell lymphoma, an aggressive lymphoma of intraepithelial T lymphocytes. Small intestinal adenocarcinoma is also more frequent in individuals with celiac disease. Thus, when symptoms such as abdominal pain, diarrhea, and weight loss develop despite a strict gluten-free diet, it is important to consider refractory sprue, a possible precursor to T-cell lymphoma in which the response to a gluten-free diet is lost, or cancer.

Environmental Enteric Dysfunction

Environmental enteric dysfunction (EED), which has also been referred to as *environmental enteropathy*, *tropical enteropathy*, or *tropical sprue*, is a disorder prevalent in areas and populations with poor sanitation and hygiene, including severely under-resourced regions in parts of sub-Saharan Africa, such as Zambia; aboriginal populations in northern Australia; and some impoverished groups in South America and Asia. Affected individuals often experience malabsorption, malnutrition, and stunted growth. In regions where EED is endemic, relatively high oral vaccine failure rates have been proposed to point to an underlying defect in mucosal immune function.

There are presently no accepted clinical, laboratory, or histopathologic criteria that allow diagnosis of EED. Intestinal biopsy specimens have been examined in a small number of cases, and reported histologic features include more pronounced immune cell infiltration and less advanced villous atrophy than is seen in advanced celiac disease.

The underlying causes of EED are unknown. Defective intestinal barrier and transport functions secondary to chronic exposure to fecal pathogens and other microbial contaminants, mucosal immune abnormalities, and repeated bouts of diarrhea within the first 2 or 3 years of life are likely involved. Although many pathogens are endemic in affected communities, no single infectious agent has been linked to EED. Moreover, oral antibiotic treatment and nutritional supplementation do not reverse or prevent EED. Irreversible failures in physical development, such as short stature and low body weight, may be accompanied by uncorrectable cognitive deficits. Thus, it is difficult to overstate the global impact of EED, which is estimated to affect more than 150 million children worldwide and may contribute to a large number of childhood deaths.

Autoimmune Enteropathy

Autoimmune enteropathy is an X-linked disorder characterized by severe persistent diarrhea and autoimmune disease that occurs most often in young children. A particularly severe familial form, termed IPEX (for immune dysregulation, polyendocrinopathy, enteropathy, and X-linkage) is caused by germline loss of function mutations in the *FOXP3* gene, which is located on the X chromosome. *FOXP3* is a transcription factor expressed in CD4⁺ regulatory T cells, and individuals with *FOXP3* mutations have defective development and function of these cells. Other defects in regulatory T-cell function have been linked to less severe forms of autoimmune enteropathy. Autoantibodies to enterocytes and goblet cells are common, and some patients have antibodies to parietal or islet cells. Within the small intestine, intraepithelial lymphocytes may be increased but not to the

extent seen in celiac disease. In contrast to celiac disease, neutrophils are often seen infiltrating the intestinal mucosa. Therapy includes immunosuppressive drugs such as cyclosporine and allogeneic hematopoietic stem cell transplantation.

Lactase (Disaccharidase) Deficiency

Lactase deficiency, which can be congenital or acquired, results in diarrhea after ingestion of foods containing lactose. Lactase is a brush-border membrane disaccharidase that is required for lactose breakdown into glucose and galactose. In the absence of lactase, lactose, which cannot be absorbed, remains in the lumen where it exerts an osmotic force that results in water accumulation. This leads to indigestion in mild cases of acquired lactase deficiency, and diarrhea in more severe cases. Because the defect is biochemical, histology is generally unremarkable. Lactase deficiency is of two types:

- *Congenital lactase deficiency*, caused by a mutation in the gene encoding lactase, is an autosomal recessive disorder. The disease is rare and presents as explosive diarrhea with watery, frothy stools and abdominal distention upon milk ingestion. Symptoms abate when exposure to milk and milk products is terminated. Congenital lactase deficiency was often fatal prior to the availability of soy-based infant formula.
- *Acquired lactase deficiency* is caused by downregulation of lactase gene expression with aging. It is common across most populations but has an increased prevalence in individuals of African American and Chinese descent. Progressive downregulation of lactase expression with increased age is so common that it can be considered a normal variant. A likely explanation for its high prevalence is that until relatively recently (in evolutionary terms), people did not ingest milk after weaning. Depending on the levels of residual lactase activity and the quantity of dairy products ingested, patients can experience abdominal fullness, flatulence due to fermentation of unabsorbed sugars by colonic bacteria, and diarrhea.
- *Microbial infections*. Lactase deficiency can also present following enteric viral or bacterial infections. Accelerated epithelial cell turnover in enteric viral or bacterial infections limits villous enterocyte differentiation and mucosal surface area, both of which can contribute to acute lactose intolerance, which typically resolves when the infection clears.

Microvillus Inclusion Disease

Microvillus inclusion disease, sometimes referred to as Davidson disease, is a rare autosomal recessive disorder of vesicular transport that leads to deficient brush-border assembly. It is caused by mutations in the *MYO5B* gene, which encodes a motor protein (myosin VB) that is required for delivery and retrieval of plasma membrane components and for normal nutrient, ion, and water transport. The disease occurs most commonly in European, Middle Eastern, and Navajo Native American populations. Severe, intractable diarrhea develops before 3 months of age (often in the first few days of life) because of defective terminal digestion and transepithelial transport functions. The defect in plasma membrane trafficking leads to the accumulation of abnormal apical vesicles containing microvilli and various membrane components. The abnormal vesicles can be identified by electron microscopy or by immunostaining for the brush border protein villin. CD10 immunohistochemistry, which labels the microvillus inclusions as well as other cytoplasmic structures, is also used for diagnosis. Total parenteral nutrition and small bowel transplantation are the only treatments available for microvillus inclusion disease.

Abetalipoproteinemia

Abetalipoproteinemia is a rare autosomal recessive disease characterized by an inability to assemble triglyceride-rich lipoproteins. It is caused by a mutation in microsomal triglyceride transfer protein (MTP), which is required for transfer of lipids to nascent apolipoprotein B polypeptide in the endoplasmic reticulum. Without MTP, lipids accumulate intracellularly. Triglyceride accumulation is evident as epithelial vacuolization and can be highlighted by special stains, such as oil red-O, particularly after a fatty meal.

Abetalipoproteinemia presents in infancy with failure to thrive, diarrhea, and steatorrhea. Because of the defect in lipoprotein assembly, the plasma is completely devoid of lipoproteins containing apolipoprotein B. Failure to absorb essential fatty acids leads to deficiencies of fat-soluble vitamins as well as lipid membrane defects that can be recognized by the presence of acanthocytes (red cells with spiky membrane protrusions) in peripheral blood smears.

KEY CONCEPTS

Congenital and Acquired (Noninfectious) Disorders of the Intestines

- Abdominal hernias may occur through any weakness or defect in the wall of the peritoneal cavity including inguinal and femoral canals, the umbilicus, and sites of surgical scars.
- Intussusception occurs when a segment of intestine telescopes into the immediately distal segment. It is the most common idiopathic cause of intestinal obstruction in children younger than 2 years but is driven by leading edge mass lesions in older individuals.
- Ischemic bowel disease of the colon is most common at the splenic flexure, sigmoid colon, and rectum; these are watershed zones where two arterial circulations terminate.
- Angiodysplasia is a malformation of submucosal and mucosal blood vessels that is a common cause of lower intestinal bleeding in those older than 60 years.
- Diarrhea can be characterized as secretory, osmotic, malabsorptive, or exudative.
- The malabsorption associated with cystic fibrosis is the result of pancreatic insufficiency, leading to inadequate pancreatic digestive enzymes and deficient luminal breakdown of nutrients.
- Celiac disease is an immune-mediated enteropathy triggered by the ingestion of gluten-containing grains. The malabsorptive diarrhea in celiac disease stems from loss of brush-border surface area due to villous atrophy and, possibly, deficient enterocyte maturation because of immune-mediated epithelial damage.
- Environmental enteric dysfunction is prevalent in areas with poor sanitation. It is estimated to affect more than 150 million children worldwide and may contribute to a large number of childhood deaths.
- Autoimmune enteropathy is characterized by persistent diarrhea associated with other forms of autoimmune disease. A particularly severe form is IPEX, caused by mutation in the *FOXP3* gene, which is necessary for regulatory T-cell development and function.
- Lactase deficiency causes osmotic diarrhea due to the inability to break down lactose. The autosomal recessive form is rare and severe; the acquired form is common and usually presents in adulthood.
- Microvillus inclusion disease is a rare autosomal recessive disorder due to mutations in *MYO5B* that impair vesicular transport and brush-border assembly.

Infectious Enterocolitis

Enterocolitis can present with a broad range of symptoms including diarrhea, abdominal pain, urgency, perianal discomfort, incontinence, and hemorrhage (Table 17.7). This global problem is responsible

TABLE 17.7 Features of Bacterial Enterocolitides

Infection Type	Geography	Reservoir	Transmission	Epidemiology	Affected GI Sites	Symptoms	Complications
Cholera	India, Africa	Shellfish	Fecal-oral, water	Sporadic, endemic, epidemic	Small intestine	Severe watery diarrhea	Dehydration, electrolyte imbalances
<i>Campylobacter</i> spp.	High-income countries	Chickens, sheep, pigs, cattle	Poultry, milk, other foods	Sporadic; children, travelers	Colon	Watery or bloody diarrhea	Reactive arthritis, Guillain-Barré syndrome
Shigellosis	Worldwide, endemic in under-resourced countries	Humans	Fecal-oral, food, water	Children, migrant workers, travelers, nursing home residents	Left colon, ileum	Bloody diarrhea	Reactive arthritis, urethritis, conjunctivitis, hemolytic uremic syndrome
Salmonellosis	Worldwide	Poultry, farm animals, reptiles	Meat, poultry, eggs, milk	Children, older adults	Colon and small intestine	Watery or bloody diarrhea	Sepsis, abscess
Enteric (typhoid) fever	India, Mexico, Philippines	Humans	Fecal-oral, water	Children, adolescents, travelers	Small intestine	Bloody diarrhea, fever	Chronic infection, carrier state, encephalopathy, myocarditis, intestinal perforation
<i>Yersinia</i> spp.	Northern and central Europe	Pigs, cows, puppies, cats	Pork, milk, water	Clustered cases	Ileum, appendix, right colon	Abdominal pain, fever, diarrhea	Reactive arthritis, erythema nodosum
<i>Escherichia coli</i> ETEC	Under-resourced countries	Unknown	Food or fecal-oral	Infants, adolescents, travelers	Small intestine	Severe watery diarrhea	Dehydration, electrolyte imbalances
EPEC	Worldwide	Humans	Fecal-oral	Infants	Small intestine	Watery diarrhea	Dehydration, electrolyte imbalances
EHEC	Worldwide	Widespread, includes cattle	Beef, milk, produce	Sporadic and epidemic	Colon	Bloody diarrhea	Hemolytic uremic syndrome
EIEC	Under-resourced countries	Unknown	Cheese, other foods, water	Young children	Colon	Bloody diarrhea	Unknown
EAEC	Worldwide	Unknown	Unknown	Children, adults, travelers	Colon	Nonbloody diarrhea, afebrile	Poorly defined
Pseudomembranous colitis (<i>C. difficile</i>)	Worldwide	Humans, hospitals	Antibiotics allow emergence	Immunosuppressed, antibiotic-treated	Colon	Watery diarrhea, fever	Relapse, toxic megacolon
Whipple disease	Rural > urban	Unknown	Unknown	Rare	Small intestine	Malabsorption	Arthritis, CNS disease
Mycobacterial infection	Worldwide	Unknown	Unknown	Immunosuppressed, endemic	Small intestine	Malabsorption	Pneumonia, infection at other sites

CNS, Central nervous system; EAEC, enteroaggregative *E. coli*; EIEC, enteroinvasive *E. coli*; EHEC, enterohemorrhagic *E. coli*; EPEC, enteropathogenic *E. coli*; ETEC, enterotoxigenic *E. coli*; GI, gastrointestinal.

for over 1 million deaths each year. Half of the mortality occurs in children under the age of 5, in whom diarrheal disease is the fourth most common cause of death worldwide. The responsible pathogens (see Table 17.7) vary widely as a function of age, nutrition, and host immune status as well as geography and other environmental influences, such as the availability of clean drinking water. Bacterial agents, such as enterotoxigenic *E. coli* and *Salmonella* spp., are frequently responsible for acute diarrheal illnesses. Rotavirus and adenovirus are common causes of death in children under 5 years. Mycobacterial infections of the GI tract are considered in detail in Chapter 8.

Bacterial Gastroenteritis

Here we will discuss the common and clinically significant bacterial infections of the intestines.

Cholera

Vibrio cholerae are comma-shaped, gram-negative bacteria that cause cholera, a diarrheal disease that has been endemic in the Ganges Valley of India and Bangladesh throughout history. Since 1817, seven great cholera pandemics have spread along trade routes to

Europe, Australia, and the Americas; after each pandemic, cholera retreated back to the Ganges Valley. Cholera also persists within the Gulf of Mexico but causes only rare cases of seafood-associated disease. This occurs because shellfish and plankton can be reservoirs of *Vibrio cholerae*. The incidence of cholera increases during summer, reflecting more rapid growth of *Vibrio* bacteria at warm temperatures. It is primarily transmitted by drinking contaminated water and may cause epidemics in areas where disasters such as earthquakes or war have destroyed sewage systems, leading to fecal contamination of drinking water supplies.

Pathogenesis

The severe secretory diarrhea that is the hallmark of cholera is caused by toxins released by the bacteria. *Vibrio* organisms are noninvasive and remain within the intestinal lumen. Proteins involved in motility and attachment are necessary for efficient colonization. Hemagglutinin, a metalloproteinase, is important for bacterial detachment and shedding into the stool, which is key to dissemination.

Following colonization, the bacteria produce cholera toxin, which is encoded by a virulence phage. It is composed of five B subunits and a single A subunit. The B subunits bind GM1 ganglioside on the surface of intestinal epithelial cells, allowing the toxin to be carried by retrograde transport to the endoplasmic reticulum (Fig. 17.30). Here, the A subunit of the toxin is reduced by protein disulfide isomerase, and a fragment of the A subunit is unfolded. This peptide fragment is then transported into the cytosol using host cell machinery that normally moves misfolded proteins from the endoplasmic reticulum to the cytosol. Once in the cytosol, the A subunit fragment refolds and interacts with cytosolic adenosine diphosphate (ADP) ribosylation factors (ARFs) to activate the stimulatory G protein $G_{s\alpha}$. This stimulates adenylate cyclase, and the resulting increase in intracellular cyclic adenosine monophosphate (cAMP) opens the CFTR, which

releases chloride ions into the lumen. Chloride and sodium absorption are also inhibited by cAMP. The resulting accumulation of chloride, bicarbonate, and sodium within the intestinal lumen creates an osmotic force that draws water into the lumen and causes massive secretory diarrhea. Remarkably, mucosal biopsies show only minimal histologic alterations.

Clinical Features

Most individuals exposed to *V. cholerae* are asymptomatic or develop only mild diarrhea. In those with severe disease there is an abrupt onset of watery diarrhea and vomiting following an incubation period of 1 to 5 days. The voluminous stools resemble rice water and are sometimes described as having a fishy odor. The rate of diarrhea may reach 1 L per hour, leading to dehydration, hypotension, muscular cramping, anuria, shock, loss of consciousness, and death. Severe cholera is fatal in about 50% of cases without treatment, but timely fluid replacement saves more than 99% of patients. Oral rehydration is often sufficient but is unfortunately underutilized. Promising investigational therapies include CFTR inhibitors, which prevent diarrhea by blocking chloride secretion. Vaccination of at-risk individuals during cholera outbreaks can limit disease spread and is recommended, along with other prevention control strategies, by the World Health Organization (WHO). Prophylactic vaccination is a long-term goal, and some data indicate that this approach will convey sufficient herd immunity to reduce the incidence of cholera.

Campylobacter Enterocolitis

***Campylobacter jejuni* is the most common bacterial enteric pathogen in high-resource countries and is an important cause of traveler's diarrhea and food poisoning.** The Centers for Disease Control and Prevention (CDC) estimate over 1.5 million cases occur each year within the United States. *Campylobacter* outbreaks are most

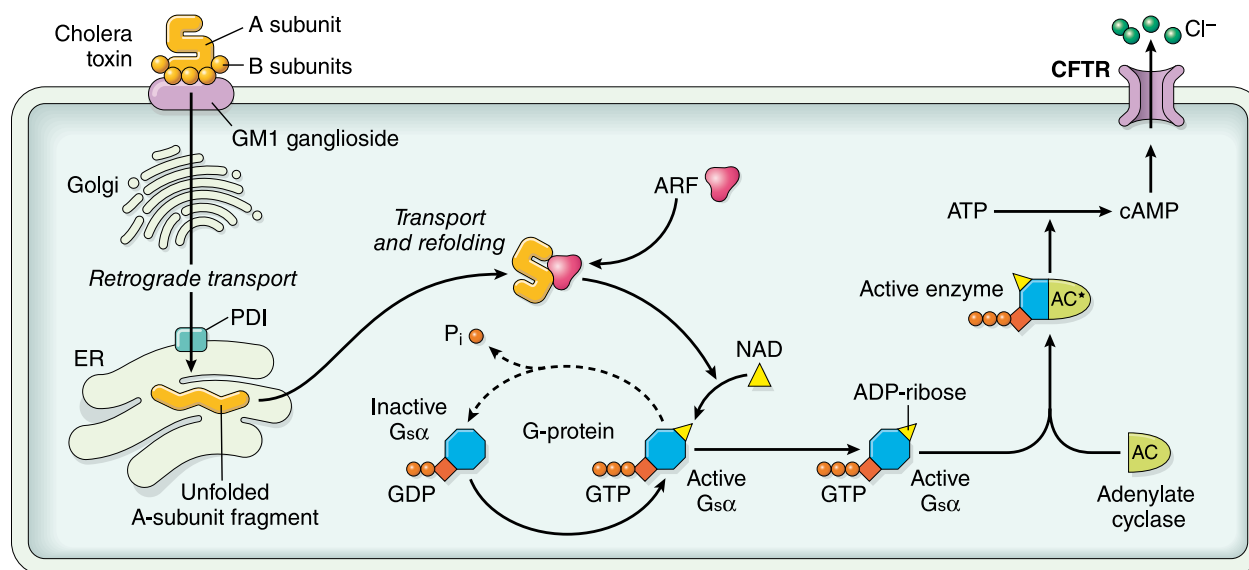


Fig. 17.30 Cholera toxin transport and signaling. After retrograde toxin transport to the endoplasmic reticulum (ER), the A subunit is released by the action of protein disulfide isomerase (PDI) and is then able to access the epithelial cell cytoplasm. In concert with an ADP-ribosylation factor, the A subunit then ADP-ribosylates $G_{s\alpha}$, which locks it in the active, GTP-bound state. This leads to adenylate cyclase (AC) activation and increased production of cyclic adenosine monophosphate (cAMP), which opens the cystic fibrosis transmembrane conductance regulator (CFTR) to drive chloride secretion and diarrhea. ATP, Adenosine triphosphate; GDP, guanosine diphosphate; GTP, guanosine triphosphate; NAD, nicotinamide adenine dinucleotide.

often associated with ingestion of improperly cooked chicken but can also be caused by drinking unpasteurized milk or contaminated water. Other important sources include direct contact with farm animals, including chickens, cattle, sheep, and pigs, and pets such as kittens and puppies. In contrast, person-to-person transmission is uncommon.

Pathogenesis

Ingestion of as few as 500 *C. jejuni* organisms can cause disease and, therefore, transmission is efficient. The four major properties that contribute to virulence are motility, adherence, toxin production, and invasion. Flagella allow *Campylobacter* to be motile, which facilitates adherence and colonization. Some *C. jejuni* isolates also release cytotoxins that cause direct epithelial damage or a cholera toxin-like enterotoxin. Dysentery (i.e., bloody diarrhea) is generally associated with bacterial invasion and is caused by only a minority of *Campylobacter* strains. Enteric fever occurs when bacteria proliferate within the lamina propria and mesenteric lymph nodes.

Campylobacter infection can result in reactive arthritis, primarily in patients with HLA-B27 genotype. Other extraintestinal complications, including erythema nodosum and Guillain-Barré syndrome, a flaccid paralysis caused by immunologically mediated inflammation of peripheral nerves (Chapter 27), are not HLA-linked. Molecular mimicry has been implicated in the pathogenesis of Guillain-Barré syndrome, as serum antibodies to *C. jejuni* lipopolysaccharide cross-react with peripheral and central nervous system gangliosides. The risk of developing Guillain-Barré syndrome following *Campylobacter* infection is estimated to be less than 0.1%, but up to 40% of Guillain-Barré cases have documented *Campylobacter* infection in the 2 weeks prior to presentation, and up to 50% have positive stool cultures or circulating

anti-*Campylobacter* antibodies. *Campylobacter* infection is also associated with postinfectious irritable bowel syndrome and a modest increase in risk of developing inflammatory bowel disease.

MORPHOLOGY

Campylobacter are comma-shaped, flagellated, gram-negative organisms. Diagnosis is primarily by stool culture, since biopsy findings are nonspecific. Mucosal and intraepithelial neutrophil infiltrates are prominent, particularly within the superficial mucosa (Fig. 17.31A); cryptitis (neutrophil infiltration of the crypt epithelium) and crypt abscesses (accumulations of neutrophils within crypt lumens) may also be present. Importantly, crypt architecture is preserved (Fig. 17.31D), although this can be difficult to assess in cases with severe mucosal damage.

Clinical Features

Campylobacter enteritis is common in young children but, unlike other enteric infections, displays a second peak in young adults. Older adults are also affected and, paradoxically, report typical symptoms such as pain less frequently than children but are more likely to have severe illness and be hospitalized. Disease develops after an incubation period of up to 8 days and may be associated with an influenza-like prodrome before onset of watery diarrhea, which is the primary symptom, but dysentery develops in more than 50% of children and ~15% of adults. Patients may shed bacteria for 1 month or more after clinical resolution. Antibiotic therapy is generally not required.

Shigellosis

***Shigella* are gram-negative unencapsulated, nonmotile, facultative anaerobes that belong to the Enterobacteriaceae family and are**

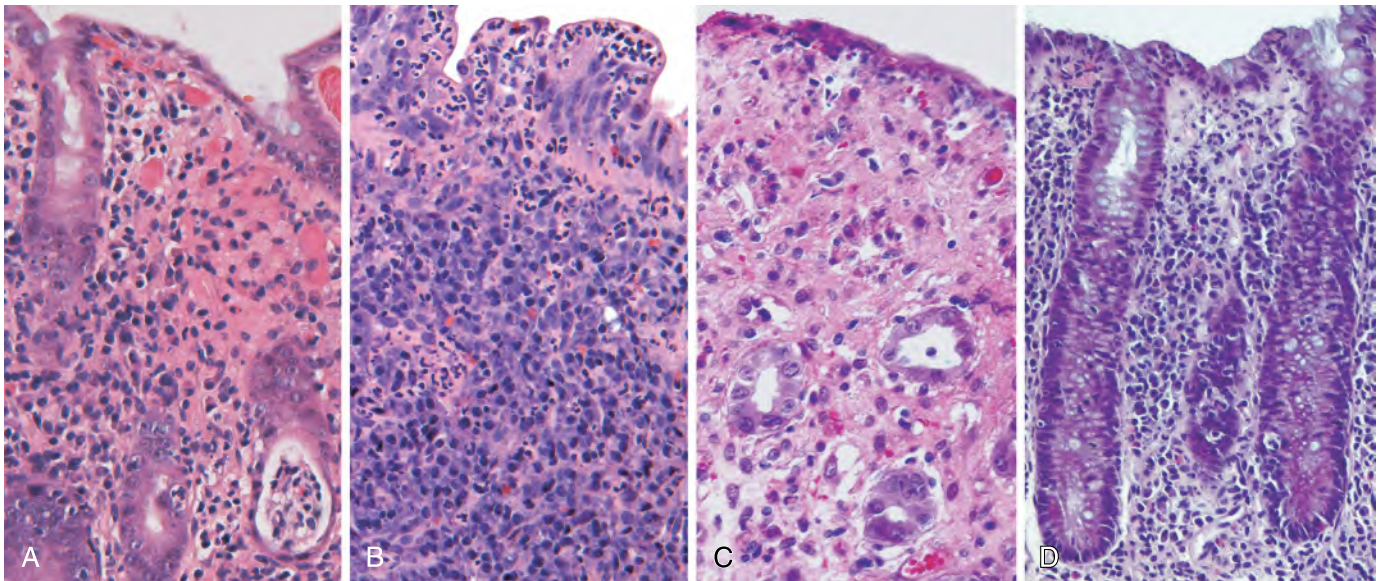


Fig. 17.31 Bacterial enterocolitis. (A) *Campylobacter jejuni* infection produces acute, self-limited colitis. Neutrophils can be seen within the surface and crypt epithelia, and a crypt abscess is present at the lower right. (B) In *Yersinia* infection the surface epithelium can be eroded by neutrophils, and the lamina propria is densely infiltrated by sheets of plasma cells admixed with lymphocytes and neutrophils. (C) Enterohemorrhagic *Escherichia coli* O157:H7 results in an ischemia-like morphology with surface atrophy and erosion. (D) Enteroinvasive *E. coli* infection is similar to other acute, self-limited colitides such as those caused by *C. jejuni*. Note the maintenance of normal crypt architecture and spacing despite abundant intraepithelial neutrophils.

closely related to enteroinvasive *E. coli*. *Shigella* spp. are one of the most common causes of bloody diarrhea, with a worldwide incidence of up to 165 million cases each year. Although humans are the only known reservoir, up to 10^9 organisms are present in each gram of stool during acute disease and as few as several hundred organisms are sufficient for infection; thus, fecal-oral transmission via contaminated water and food is extremely efficient.

In the United States and Europe, children in daycare centers, migrant workers, travelers to low-resource countries, and individuals in nursing homes are most commonly affected. Deaths are generally limited to children younger than 5 years of age. *Shigella* is endemic in countries with poor sanitation, where it is responsible for 10% of pediatric diarrheal disease and up to 75% of diarrheal deaths.

Pathogenesis

Shigella are resistant to gastric acid, which partly explains the low infective dose. Once in the intestine, organisms are taken up by M (microfold) cells, which are specialized for luminal antigen sampling. The bacteria proliferate within the M cells and then escape into the lamina propria, where they are phagocytosed by and induce apoptosis of macrophages. The inflammatory response damages the surface epithelium and allows *Shigella* within the intestinal lumen as well as those within the lamina propria to gain access to the basolateral membranes of colonic epithelial cells, across which bacteria invade the cytoplasm more efficiently than from the apical surface. All *Shigella* spp. carry virulence plasmids, some of which encode a type III secretion system capable of directly injecting bacterial proteins into the host cytoplasm. *Shigella dysenteriae* serotype 1 also release the Shiga toxin Stx, which inhibits eukaryotic protein synthesis, resulting in host cell damage and death.

MORPHOLOGY

Shigella infections are most prominent in the left colon, but the ileum may also be involved, perhaps reflecting the abundance of M cells in the **dome epithelium** overlying Peyer patches. The mucosa is hemorrhagic and ulcerated, and pseudomembranes may be present. The histology in early cases is like other acute self-limited forms of colitis such as *Campylobacter* colitis, but because of the tropism for M cells, aphthous ulcers similar to those seen in Crohn disease may occur. This and other factors may lead to confusion with inflammatory bowel disease.

Clinical Features

After an incubation period of up to 1 week, *Shigella* causes self-limited disease characterized by 7 to 10 days of diarrhea, fever, and abdominal pain. Watery diarrhea progresses to dysentery in approximately 50% of patients, and constitutional symptoms can persist for as long as 1 month. Symptom duration is typically short in children, but severity is often much greater in adults. An uncommon subacute presentation is weeks of waxing and waning diarrhea can mimic new-onset ulcerative colitis.

Confirmation of *Shigella* infection requires stool culture. Antibiotic treatment shortens both the clinical course and the duration of bacterial shedding in stools. Conversely, antidiarrheal medications can prolong symptoms and delay *Shigella* clearance and are therefore contraindicated.

Extraintestinal complications of *Shigella* infection are uncommon but include a triad of sterile reactive arthritis, urethritis, and conjunctivitis that preferentially affects HLA-B27–positive males between 20 and 40 years of age. *S. dysenteriae* serotype 1 that secrete

Shiga toxin may also trigger hemolytic uremic syndrome, although enterohemorrhagic *E. coli* (EHEC) are more commonly responsible (Chapter 20). Toxic megacolon and intestinal obstruction are uncommon complications.

Salmonella

There are two groups of *Salmonella*: *S. typhi*, the causative agent of typhoid fever (discussed in the next section), **and nontyphoid *Salmonella* species, such as *S. enteritidis*.** The latter are the most common cause of salmonellosis, of which more than 1 million cases occur each year in the United States; the prevalence is even greater in under-resourced countries. Infection is most common in young children and older adults, with peak incidence in the summer and fall. *Salmonella* is transmitted via contaminated food, particularly raw or undercooked meat, poultry, eggs, and milk. As a result, gatherings where food is left unrefrigerated for extended times can lead to local outbreaks. Larger outbreaks can be caused by centralized processing of many foods, including cantaloupes, fresh diced onions, ground beef, and raw cookie dough. Vaccines are available for both humans and farm animals (e.g., egg-laying hens).

Pathogenesis

Very few viable *Salmonella* are necessary to cause infection; the absence of gastric acid, in individuals with atrophic gastritis or those on acid-suppressive therapy, further reduces the required inoculum. *Salmonella* possess virulence genes that encode a type III secretion system capable of transferring bacterial proteins into M cells and enterocytes. The transferred proteins activate host Rho guanosine triphosphatases (GTPases), thereby triggering actin rearrangement and bacterial endocytosis which, in turn, facilitates bacterial growth within endosomes. *Salmonella* also secrete a molecule that induces epithelial cell release of the eicosanoid heparin A3, a potent chemoattractant that stimulates neutrophil chemotaxis into the intestinal lumen and potentiates mucosal damage. Flagellin, the core protein of bacterial flagella, and bacterial lipopolysaccharide may activate the innate immune receptors TLR5 and TLR4, respectively, further augmenting inflammation and tissue damage. Both Th1 and Th17 T cells contribute to pathogen clearance, explaining why those with genetic defects in Th1 or Th17 immunity are at risk for disseminated salmonellosis.

Clinical Features

Salmonella infections are clinically indistinguishable from those caused by other enteric pathogens. Stool cultures are therefore essential for diagnosis. Symptoms range from loose stools to cholera-like profuse diarrhea to dysentery. Fever often resolves within 2 days, but diarrhea can persist for a week, and organisms can be shed in the stool for several weeks after resolution. Antibiotic therapy is not generally recommended because it can prolong the carrier state or even cause relapse and does not shorten the duration of diarrhea. Although *Salmonella* infections are typically self-limited, some complications, including reactive arthritis, meningitis, and even death may occur, particularly in patients with malignancies, immunosuppression, alcohol use disorder, cardiovascular dysfunction, sickle cell disease, or hemolytic anemia. Like *Campylobacter* infection, *Salmonella* infection is associated with postinfectious irritable bowel syndrome and an increased risk of inflammatory bowel disease.

Typhoid Fever

Typhoid fever, also referred to as enteric fever, is caused by *Salmonella enterica* and its two subtypes, *S. typhi* and *S. paratyphi*. It has an annual worldwide incidence of 30 million individuals. Most cases in

endemic countries are due to *S. typhi*, while infection by *S. paratyphi* is more common among travelers, perhaps because they are vaccinated against *S. typhi*. In endemic areas, children and adolescents are affected most often, but there is no age preference in nonendemic countries. Infection is strongly associated with travel to India, Mexico, the Philippines, Pakistan, El Salvador, and Haiti. Humans are the sole reservoir for *S. typhi* and *S. paratyphi*; transmission occurs from person to person or via food or contaminated water. Gallbladder colonization with *S. typhi* or *S. paratyphi* may be associated with gallstones and a chronic carrier state.

Pathogenesis

In contrast to *S. enteritidis*, *S. typhi* disseminate via lymphatic and blood vessels, which causes systemic reactive hyperplasia of phagocytes and lymphoid tissues. As with *Shigella* spp., *S. typhi* are resistant to gastric acid and initially invade via small intestinal M cells.

MORPHOLOGY

Salmonella infection causes Peyer patches in the terminal ileum to enlarge into sharply delineated, plateau-like elevations up to 8 cm in diameter. Draining mesenteric lymph nodes are also enlarged. Neutrophils accumulate within the superficial lamina propria together with macrophages containing bacteria, red cells, nuclear debris, lymphocytes, and plasma cells. Mucosal damage creates oval ulcers, often oriented longitudinally along the axis of the ileum, that may perforate.

In disseminated *S. typhi* infection, the spleen is enlarged and soft, with uniformly pale red pulp, obliterated follicular markings, and prominent phagocyte hyperplasia. The liver contains small, randomly scattered foci of parenchymal necrosis in which hepatocytes are replaced by macrophage aggregates, called **typhoid nodules**, which may also be found in the bone marrow and lymph nodes.

Clinical Features

Acute infection results in anorexia, abdominal pain, bloating, nausea, vomiting, and bloody diarrhea followed by a short asymptomatic phase that gives way to bacteremia and fever with flu-like symptoms. Abdominal tenderness may mimic appendicitis. Rose spots, small erythematous maculopapular lesions, develop on the chest and abdomen. Symptoms abate after several weeks but relapse can occur. Systemic dissemination may cause extraintestinal complications including encephalopathy, meningitis, seizures, endocarditis, myocarditis, pneumonia, and cholecystitis. Patients with sickle cell disease are particularly susceptible to *Salmonella* osteomyelitis. Blood cultures are positive in more than 90% of affected individuals during the febrile phase. Antibiotic treatment can prevent disease progression.

Yersinia

Three *Yersinia* species are human pathogens. *Y. enterocolitica* and *Y. pseudotuberculosis* cause GI disease and are discussed here; *Y. pestis*, the agent of pulmonic and bubonic plague, is discussed in Chapter 8. The incidence of GI *Yersinia* infections in Europe is greater than in North America, possibly due to different dietary practices. Infections are most frequently linked to ingestion of pork, raw milk, and contaminated water and tend to cluster in the winter.

Pathogenesis

Yersinia invade M cells and use specialized bacterial proteins, called adhesins, to bind to host cell β_1 integrins. A pathogenicity island encodes proteins that mediate iron capture and transport; similar

iron transport systems are also present in *E. coli*, *Klebsiella*, *Salmonella*, and enterobacteria. Iron enhances virulence of *Yersinia* and stimulates systemic dissemination, explaining why individuals with increased levels of nonheme iron, such as those with certain chronic forms of anemia or hemochromatosis, are at greater risk for sepsis and death.

MORPHOLOGY

Yersinia infections preferentially involve the ileum, appendix, and right colon (Fig. 17.31B). The organisms proliferate extracellularly in lymphoid tissue, resulting in regional lymph node and Peyer patch hyperplasia as well as bowel wall thickening. The mucosa overlying lymphoid tissue may become hemorrhagic, and aphthous ulcers may appear along with neutrophil infiltrates (see Fig. 17.31B) and granulomas. As with *Shigella*, these features can cause diagnostic confusion with inflammatory bowel disease.

Clinical Features

Yersinia infection generally presents as abdominal pain, nausea, vomiting, and abdominal tenderness. Extraintestinal symptoms of pharyngitis, arthralgia, and erythema nodosum occur frequently. Fever and diarrhea are less common. Peyer patch invasion with subsequent involvement of regional lymphatics can mimic acute appendicitis in teenagers and young adults, but enteritis and colitis predominate in younger children. *Yersinia* can be detected by stool culture on *Yersinia*-selective agar. In cases with extraintestinal disease, lymph nodes or blood cultures may also be positive. Postinfectious complications include reactive arthritis with urethritis and conjunctivitis, myocarditis, erythema nodosum, and kidney disease.

Escherichia coli

***E. coli* are gram-negative bacilli that colonize the healthy GI tract; most are nonpathogenic, but a subset cause human disease.** The latter are classified according to morphology, pathogenesis, and in vitro behavior. Subgroups with major clinical relevance include enterotoxigenic *E. coli* (ETEC), enteropathogenic *E. coli* (EPEC), enterohemorrhagic *E. coli* (EHEC), enteroinvasive *E. coli* (EIEC), and enteroaggregative *E. coli* (EAEC).

Enterotoxigenic *Escherichia coli*. ETEC are the principal cause of traveler's diarrhea and spread via contaminated food or water. In under-resourced countries, children younger than 2 years are particularly susceptible. ETEC are noninvasive but produce heat-labile toxin (LT) and heat-stable toxin (ST). LT is similar to cholera toxin and activates adenylate cyclase, resulting in increased intracellular cAMP and chloride secretion. ST binds to guanylate cyclase and increases intracellular cyclic guanosine monophosphate (cGMP); the effects on epithelial ion transport are like those induced by LT. Like cholera, ETEC infection results in only mild histologic changes.

Enteropathogenic *Escherichia coli*. EPEC are prevalent worldwide and are an important cause of endemic diarrhea as well as diarrheal outbreaks, particularly in children less than 2 years. EPEC are characterized by their ability to produce attaching and effacing (A/E) lesions in which bacteria attach tightly to small intestinal enterocyte apical membranes and cause local effacement of the microvilli. The lesions are caused by bacterial proteins that insert into the intestinal epithelial cell plasma membrane. Assays for these proteins can be used for molecular diagnosis of EPEC infection.

Enterohemorrhagic *Escherichia coli*. EHEC consists of *E. coli* O157:H7 and non-O157:H7 serotypes. *E. coli* O157:H7 outbreaks are often caused by consumption of inadequately cooked ground beef, reflecting a natural reservoir in cows. Contaminated milk and vegetables are also vehicles for infection. Both O157:H7 and non-O157:H7 serotypes produce Shiga-like toxins, and therefore lesions (see Fig. 17.31C) and clinical symptoms are similar to those resulting from *S. dysenteriae* infection. O157:H7 serotypes are more likely than non-O157:H7 serotypes to cause outbreaks of bloody diarrhea, hemolytic uremic syndrome, and ischemic colitis. Antibiotic treatment can lead to release of Shiga-like toxins and increased risk of hemolytic uremic syndrome, especially in children.

Enteroinvasive *Escherichia coli*. EIEC organisms are bacteriologically similar to *Shigella* and are transmitted via food or water or by person-to-person contact. While EIEC do not produce toxins, they invade epithelial cells and cause nonspecific acute self-limited colitis (see Fig. 17.31D). EIEC infections are most common among young children in under-resourced countries.

Enteraggregative *Escherichia coli*. These organisms cause diarrhea in children and adults worldwide, including traveler's diarrhea. The organisms attach to enterocytes via adherence fimbriae and are aided by dispersin, a bacterial surface protein that neutralizes the negative surface charge of lipopolysaccharide. EAEC organisms produce minimal histologic changes and cause nonbloody diarrhea that may be prolonged in individuals with immunodeficiency.

Pseudomembranous Colitis

Pseudomembranous colitis is generally caused by *Clostridioides* (formerly called *Clostridium*) *difficile* and is also referred to as antibiotic-associated colitis or antibiotic-associated diarrhea. While other organisms such as *Salmonella*, *Clostridium perfringens* type A, or *Staphylococcus aureus* may also produce diarrhea in the context of antibiotic therapy, only *C. difficile* causes pseudomembranous colitis.

Pathogenesis

Disruption of the normal colonic microbiota by antibiotics allows *C. difficile* overgrowth. Almost any antibiotic may be responsible; the most important determinants of the disease are frequency of use and the effect on colonic microbiota. Toxins released by *C. difficile* cause the ribosylation of small GTPases such as Rho, leading to disruption of the epithelial cytoskeleton, tight junction barrier loss, cytokine release, and apoptosis. Toxin-mediated cytoskeletal disruption in other cell types, particularly neutrophils, may also contribute to development of purulent pseudomembranes, but the processes that produce pseudomembranous colitis remain incompletely understood.

MORPHOLOGY

The surface epithelium is denuded, and the superficial lamina propria contains a dense infiltrate of neutrophils and occasional fibrin thrombi within capillaries.

Pseudomembranes (Fig. 17.32A,B), made up of an adherent layer of inflammatory cells and debris, are not specific and may also be present in ischemia or necrotizing infections. However, the histopathology of *C. difficile*-associated colitis is so distinct that it is pathognomonic. Specific features include a mucopurulent exudate that characteristically “erupts” from damaged crypts to form “volcano” lesions (Fig. 17.32C) and coalesces to form pseudomembranes.

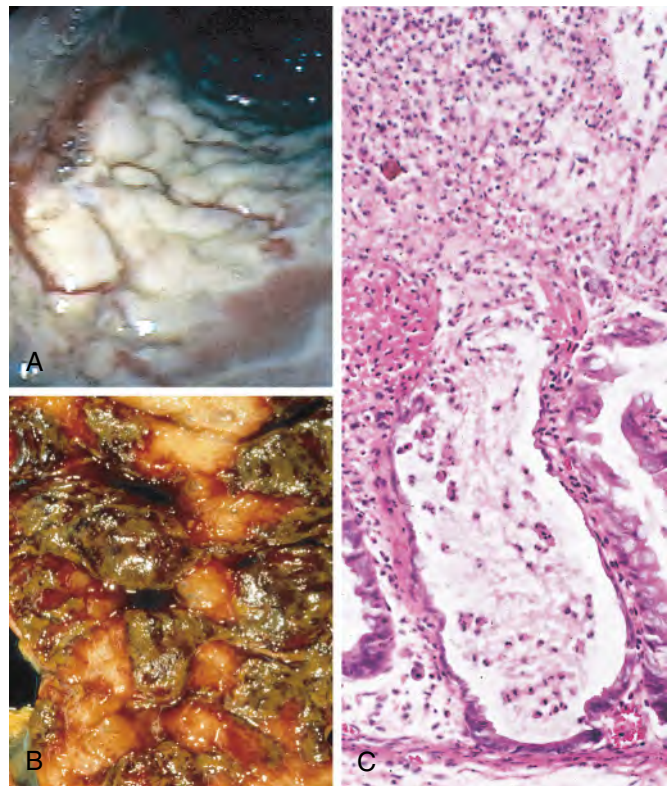


Fig. 17.32 *Clostridioides difficile* colitis. (A) The colon is coated by gray pseudomembranes composed of neutrophils, dead epithelial cells, and inflammatory debris (endoscopic view). (B) Pseudomembranes are easily appreciated on gross examination. (C) Typical pattern of neutrophils emanating from a damaged, dilated crypt (a so-called volcano lesion).

Clinical Features

Risk factors for *C. difficile*-associated colitis include antibiotic treatment, advanced age, hospitalization, and immunosuppression. The organism is particularly prevalent in hospitals; as many as 30% of hospitalized adults are colonized with *C. difficile* (a rate 10-fold greater than the general population), but most colonized patients are free of disease. Individuals with *C. difficile*-associated colitis present with fever, leukocytosis, abdominal pain, cramps, watery diarrhea, and dehydration. Protein loss can give rise to hypoalbuminemia. Fecal leukocytes and occult blood may be present, but grossly bloody diarrhea is uncommon. **Toxic megacolon** is a potentially fatal complication resulting from marked injury to the colonic wall. The diagnosis is usually made by detection of *C. difficile* toxin, rather than by culture, and is supported by the characteristic histopathology. Metronidazole and vancomycin are generally effective therapies, but the prevalence of antibiotic-resistant and hypervirulent *C. difficile* strains is increasing. Recurrent *C. difficile*-associated colitis occurs in up to 40% of patients. Fecal microbial transplantation successfully prevents recurrent infection in some patients and is becoming an accepted therapy. Many investigators are working to identify specific microbial populations that can be administered in place of fecal preparations. New antibiotics as well as monoclonal antibodies against toxins A and B can also be effective but are not yet in widespread use.

Whipple Disease

Whipple disease is a rare, multivisceral chronic disease that causes malabsorptive diarrhea. It was first described by George Hoyt

Whipple, a pathologist who shared the 1934 Nobel Prize for his work on pernicious anemia.

Pathogenesis

Whipple's original case report described an individual with malabsorption, lymphadenopathy, and arthritis of undefined origin. Postmortem examination demonstrated the presence of foamy macrophages filled with gram-positive microbes. The organism was identified as an actinomycete in 1992 and named *Tropheryma whipplei*. Clinical symptoms occur because bacteria-laden macrophages accumulate within the small intestinal lamina propria and mesenteric lymph nodes, causing lymphatic obstruction. Thus, **the malabsorptive diarrhea of Whipple disease is due to impaired lymph drainage.**

MORPHOLOGY

The morphologic hallmark of Whipple disease is a **dense accumulation of distended, foamy macrophages in the small intestinal lamina propria** (Fig. 17.33A). The macrophages contain periodic acid–Schiff (PAS)–positive, diastase-resistant granules that represent partially digested bacteria within lysosomes (Fig. 17.33B). Intact rod-shaped bacilli can also be identified by electron microscopy (Fig. 17.33C). A similar infiltrate of foamy macrophages is present in intestinal mycobacterial infections (Fig. 17.33D), and the organisms are PAS-positive in both diseases. The acid-fast stain can be helpful, since mycobacteria stain positively (Fig. 17.33E), while *T. whipplei* does not.

Villous expansion caused by dense macrophage infiltrates imparts a “shaggy” appearance to the mucosal surface. Lymphatic dilation and mucosal lipid deposition produce endoscopically visible white to yellow mucosal plaques. In Whipple disease, bacteria-laden macrophages can accumulate within **mesenteric lymph nodes, synovial membranes, cardiac valves, brain,** and other sites.

Clinical Features

Whipple disease is most common in males of Northern European descent, particularly farmers and others with occupational exposure to soil or animals. While there is no consistent familial clustering, the rarity of infection despite a large number of healthy carriers suggests that genetic risk factors exist.

The clinical presentation of Whipple disease is usually a triad of diarrhea, weight loss, and arthralgia. Extraintestinal symptoms may precede malabsorption by months or years and include arthritis; arthralgia; fever; lymphadenopathy; and neurologic, cardiac, or pulmonary disease.

Viral Gastroenteritis

Symptomatic human infection is caused by many viruses. The most common are discussed here.

Norovirus

This icosahedral virus with a single-stranded RNA genome was previously known as Norwalk-like virus. With increasing use of the rotavirus vaccine, **norovirus has become the most common cause of diarrhea worldwide**, being responsible for ~20% of acute gastroenteritis with diarrhea and vomiting and 50,000 pediatric deaths annually. In the United States norovirus outbreaks occur in the winter and affect ~20 million people annually.

Humans are the only known reservoir of norovirus. Fecal-oral transmission is responsible of most sporadic cases. Outbreaks spread easily within schools, hospitals, nursing homes, and other large groups in close quarters, such as on cruise ships. In these environments, vehicles for transmission include airborne droplets, environmental surfaces, and fomites. Enforcement of personal hygiene, enteric precautions, and regular

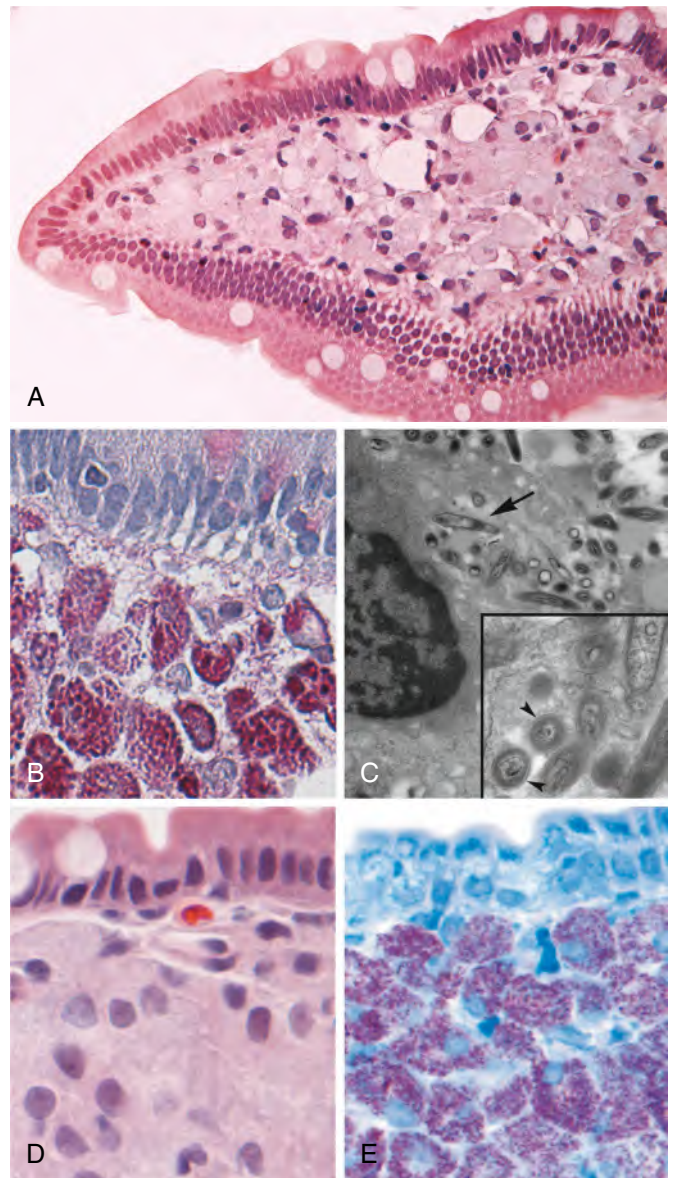


Fig. 17.33 Whipple disease and mycobacterial infection. (A) Hematoxylin and eosin staining shows effacement of normal lamina propria by a sheet of distended macrophages. (B) Periodic acid–Schiff stain highlights macrophage lysosomes full of bacilli. (C) Electron micrograph of part of a macrophage shows bacilli within the cell (top arrow); also seen at higher magnification (arrowheads in inset). (D) The morphology of mycobacterial infection can be similar to Whipple disease, particularly in the immunocompromised host. Compare with (A). (E) Unlike *T. whipplei*, mycobacteria are positive with stains for acid-fast bacteria. (C, Courtesy George Kasnic and Dr. William Clapp, University of Florida, Gainesville.)

environmental decontamination with alcohol-based hand sanitizers and quaternary ammonium wipes may reduce transmission, but prevention is difficult to achieve due to the resistance of the virus to a wide range of temperatures, from freezing to 60°C, and the extraordinarily low infective dose of less than 20 viral particles.

Following an incubation period of up to 2 days, affected individuals develop vomiting; cramping abdominal pain; watery diarrhea; and nonspecific symptoms including headache, chills, and myalgias. The symptoms resolve within 2 to 3 days in most cases. Infectious viral particles can be shed prior to symptoms and for up to 8 weeks after infection in healthy, immunocompetent individuals.

Norovirus infection in immunocompromised patients is a significant problem. Some data suggest that nearly 20% of patients on immunosuppression after renal transplantation or as treatment for graft-versus-host disease after allogeneic hematopoietic stem cell transplantation are infected with norovirus. Many of these patients are unable to clear the infection, experience chronic diarrhea, and shed infectious virus for up to 1 year.

Some individuals are naturally resistant to norovirus infection due to mutations that inactivate galactoside 2- α -L-fucosyltransferase (FUT2), an enzyme that fucosylates proteins in the Golgi apparatus that are destined for the plasma membrane, including blood group antigens. These glycoproteins serve as binding sites for the virus, and in their absence cells cannot be infected. Approximately 20% of individuals of European and African descent are homozygous for these mutations. These individuals, termed nonsecretors, neither shed blood group antigens in saliva nor express them on epithelial cells at mucosal sites including the GI tract. In others, acquired immunity is difficult to achieve because there is a great diversity of norovirus strains, which limits cross protection, and because the viral genome regularly undergoes recombination that results in antigenic shifts. As a result, a new pandemic strain emerges every 2 to 4 years.

MORPHOLOGY

Morphologic changes are nonspecific but include mild villous shortening, epithelial vacuolization, crypt hypertrophy, and lamina propria infiltration by neutrophils, lymphocytes, and monocytes (Fig. 17.34A). The microvillus brush border is also compromised, and reduced expression of digestive enzymes can lead to mild steatorrhea and carbohydrate malabsorption.

Rotavirus

This encapsulated virus with a segmented, double-stranded RNA genome is highly prevalent and a significant cause of diarrheal deaths worldwide. Children between 6 and 24 months of age are most vulnerable, probably because antibodies in breast milk confer protection during the first 6 months of life. Rotavirus vaccines are becoming widely used but are incompletely effective in under-resourced regions, possibly as a consequence of malnutrition. The WHO recommends that the first dose be given at 6 weeks of age.

Rotavirus outbreaks in hospitals and daycare centers are common, and infection spreads easily; the estimated minimal infective inoculum is only 10 viral particles. Rotavirus selectively infects and destroys mature enterocytes in the small intestine, and the villus surface is

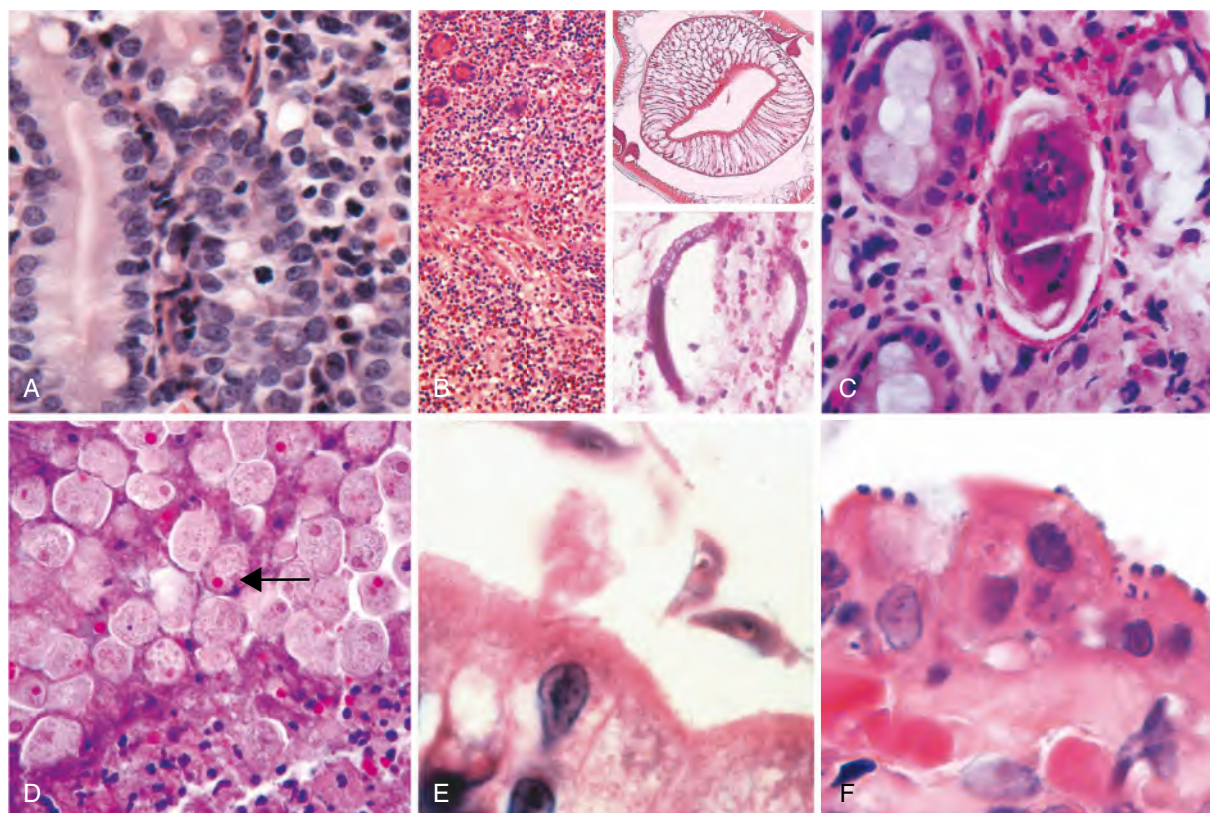


Fig. 17.34 Infectious enteritis. (A) Histologic features of viral enteritis include increased numbers of intraepithelial and lamina propria lymphocytes and crypt hypertrophy. (B) Diffuse eosinophilic infiltrates in parasitic infection. This case was caused by *Ascaris* (upper inset), but a similar tissue reaction could be caused by *Strongyloides* (lower inset). (C) Schistosomiasis can induce an inflammatory reaction to eggs trapped within the lamina propria. (D) *Entamoeba histolytica* in a colon biopsy specimen. Note some organisms ingesting red blood cells (arrow). (E) *Giardia lamblia*, which are present in the luminal space over nearly normal-appearing villi, are easily overlooked. (F) *Cryptosporidium* organisms are seen as small blue spheres that appear to lie on top of the brush border but are actually enveloped by a thin layer of host cell cytoplasm.

repopulated by immature epithelial cells. Enterocyte damage may be mediated by a viral factor called nonstructural protein 4 (NSP4), which can induce apoptosis. The loss of absorptive function and net secretion of water and electrolytes is compounded by an osmotic diarrhea caused by the incomplete absorption of nutrients. Like norovirus, rotavirus has a short incubation period followed by several days of vomiting and watery diarrhea.

Adenovirus

Adenovirus is a common cause of pediatric diarrhea that also affects older immunocompromised patients. Small intestinal biopsy specimens can show epithelial degeneration but more often exhibit nonspecific villous atrophy and compensatory crypt hyperplasia. Viral nuclear inclusions are uncommon. Disease typically presents after an incubation period of 1 week with symptoms that include diarrhea, vomiting, and abdominal pain. Fever and weight loss may also be present. Symptoms generally resolve within 10 days.

SARS-CoV-2

Coronavirus disease 2019 (COVID-19), caused by SARS-CoV-2, most often presents as pulmonary disease, ranging from mild upper respiratory infection to severe pneumonia, acute respiratory distress syndrome, and death (Chapter 8). However, ~50% of patients report gastrointestinal symptoms, typically anorexia, nausea, diarrhea, and vomiting, and gastrointestinal symptoms are the chief complaint in up to 15% of patients. COVID-19 is most severe in older adults and those with preexisting comorbidities. Overall COVID-19 prevalence in those with chronic intestinal disease is not higher than that of the general population, but risk of severe disease is increased by use of glucocorticoids in those with inflammatory bowel disease.

Gastrointestinal disease may reflect expression of angiotensin-converting enzyme 2 (ACE2) and transmembrane serine protease 2 (TMPRSS2), which serve as SARS-CoV-2 receptors, in intestinal epithelial cells. Gastrointestinal disease may, therefore, result from direct virus-mediated cytotoxicity as well as subsequent damage caused by the immune response to SARS-CoV-2. Consistent with this, viral proteins have been detected in gastric, small intestinal, and colorectal glandular epithelial cells.

Parasitic Enterocolitis

Although viruses and bacteria are the predominant enteric pathogens in the United States, parasitic disease and protozoal infections affect more than one-half of the world's population on a chronic or recurrent basis. The small intestine provides a home for many as 20 species of parasites, including nematodes such as the roundworms *Ascaris* and *Strongyloides*; hookworms and pinworms; cestodes, including flatworms and tapeworms; trematodes, or flukes; and protozoa. Parasitic infections are covered in Chapter 8; those that are common in the intestinal tract are discussed briefly here.

Ascaris lumbricoides

This nematode infects 1 billion individuals worldwide. *Ascaris* eggs are shed in the feces of infected individuals, allowing spread via the direct fecal-oral route. Eggs may also be present in improperly cleaned and peeled fruits and vegetables that have grown in contaminated soil, explaining why *Ascaris*, *Strongyloides*, hookworm (*Necator duodenale* and *Ancylostoma duodenale*), and whipworm (*Trichuris trichiura*) are known as soil-transmitted helminths. Ingested eggs hatch in the intestine, and larvae penetrate the intestinal mucosa. Larvae then migrate from the splanchnic to the systemic circulation and enter the lungs where they grow within alveoli. Approximately 3 weeks later, the larvae are coughed up and swallowed. Upon return to the small intestine, the larvae mature

into adult worms, which induce an eosinophil-rich inflammatory reaction (Fig. 17.34B) that can cause obstruct the intestine or biliary tree. Larvae can also form hepatic abscesses and cause pneumonitis. Diagnosis is usually made by detection of eggs in stool samples.

Strongyloides

The larvae of *Strongyloides* live in fecally contaminated ground soil and can penetrate unbroken skin. They migrate through the lungs, where they induce inflammatory infiltrates, and then reside in the intestine while maturing into adult worms. Unlike other intestinal worms, which require an ova or larval stage outside the human, the eggs of *Strongyloides* can hatch within the intestine and release larvae that penetrate the mucosa, causing autoinfection (see Fig. 17.34B). Hence, *Strongyloides* infection can persist for life, and immunosuppressed individuals can develop overwhelming autoinfection. *Strongyloides* incite a strong tissue reaction and peripheral eosinophilia.

Necator duodenale and *Ancylostoma duodenale*

These hookworms infect 1 billion people worldwide and cause significant morbidity. Infection is initiated by larval penetration through the skin. After further development in the lungs, they migrate up the trachea and are swallowed. Worms attach to the duodenal mucosa, suck blood, and reproduce. This causes multiple superficial erosions, focal hemorrhage, inflammatory infiltrates, and, in chronic infection, iron deficiency anemia. Diagnosis can be made by detection of eggs in fecal smears.

Trichuris trichiura

Whipworms primarily infect young children. Like *Enterobius vermicularis*, *T. trichiura* does not penetrate the intestinal mucosa and rarely causes serious disease. Heavy infections may cause bloody diarrhea and rectal prolapse.

Enterobius vermicularis and *Enterobius gregorii*

These organisms, also known as pinworms, infect people in both high-resource and low-resource countries; over 1 billion are infected worldwide. *E. vermicularis* is the more common species, but *E. gregorii* has been reported in Europe, Africa, and Asia. Because they do not invade host tissues and live their entire life within the intestinal lumen, pinworms rarely cause serious illness. Infection is primarily by the fecal-oral route. Adult worms living in the intestine migrate to the anal orifice at night, where the female deposits eggs on the perirectal mucosa. The eggs cause irritation and rectal and perineal pruritus. Scratching leads to contamination of the fingers, which promotes human-to-human transmission. Both eggs and adult pinworms remain viable outside the body, and repeat infection is common. Diagnosis can be made by applying cellophane tape to the perianal skin and examining the tape under a microscope to identify eggs.

Schistosomiasis

This disease involving the intestines most commonly takes the form of adult worms residing within the mesenteric veins. Eggs are trapped within the mucosa and submucosa (Fig. 17.34C). The resulting immune reaction is often granulomatous and can cause bleeding and even obstruction. More details are presented in Chapter 8.

Intestinal cestodes

The three main species of cestodes that infect humans are *Diphyllobothrium latum*, fish tapeworms; *Taenia solium*, pork tapeworms; and *Hymenolepis nana*, dwarf tapeworms. They reside exclusively within the intestinal lumen and are transmitted by ingestion of raw or undercooked fish, meat, or pork that contains encysted larvae. Release of the larva allows attachment to the intestinal mucosa through its head, or scolex. The

worm derives its nutrients from the food stream and enlarges by formation of egg-filled segments termed proglottids. Humans are usually infected by a single worm, and since the worm does not penetrate the intestinal mucosa, peripheral eosinophilia does not generally occur. Nevertheless, the parasite burden can be staggering. Adult worms can grow to many meters in length and shed large numbers of proglottids and eggs into the feces. Clinical symptoms include abdominal pain, diarrhea, and nausea, but most cases are asymptomatic. Occasionally, *D. latum* causes vitamin B₁₂ deficiency and megaloblastic anemia because it competes with the host for dietary B₁₂. Identification of proglottids and eggs in stools is the most efficient method of diagnosis.

Entamoeba histolytica

This protozoan causes amebiasis and is spread by fecal-oral transmission. *E. histolytica* infects approximately 500 million people in countries such as India, Mexico, and Colombia and causes 40 million cases of dysentery and liver abscess annually. The cysts have a chitin wall that is resistant to gastric acid, a characteristic that allows them to pass through the stomach without harm. *E. histolytica* then colonizes the epithelial surface of the colon and releases trophozoites, amoeboid forms that reproduce under anaerobic conditions.

Amebiasis affects the cecum and ascending colon most often, but sigmoid colon, rectum, and appendix can also be involved. Dysentery develops when the amoebas attach to the colonic epithelium, induce apoptosis, invade crypts, and burrow laterally into the lamina propria. The infection results in recruitment of neutrophils, tissue damage, and formation of flask-shaped ulcers with a narrow neck and broad base. Histologic diagnosis can be difficult because amoebas resemble macrophages in size and appearance (Fig. 17.34D). Parasites penetrate splanchnic vessels and embolize to the liver to produce abscesses in about 40% of patients with amoebic dysentery. Amebic liver abscesses can exceed 10 cm in diameter and have a scant inflammatory reaction at their margins and a shaggy fibrinous lining. The abscesses persist after the acute intestinal illness has passed and may, rarely, reach the lung and the heart by direct extension. Amoebas may also spread to the kidneys and brain via the bloodstream.

Individuals with amebiasis may present with abdominal pain, bloody diarrhea, or weight loss. Occasionally, acute necrotizing colitis and megacolon occur; both are associated with significant mortality. The parasites lack mitochondria or Krebs cycle enzymes and are thus obligate fermenters of glucose. Metronidazole, which inhibits pyruvate oxidoreductase, an enzyme required for fermentation, is the most effective treatment for systemic disease.

Giardia lamblia

These organisms, also referred to as *G. duodenalis* or *G. intestinalis*, were initially described by van Leeuwenhoek, the inventor of the microscope, who discovered the pathogen in his own stool. *G. lamblia* is the most common parasitic pathogen in humans and is spread by fecally contaminated water or food. Infection may occur after ingestion of as few as 10 cysts. Because cysts are resistant to chlorine, *Giardia* are endemic in unfiltered public water supplies. They are also common in rural streams, explaining infection in campers who use these as a water source. Infection may occur by the fecal-oral route, and because the cysts are stable, they may be accidentally swallowed while swimming in contaminated water.

Giardia are flagellated protozoa that cause decreased expression of brush-border enzymes, microvillous damage, and apoptosis of small intestinal epithelial cells. Secretory IgA and mucosal IL-6 responses are important for clearance of *Giardia* infections. Immunosuppressed, agammaglobulinemic, or malnourished individuals are often severely affected. *Giardia* can evade immune clearance through continuous modification of their major surface antigen, variant surface protein, and can persist for months or years, causing intermittent symptoms.

Giardia trophozoites can be identified in duodenal biopsies based on their characteristic pear shape and the presence of two equally sized nuclei. Despite large numbers of trophozoites, some of which are tightly bound to the brush border of villous enterocytes, there is no invasion, and small intestinal morphology may be normal (Fig. 17.34E); villous blunting with increased numbers of intraepithelial lymphocytes and mixed lamina propria inflammatory infiltrates can accompany heavy infections.

Giardiasis may be subclinical or accompanied by acute or chronic diarrhea, malabsorption, and weight loss. Infection is usually documented by immunofluorescent detection of cysts in stool samples. Oral antimicrobial therapy is effective, but recurrence is common. Infants and young children, travelers, and immunocompromised patients have an increased risk of severe disease.

Cryptosporidium

Like *Giardia*, cryptosporidia are an important cause of diarrhea worldwide. They cause an acute, self-limited disease in both immunocompromised and immunologically normal hosts. Cryptosporidiosis also causes persistent diarrhea in residents of under-resourced countries. Oocysts are resistant to chlorine and may therefore persist in treated, but unfiltered, water. Contaminated drinking water continues to be the most common means of transmission. The largest documented outbreak, a result of inadequate water purification, occurred in 1993 in Milwaukee, Wisconsin, and affected more than 400,000 people. Like giardiasis, cryptosporidiosis is readily spread to water sport enthusiasts. Food-borne infection occurs less frequently.

Humans are infected by several different *Cryptosporidium* species, including *C. hominis* and *C. parvum*. All go through an entire life cycle, with asexual and sexual reproductive phases, in a single host. Once ingested, as few as 10 encysted oocytes are sufficient to cause symptomatic infection. Sporozoites are released following protease activation by gastric acid. They are motile and have a specialized organelle that attaches to the brush border and causes changes in the enterocyte cytoskeleton. These changes induce the enterocyte to engulf the parasite, which takes up residence in an endocytic vacuole. Sodium malabsorption, chloride secretion, and increased tight junction permeability cause the nonbloody, watery diarrhea that ensues.

Mucosal histology is often only minimally altered, but persistent cryptosporidiosis in children and heavy infection in immunosuppressed patients can result in villous atrophy, crypt hyperplasia, and inflammatory infiltrates. Although the sporozoite is intracellular, it appears, by light microscopy, to sit on top of the epithelial apical membrane (Fig. 17.34F). Organisms are typically most concentrated in the terminal ileum and proximal colon, but can be present throughout the gut, biliary tract, and even the respiratory tract of immunodeficient hosts. Diagnosis is based on finding oocysts in the stool.

KEY CONCEPTS

Infectious Enterocolitis

- *V. cholerae* secrete a preformed toxin that causes massive chloride secretion. Water follows the resulting osmotic gradient, leading to high volume secretory diarrhea.
- *C. jejuni* is the most common bacterial enteric pathogen in high-income countries and also causes traveler's diarrhea. Most isolates are noninvasive.
- *Salmonella* and *Shigella* spp. are invasive and associated with exudative bloody diarrhea (dysentery).
- *Salmonella enteritidis* infection is a common cause of food poisoning.
- *S. typhi* cause systemic disease (typhoid fever).
- Pseudomembranous colitis is often triggered by antibiotic therapy. The responsible organism, *C. difficile*, releases toxins that disrupt epithelial function and induce an inflammatory response that includes characteristic volcano-like neutrophil eruptions that spread to form mucopurulent pseudomembranes.

- Norovirus is a common cause of self-limited diarrhea in adults and children. It spreads from person to person in sporadic cases and via contaminated water in epidemics.
- Rotavirus causes severe childhood diarrhea and significant mortality worldwide. Widespread vaccination has reduced the incidence of rotavirus infection.
- Epithelial expressions of proteins that serve as SARS-CoV-2 receptors may explain gastrointestinal symptoms in ~50% of patients with COVID-19.
- Parasitic and protozoal infections affect more than one-half of the world's population on a chronic or recurrent basis. Each parasite has a distinctive life cycle and tissue reaction. Those that invade are typically associated with tissue and systemic eosinophilia.

IRRITABLE BOWEL SYNDROME

Irritable bowel syndrome (IBS) is characterized by chronic, relapsing abdominal pain, bloating, and changes in bowel habits. Despite real symptoms, endoscopic and microscopic evaluations are normal in IBS patients. Thus, the diagnosis depends on clinical symptoms and functional testing. It should be recognized that IBS is a syndrome consisting of a spectrum of multiple illnesses, and that multiple illnesses may fall under this broad descriptor. IBS is currently divided into diarrhea-predominant, constipation-predominant, and mixed subtypes.

Pathogenesis

The pathogenesis of IBS remains poorly defined, although there is clearly interplay between psychologic stressors, diet, perturbation of the gut microbiome, increased enteric sensory responses to GI stimuli, and abnormal GI motility. For example, patients with constipation-predominant or diarrhea-predominant IBS tend to have decreased or increased colonic contractions and transit rates, respectively. Excess bile acid synthesis or bile acid malabsorption has been identified as one cause of diarrhea-predominant IBS, likely due to the effects of bile acids on intestinal motility and epithelial ion transport.

Other data link disturbances in enteric nervous system function to IBS, suggesting a role for defective brain-gut axis signaling. Consistent with this, deep sequencing and genome-wide association studies have linked several candidate genes to IBS including serotonin reuptake transporters, cannabinoid receptors, and TNF-related inflammatory mediators. Further, 5-HT₃ receptor antagonists are

effective in many cases of diarrhea-predominant IBS. Opioids and psychoactive drugs with anticholinergic effects are also used to treat diarrhea-predominant IBS.

A separate group of IBS patients relate onset to a bout of infectious gastroenteritis, suggesting that immune activation or, alternatively, a shift in the gut microbiome may trigger disease.

Clinical Features

The peak prevalence of IBS is between 20 and 40 years of age, and there is a significant female predominance. Variability in diagnostic criteria makes it difficult to establish the incidence, but most authors report a prevalence in high-income countries of between 5% and 10%. IBS is presently diagnosed using clinical criteria that require the occurrence of abdominal pain or discomfort at least 3 days per month over 3 months with improvement following defecation and a change in stool frequency or form. Other causes, such as enteric infection or inflammatory bowel disease, must be excluded.

INFLAMMATORY BOWEL DISEASE

Inflammatory bowel disease (IBD) is a chronic condition comprised of two disorders, ulcerative colitis and Crohn disease. Descriptions of ulcerative colitis and Crohn disease date back to antiquity and at least the sixteenth century, respectively, but it took modern microbiologic techniques to exclude more common infectious etiologies for these diseases. As will be discussed later, however, luminal microbiota likely play an important role in IBD pathogenesis.

The distinction between ulcerative colitis and Crohn disease is based on the distribution of affected sites ([Fig. 17.35](#)) and the morphologic expression of disease ([Table 17.8](#)) at those sites. In simplest terms, **ulcerative colitis involves only the colon and rectum and is limited to the mucosa and submucosa. In contrast, Crohn disease, which was previously termed ileal enteritis due to frequent involvement of the ileum, may involve any area of the GI tract and is typically transmural.**

Epidemiology

Ulcerative colitis and Crohn disease most frequently present in the teenage years and early 20s but can develop at any age. IBD is most common among people of European descent, and in the United States

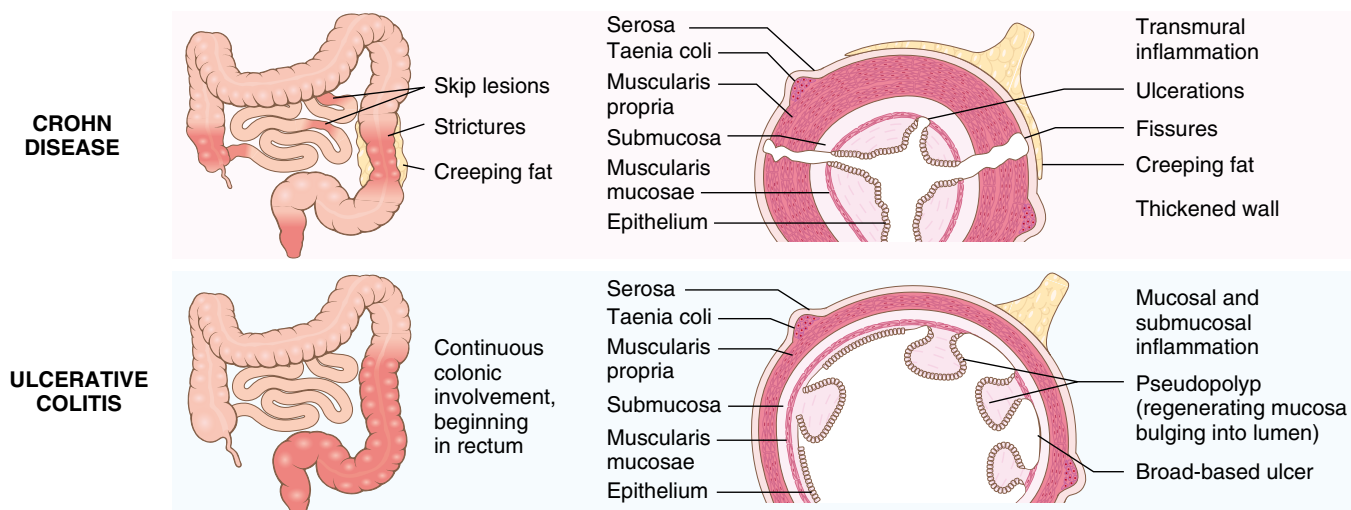


Fig. 17.35 Distribution of lesions in inflammatory bowel disease. The distinction between Crohn disease and ulcerative colitis is primarily based on gross and microscopic morphologies.

TABLE 17.8 Features that Differ Between Crohn Disease and Ulcerative Colitis

Feature	Crohn's Disease	Ulcerative Colitis
Macroscopic		
Bowel region	Ileum ± colon	Colon only
Distribution	Skip lesions	Diffuse
Stricture	Yes	Rare
Wall appearance	Thick	Normal
Microscopic		
Inflammation	Transmural	Limited to mucosa
Pseudopolyps	Moderate	Marked
Ulcers	Deep, knife-like	Superficial, broad-based
Lymphoid reaction	Marked	Moderate
Fibrosis	Marked	Mild to none
Serositis	Marked	Mild to none
Granulomas	Yes (~35%)	No
Fistulae/sinuses	Yes	No
Clinical		
Perianal fistula	Yes (in colonic disease)	No
Fat/vitamin malabsorption	Yes	No
Malignant potential	With colonic involvement	Yes
Recurrence after surgery	Common	No
Toxic megacolon	No	Yes

All features may not be present in a single case.

occurs three to five times more often among eastern European (Ashkenazi) Jewish people than the general population. This is at least partly due to genetic factors, as discussed later. IBD is most common in North America, northern Europe, and Australia, but its incidence is on the rise in Africa, South America, and Asia, where prevalence was historically low. The hygiene hypothesis suggests that this increasing incidence is related to improved food storage conditions, decreased food contamination, and changes in gut microbiome composition that result in inadequate development of regulatory processes that limit mucosal immune responses. This in turn allows some mucosa-associated microbes to trigger persistent and chronic inflammation in susceptible hosts. Other potential explanations for increased IBD prevalence include the idea that preservatives and other materials added to processed foods induce low-grade mucosal damage that predisposes to IBD.

Pathogenesis

IBD results from a combination of abnormalities in immune regulation, host-microbe interactions, and epithelial functions in genetically susceptible individuals (Fig. 17.36).

Genetics. Over 200 IBD-associated risk alleles have been identified, but these account for less than 20% of disease risk in Crohn disease and make even smaller contributions to ulcerative colitis. For example, in European populations, *NOD2*, which encodes an intracellular sensor of muramyl dipeptide, a bacterial cell wall component, is most strongly associated with Crohn disease. Yet, the three major *NOD2* risk alleles are present in only ~30% of Crohn disease patients versus 5% of

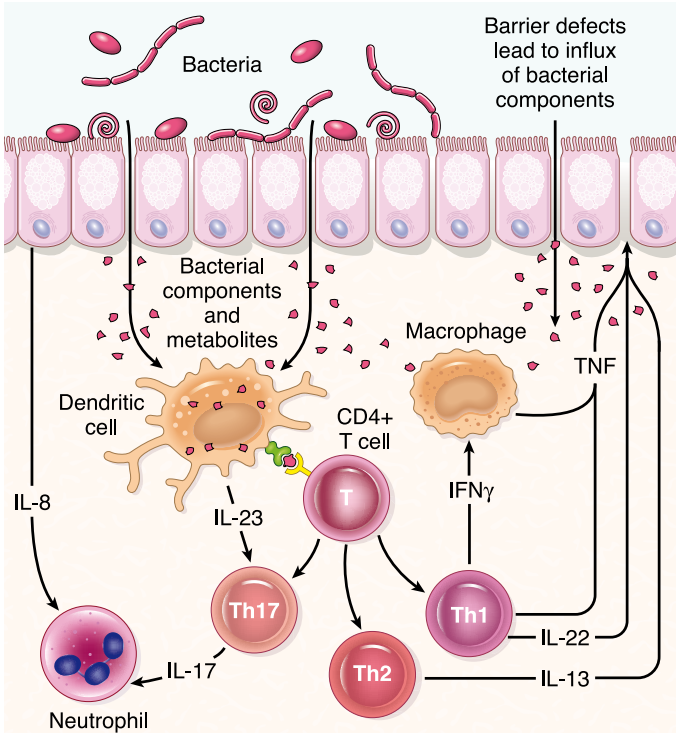


Fig. 17.36 Inflammatory bowel disease is idiopathic and mechanisms are incompletely understood. This model of pathogenesis incorporates features thought to be critical, including inflammatory responses mediated by dendritic cell, macrophage, CD4+ T cell, and neutrophils, cytokine signaling, the gut microbiome, and the intestinal epithelial barrier, which can be regulated by many cytokines. *IL*, Interleukin; *TNF*, tumor necrosis factor.

healthy individuals. Moreover, individuals with one risk allele have only a ~five-fold increased Crohn disease risk. Even those in whom both copies of *NOD2* are risk alleles, Crohn disease risk is increased only ~20-fold. In addition, both frequencies and risk associations of *NOD2* and other IBD-associated loci vary widely across populations; for example, *NOD2* risk alleles are rare in Asian populations. Thus, IBD is best understood as a polygenic disorder in which the accumulated effects of multiple polymorphisms impact disease susceptibility. Consistent with this, studies assessing all IBD genetic associations in aggregate suggested some genotype-phenotype associations, including age at diagnosis, location and stricture formation versus penetrating behavior in Crohn disease, and extent of colonic involvement in ulcerative colitis.

Most IBD-associated polymorphisms can increase risk of both Crohn disease and ulcerative colitis, and ~30% have indistinguishable effect sizes in the two diseases. Two notable exceptions to this generalization are *NOD2* and *PTPN22* variants that increase Crohn disease risk but are protective in ulcerative colitis. Interestingly, these genes are also among the ~10% of IBD-associated polymorphisms that involve coding sequences. Another ~30% of IBD loci are in regulatory regions of genes, but neither coding nor gene expression changes are present in over 50% of IBD-associated loci.

Despite some differences, the extensive overlap of Crohn disease and ulcerative colitis genetic risk factors suggests that nearly all mechanisms involved in one disease have a role in the other. When assessed as clusters, IBD-associated genes are most closely linked to signaling in immune cells, responses to bacterial products, and host defense. Many

IBD risk alleles overlap with those associated with other immune-mediated diseases, including type I diabetes, ankylosing spondylitis and psoriasis. In most cases, specific polymorphisms have consistent effects across diseases, but some that are associated with increased IBD risk are protective in other diseases or vice versa.

A notable exception to the polygenic nature of IBD is very early onset IBD (so-called *infantile colitis*), which can be monogenic, with mutations of *IL10* and the genes encoding the IL-10 receptor constituents *IL10RA* and *IL10RB* being the most common. Of the over 100 genes associated with monogenic IBD, the majority encode proteins expressed in T cells and phagocytes. A smaller group of genes encode epithelial cell proteins that contribute to epithelial barrier function, intercellular adhesion, brush border digestion, ion and solute transport, mucin secretion, and Paneth cell secretion of antimicrobial peptides. Other mutations that cause monogenic IBD affect widely expressed genes involved in NOD signaling, NF- κ B signaling, epithelial death and survival, and endoplasmic reticulum (ER) stress (i.e., the unfolded protein response). As might be expected, both functional and genetic data implicate all these cell type specific and ubiquitous processes in polygenic IBD.

Mucosal Immunity. Crohn disease and ulcerative colitis are both characterized by mixed inflammatory reactions marked by activation of Th17 and Th1 cells as well as more limited Th2 cell activation, particularly in ulcerative colitis. Consistent with this, all current IBD therapies, from corticosteroids to biologics, target the immune system; the efficacies of specific agents are generally similar in Crohn disease and ulcerative colitis. Anti-TNF therapies are effective in IBD, presumably by inhibiting inflammation. Up to 50% of patients ultimately become unresponsive to their first anti-TNF biologic. Some of these patients respond to other anti-TNF biologics, but they may also be treated with other biologics, including anti-IL-12/23p40 antibodies, which inhibit the activation of Th1 and Th17 cells, and anti-IL-23, which inhibits Th17 cells. In addition to biologic therapies, cytokine signaling has been targeted in IBD using JAK inhibitors. Finally, disruption of lymphocyte trafficking using anti- α 4- or anti- α 4 β 7-integrin antibodies or sphingosine-1-phosphate receptor antagonists has proven effective in IBD. In summary, characterization of immune mechanisms of IBD have led to, and been validated by, targeted therapies, but substantial gaps in our understanding of IBD immunopathogenesis remain.

Autophagy and Cellular Stress Responses. Genetic associations as well as molecular analyses indicate that defects in autophagy and cellular stress responses contribute to IBD pathogenesis. The most studied of these genes include *ATG16L1* and *IRGM*, both of which are involved in autophagosome formation. Autophagy is a normal homeostatic mechanism that eliminates damaged organelles and is a means of clearing sources of reactive oxygen species and intracellular pathogens. The *ATG16L1* mutation that is linked to Crohn disease promotes degradation of the autophagy-related protein ATG16L1, thereby limiting autophagy. The precise pathways by which loss of ATG16L1 function leads to disease are still being defined, but it is noteworthy that Paneth cell granules, which contain antibacterial peptides that are released into the crypt lumen, are structurally and functionally abnormal in patients with *ATG16L1* defects.

Epithelial Barrier Dysfunction. Partial loss of the intestinal epithelial barrier, which limits interactions between the luminal milieu and the mucosal immune system, has also been linked to IBD. Results are clearest in Crohn disease, where the ~10% of healthy relatives of those with Crohn disease with modest barrier loss are at greater risk of developing IBD than are healthy relatives without

barrier loss. Similarly, barrier loss during remission is associated with relapse of active Crohn disease. Finally, preclinical data indicate that barrier restoration can attenuate experimental IBD severity. Thus, compromise of a range of epithelial functions contributes to IBD pathogenesis. No epithelial-targeted therapies are currently available.

Host-Microbial Interactions. The gut microbiome, composed of bacteria, fungi, and viruses, and its role in IBD have been the subject of intense investigation over the last decade. The microbiome is populated by a small number of species at birth and evolves to become a far more complex ecosystem with many more species appearing after weaning and through childhood. Patients with IBD often have shifts in microbial populations and reduced species diversity, called *dysbiosis*. Although dysbiosis is often established at disease presentation, it is difficult to differentiate between microbial changes that trigger disease and those that are caused by disease. Unregulated immune responses to bacterial species in the microbiome have been postulated to underlie IBD, but neither the microbial triggers nor the specific type of immune dysregulation have been defined. Recent studies have focused on understanding the metabolites generated by microbes and how these may impact immune and epithelial function, and to use this knowledge to develop new therapies.

Crohn Disease

The eponym Crohn disease is based on a 1932 publication, but the entity was described centuries earlier. For example, Louis XIII of France (1601–1643) experienced relapsing bloody diarrhea, fever, rectal abscess, small intestinal and colonic ulcers, and fistulae beginning at age 20 years, all of which are classic features of Crohn disease.

MORPHOLOGY

Crohn disease may occur in any area of the GI tract but most commonly involves the terminal ileum, ileocecal valve, and cecum. Disease is limited to the small intestine in about 40% of cases; the small intestine and colon are both involved in 30% of patients; the remainder have only colonic involvement. Pancolitis can be difficult to distinguish from ulcerative colitis. The presence of multiple, separate, sharply delineated areas of disease, resulting in **skip lesions**, is characteristic, and when present, differentiates Crohn disease from ulcerative colitis. The presence of strictures, which occur commonly in Crohn disease but only rarely in long-standing ulcerative colitis, may also be helpful in diagnoses (Fig. 17.37A).

The earliest lesion of Crohn disease, the **aphthous ulcer**, may progress, and multiple lesions often coalesce into elongated, serpentine ulcers oriented along the long axis of the bowel (Fig. 17.37B). Edema and loss of the normal mucosal folds are also common. Ulceration with sparing of interspersed mucosa, a result of the patchy distribution of Crohn disease, can create an irregular, **cobblestone** appearance of the mucosa (see Fig. 17.37B). **Fissures** frequently develop and may extend deeply to become peritoneal abscesses and perforation sites (Fig. 17.37C). Alternatively, fissures may become fistula tracts between loops of bowel or intestine and urinary bladder, vagina, or abdominal or perianal skin. The intestinal wall is thickened and rubbery as a consequence of transmural edema, inflammation, submucosal fibrosis, and hypertrophy of the muscularis propria, all of which contribute to stricture formation (see Fig. 17.37A). In cases with extensive transmural disease, mesenteric adipose tissue frequently extends over the serosal surface (**creeping fat**) (Fig. 17.37D).

The microscopic features of active Crohn disease include abundant neutrophils that infiltrate and damage crypt epithelium. Clusters of neutrophils within

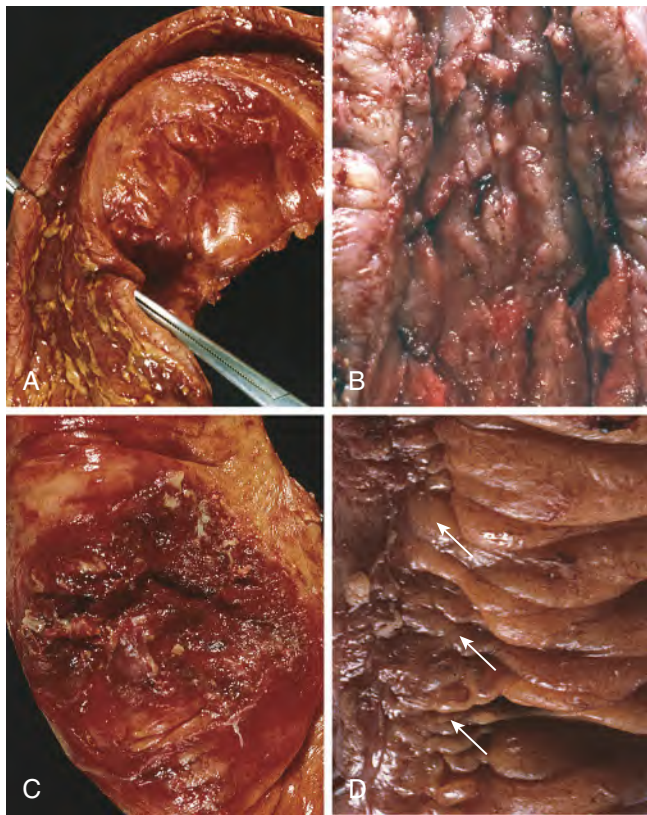


Fig. 17.37 Gross pathology of Crohn disease. (A) Small intestinal stricture. (B) Linear mucosal ulcers, which impart a cobblestone appearance to the mucosa, and thickened intestinal wall. (C) Perforation and associated serositis. (D) Creeping fat (arrows).

a crypt are referred to as **crypt abscesses** and are often associated with crypt destruction. Ulceration is common in Crohn disease, and there may be an abrupt transition between ulcerated and adjacent normal mucosa. Even in areas where gross examination suggests diffuse disease, microscopic pathology can appear patchy. Repeated cycles of crypt destruction and regeneration lead to **distortion and disorganization of mucosal glands** in which the normally straight and parallel crypts take on bizarre branching shapes and unusual orientations relative to one another and the muscularis mucosae (Fig. 17.38A). Epithelial metaplasia is another consequence of chronic relapsing injury. One form, **pseudopyloric metaplasia**, which refers to the presence of gastric antral-appearing glands, can occur in the small intestine or colon. **Paneth cell metaplasia** may also occur in the left colon, where Paneth cells are normally absent. These architectural and metaplastic changes may persist even when active inflammation has ceased. Mucosal atrophy, with loss of crypts, may also occur after years of disease. **Noncaseating granulomas** (Fig. 17.38C), a hallmark of Crohn disease, are found in approximately 35% of cases and may occur in areas of active disease (see Fig. 17.38C) or uninvolved regions within any layer of the intestinal wall (Fig. 17.38D). Granulomas may also be present in draining mesenteric lymph nodes and at extraintestinal sites.

Clinical Features

The clinical manifestations of Crohn disease are extremely variable. In most patients, disease begins with intermittent attacks of relatively mild diarrhea, abdominal pain, and fever. Approximately 20% of patients present acutely with right lower quadrant pain, fever, and bloody diarrhea that may mimic acute appendicitis or bowel perforation. Periods of active disease are typically interrupted by asymptomatic periods that last for weeks to many months. Disease reactivation can be associated with a variety of external

triggers, including physical or emotional stress, specific dietary items, and cigarette smoking. The latter is a strong risk factor for development of Crohn disease, and in some cases, disease onset is associated with initiation of smoking. Unfortunately, smoking cessation does not result in disease remission. Intestinal barrier loss, which is present in ~10% of healthy first-degree relatives of Crohn disease patients, is another risk factor for disease development. Similarly, barrier dysfunction during remission is a marker of impending relapse in those with established Crohn disease.

Iron deficiency anemia due to blood loss may develop in individuals with colonic disease, while extensive small bowel disease may result in protein loss sufficient to cause hypoalbuminemia and malabsorption of nutrients, vitamin B₁₂, and bile salts. Fibrosing strictures, particularly of the terminal ileum, are common and require surgical resection. Disease often recurs at the site of anastomosis and necessitates additional resections within 10 years in up to 40% of patients. Bowel preservation is, therefore, an important consideration when planning resections. As discussed above, therapies focus on immunosuppression or immunomodulation and include antiinflammatory agents (e.g., salicylates); immunosuppressive drugs (e.g., corticosteroids); and biologic therapies discussed earlier. Some current efforts aim to develop personalized treatment algorithms based on clinical phenotype with genetic, immune and microbiome data.

Crohn disease patients often have anti-*Saccharomyces cerevisiae* antibodies, which are typically absent in ulcerative colitis. Other manifestations of Crohn disease include cutaneous nodules formed by granulomas, uveitis, migratory polyarthritis, sacroiliitis, ankylosing spondylitis, erythema nodosum, and clubbing of the fingertips, any of which may precede clinically evident intestinal disease. Crohn disease patients are also at increased risk of developing pericholangitis and primary sclerosing cholangitis, although the association is much stronger in ulcerative colitis (see later and Chapter 18). As discussed later, risk of colonic adenocarcinoma is increased in patients with long-standing colonic disease.

Ulcerative Colitis

Ulcerative colitis is closely related to Crohn disease, but intestinal involvement is continuous and limited to the colon and rectum.

Although the implication for the pathogenesis is unclear, it is notable that up to 75% of ulcerative colitis patients, but only ~10% of those with Crohn disease, have perinuclear antineutrophil cytoplasmic antibodies (p-ANCA). Extraintestinal manifestations of ulcerative colitis overlap with those of Crohn disease and include migratory polyarthritis, sacroiliitis, ankylosing spondylitis, uveitis, and skin lesions, but neither intestinal nor extraintestinal granulomas are present. As noted above, risk of primary sclerosing cholangitis is markedly increased in ulcerative colitis and affects ~5% of patients (Chapter 18). Conversely, the majority of those with primary sclerosing cholangitis also have ulcerative colitis. The long-term outlook for ulcerative colitis patients depends on the severity of active disease and disease duration. In general, therapies, including anti-TNF antibodies, that are effective in Crohn disease are also effective in ulcerative colitis.

MORPHOLOGY

Grossly, ulcerative colitis always involves the rectum and extends proximally in a continuous fashion to involve part or all of the colon. Disease of the entire colon is termed **pancolitis** (Fig. 17.39A). Limited distal disease may be referred to descriptively as left-sided ulcerative colitis, **ulcerative proctitis**, or **ulcerative proctosigmoiditis**. The small intestine is not affected, although mild mucosal inflammation of the distal ileum, termed **backwash ileitis**, may be present in severe cases of pancolitis. Although there are no skip lesions, focal appendiceal or cecal inflammation, termed a **cecal red spot**, may be present in left-sided ulcerative colitis.

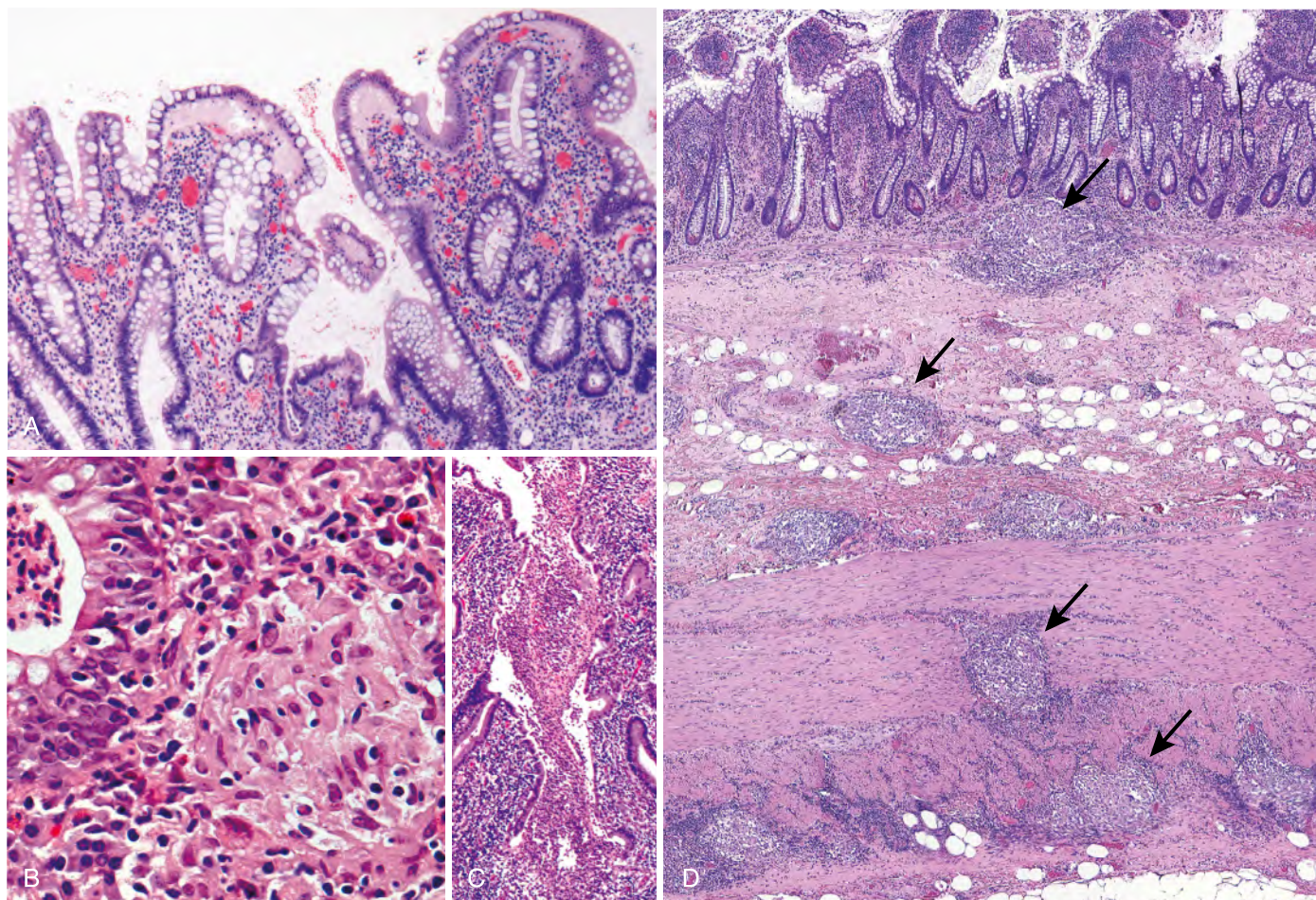


Fig. 17.38 Microscopic pathology of Crohn disease. (A) Haphazard crypt organization results from repeated injury and regeneration. (B) Noncaseating granuloma. (C) Active Crohn's disease, with ulceration and purulent exudate. (D) Transmural Crohn's disease with submucosal and serosal granulomas (arrows).

Grossly, the appearance of involved colonic mucosa ranges from slightly red and granular to extensive, broad-based ulcers. The transition between diseased and uninvolved colon can be abrupt (Fig. 17.39B). Ulcers are aligned along the long axis of the colon but do not typically replicate the serpentine ulcers of Crohn disease. Isolated islands of regenerating mucosa often bulge into the lumen to create **pseudopolyps** (Fig. 17.39C), and the tips of these polyps may fuse to create **mucosal bridges** (Fig. 17.39D). Chronic disease may lead to **mucosal atrophy** with a smooth mucosal surface that lacks normal folds. Unlike Crohn disease, ulcerative colitis is not transmural. Therefore, **the colon wall is not thickened, the serosal surface is normal, and strictures do not occur**. Perforations are also uncommon in ulcerative colitis and only are associated with severe cases that progress to **toxic megacolon**.

Histologic features of mucosal disease in ulcerative colitis are like colonic Crohn disease and include inflammatory infiltrates, crypt abscesses (Fig. 17.40A), crypt distortion, pseudopyloric metaplasia (Fig. 17.40B), and Paneth cell metaplasia. However, unlike Crohn disease, granulomas are not present in ulcerative colitis and inflammation is generally limited to the mucosa and superficial submucosa (Fig. 17.40C). In severe cases, extensive mucosal destruction may be accompanied by ulcers that extend into the submucosa, but involvement of the muscularis propria is rare. Submucosal fibrosis, mucosal atrophy, and distorted mucosal architecture remain as

residua of healed disease, but histology may revert to near normal after prolonged remission.

Clinical Features

Ulcerative colitis is a relapsing disorder characterized by attacks of bloody diarrhea with stringy mucoid material, lower abdominal pain, and cramps that are temporarily relieved by defecation. These symptoms may persist for days, weeks, or months before they subside. The initial attack may, in some cases, be severe enough to constitute a medical or surgical emergency. More than half of patients have clinically mild disease, although almost all experience at least one relapse during a 10-year period. Historically, up to 30% of those affected required colectomy within the first 3 years after presentation because of uncontrollable symptoms, but the incidence of colectomy has fallen sharply over the past 20 years due to biologics and other improvements in medical management. Although colectomy cures intestinal disease in ulcerative colitis, extraintestinal manifestations may persist.

The factors that trigger development of ulcerative colitis in previously healthy individuals are not known. However, infectious enteritis precedes disease onset in some cases of ulcerative colitis and Crohn disease. It has been hypothesized that enteritis triggers mucosal immune activation and microbial changes that lead to disease in susceptible individuals. In other

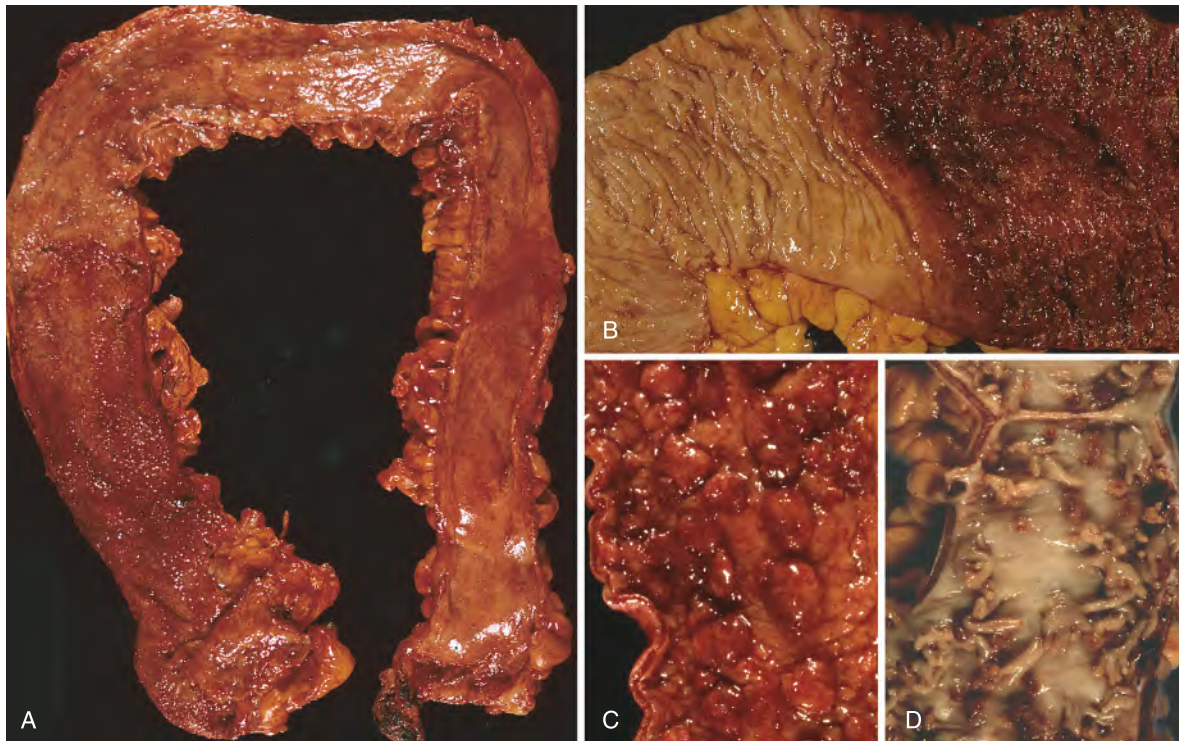


Fig. 17.39 Gross pathology of ulcerative colitis. (A) Total colectomy with pancolitis showing active disease, with red, granular mucosa in the cecum (*left*) and smooth, atrophic mucosa distally (*right*). (B) Sharp demarcation between active ulcerative colitis (*right*) and normal mucosa (*left*). (C) Inflammatory polyps. (D) Mucosal bridges can join inflammatory polyps.

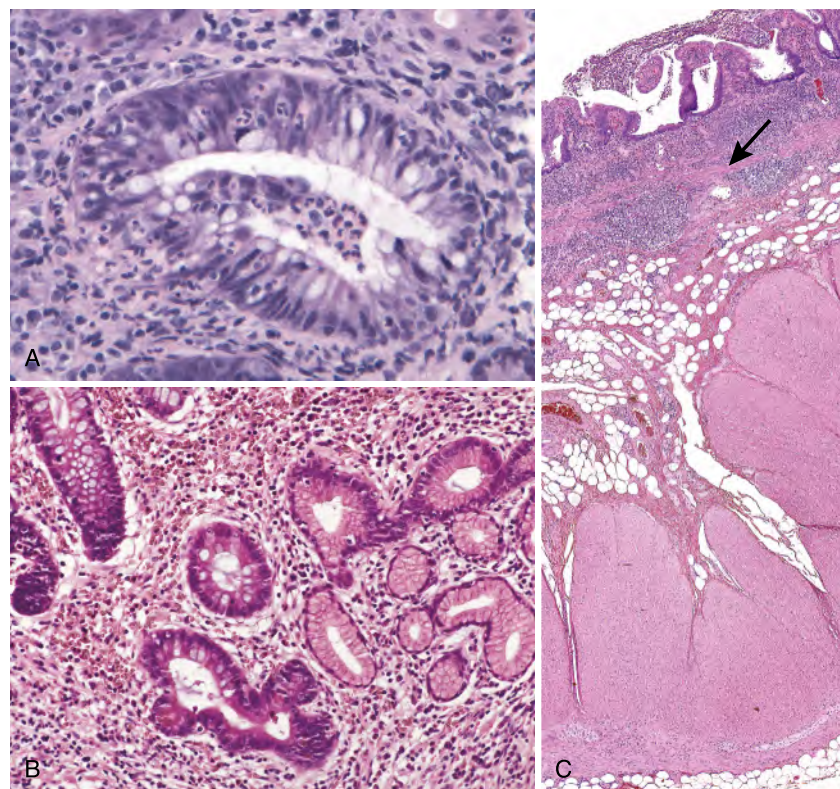


Fig. 17.40 Histopathology of ulcerative colitis. (A) Crypt abscess. (B) Pseudopyloric metaplasia (*right*). (C) Disease is limited to the mucosa (*above the arrow*). Compare with [Fig. 17.38D](#).

patients the first episode of disease is preceded by psychological stress, which may also be linked to relapse during remission. In contrast to Crohn disease, initial onset of symptoms can be associated with smoking cessation. In these patients, smoking may provide partial symptomatic relief.

Indeterminate Colitis

In up to 10% of IBD patients, it is not possible to establish a diagnosis of Crohn disease or ulcerative colitis. These cases, termed indeterminate colitis, do not involve the small bowel and have colonic disease in a continuous pattern typical of ulcerative colitis but also have features suggestive of Crohn disease, such as patchy histologic disease, a family history of Crohn disease, or perianal lesions. Fortunately, overlap in medical management allows indeterminate colitis to be treated effectively.

Colitis-Associated Neoplasia

One of the most feared long-term complications of ulcerative colitis and colonic Crohn disease is the development of neoplasia. The risk of cancer is related to several factors:

- *Duration of the disease.* Risk increases sharply 8 to 10 years after disease onset.
- *Extent of the disease.* Patients with pancolitis are at greater risk than those with only left-sided disease. Crohn disease patients without colonic involvement are not at increased risk.
- *Nature of the inflammatory response.* Greater frequency and severity of active inflammation (characterized by the presence of neutrophils) confers increased risk.

To facilitate early detection of neoplasia, patients are typically enrolled in surveillance programs approximately 8 years after diagnosis of IBD. The major exception to this is patients with IBD and primary sclerosing cholangitis, who have a greater risk of developing cancer and are generally enrolled for surveillance at the time of diagnosis.

The goal of surveillance biopsies is to identify dysplastic epithelium, which is a precursor of colitis-associated carcinoma. Dysplasia can develop in flat areas of mucosa that are not grossly recognized as abnormal. High-resolution endoscopes and specialized endoscopic techniques, including chromoendoscopy, have helped target biopsies toward areas suspicious for dysplasia, but multiple surveillance biopsies and histological examination are still required. IBD-associated dysplasia is classified histologically as low grade or high grade (Fig. 17.41A,B) and may be multifocal. High-grade dysplasia may be associated with invasive carcinoma at the same site (Fig. 17.41C) or elsewhere in the colon and therefore often prompts colectomy. Low-grade dysplasia may be followed closely with frequent endoscopic surveillance, but colectomy is also common, particularly when multiple foci of flat (nonpolypoid) dysplasia are present, in those with extensive or long-standing disease, and in older patients. Cases of invasive adenocarcinoma beneath superficial areas with only low-grade dysplasia have been reported. IBD patients are susceptible to sporadic adenomas, which can be difficult or impossible to differentiate from polypoid foci of IBD-associated dysplasia. Fortunately, when solitary, polypoid dysplastic lesions in the absence of flat dysplasia can often be managed by endoscopic resection and careful follow-up.

Other Causes of Colitis

Diversion Colitis

Surgical treatment of ulcerative colitis, Hirschsprung disease, colon cancer, and other intestinal disorders sometimes requires creation of

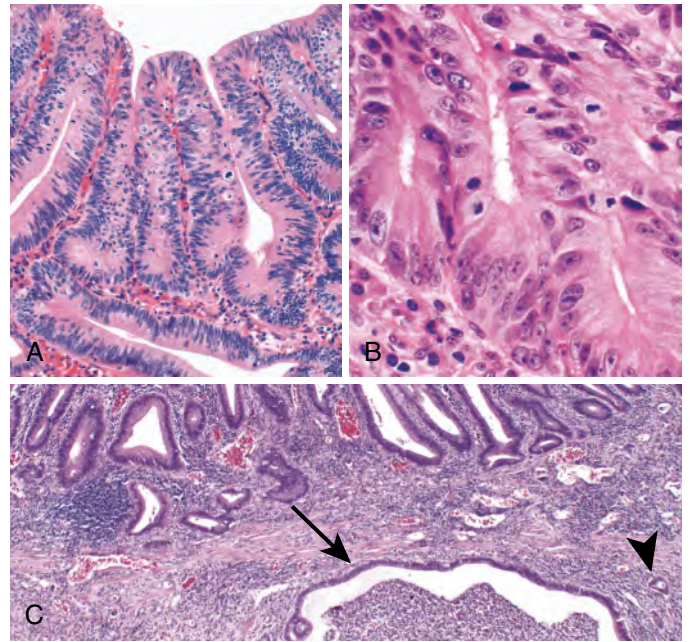


Fig. 17.41 Colitis-associated dysplasia. (A) Dysplasia with extensive nuclear stratification and marked nuclear hyperchromasia. (B) Cribriform glandular arrangement in high-grade dysplasia. (C) Colectomy specimen with high-grade dysplasia on the surface and underlying invasive adenocarcinoma. A large cystic space containing abundant neutrophils that is lined by invasive adenocarcinoma (arrow) is present beneath the muscularis mucosae. Also seen are small invasive glands (arrowhead).

a temporary or permanent ostomy, resulting in a blind distal segment of colon from which the normal fecal flow is diverted. Colitis can develop within the diverted segment. Besides mucosal erythema and friability, the most striking feature of diversion colitis is the development of numerous mucosal lymphoid follicles (Fig. 17.42A). Large numbers of lymphocytes, monocytes, macrophages, and plasma cells may also be present in the lamina propria. In severe cases the histopathology may resemble IBD and include crypt abscesses, mucosal architectural distortion, or, rarely, granulomas. The mechanisms responsible for diversion colitis are not well understood, but changes in the luminal microbiota and diversion of the fecal stream, which provides nutrients to colonic epithelial cells, have been proposed. Consistent with this, enemas containing short-chain fatty acids, a product of colonic bacterial metabolism that signals to and serves as an energy source for colonic epithelial cells, can promote mucosal recovery. Such enemas are, however, used infrequently because they produce a foul odor. Diversion colitis resolves after anastomosis and restoration of fecal flow.

Microscopic Colitis

Microscopic colitis encompasses two entities, *collagenous colitis* and *lymphocytic colitis*. These idiopathic diseases both present with chronic, nonbloody, watery diarrhea without weight loss. Radiologic and endoscopic studies are typically normal. *Collagenous colitis*, which occurs primarily in middle-aged and older females, is characterized by the presence of a dense subepithelial collagen layer (more than 10 μm), increased numbers of intraepithelial T lymphocytes, and a mixed inflammatory infiltrate within the lamina propria (Fig. 17.42B). *Lymphocytic colitis* is histologically similar, but the subepithelial collagen layer is of normal thickness and the increase in intraepithelial T lymphocytes is greater, frequently exceeding 20 lymphocytes for

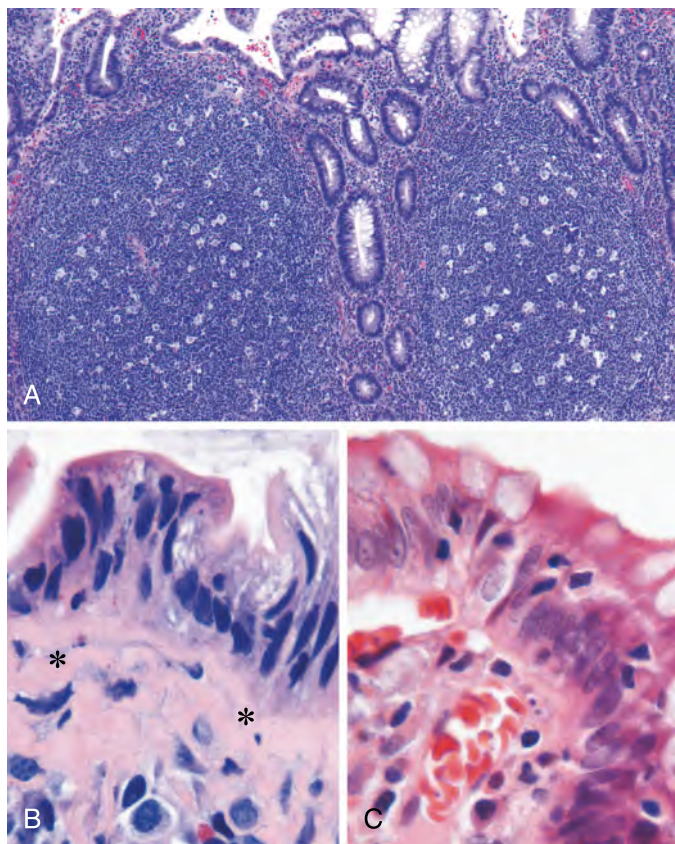


Fig. 17.42 Uncommon causes of colitis. (A) Diversion colitis. Note the large lymphoid follicles with germinal centers. (B) Collagenous colitis with a dense subepithelial collagen band (*asterisks*). (C) Lymphocytic colitis. Intraepithelial lymphocytes can be recognized by their densely stained, small nuclei.

every 100 colonocytes (Fig. 17.42C). Lymphocytic colitis shows a strong association with celiac disease and autoimmune diseases including Graves disease, rheumatoid arthritis, and autoimmune or lymphocytic gastritis. Similar histologic changes can be induced by NSAIDs and, possibly, proton pump inhibitors.

Graft-Versus-Host Disease

Graft-versus-host disease (GVHD) occurs following allogeneic hematopoietic stem cell transplantation. The small bowel and colon are involved in most cases. Although graft-versus-host disease is secondary to donor T cells targeting antigens on the recipient's GI epithelial cells, the lamina propria lymphocytic infiltrate is typically sparse. Epithelial apoptosis, particularly of crypt cells, is the most common histologic finding. The crypts may be completely destroyed in severe cases. Intestinal graft-versus-host disease often presents with watery diarrhea that may become bloody in severe cases.

Apoptotic Enterocolitis

Apoptotic enterocolitis is characterized by an increase in the number of apoptotic cells in the GI epithelium stemming from a myriad of conditions. Enterocolitis and diarrhea are becoming increasingly common in oncology patients treated with immune checkpoint inhibitors. In some cases, disease severity requires discontinuation of treatment. The most common morphology associated within immune checkpoint inhibitor-induced enterocolitis, epithelial apoptosis, is

similar to graft-versus-host disease. Consistent with this, although incompletely defined, the mechanisms of checkpoint inhibitor-induced enterocolitis likely overlap, at least partially, with those of graft-versus-host disease. Other forms of apoptotic enteritis and colitis are associated with treatment with the immunosuppressive agent mycophenolate mofetil.

SIGMOID DIVERTICULAR DISEASE

Diverticular disease is due to pseudodiverticular outpouchings of the colonic mucosa and submucosa. Unlike true congenital diverticula, such as Meckel diverticulum, colonic diverticula are acquired and do not involve all layers of the bowel wall. They are rare in persons younger than age 30 years, but the prevalence approaches 50% in Western adult populations older than age 60 years. Diverticulosis is less common in Japan as well as under-resourced countries, probably because of dietary differences.

Pathogenesis

Colonic diverticula are the product of the unique structure of the muscularis propria and elevated intraluminal pressure. There are focal discontinuities in the inner circular muscle layer at sites where nerves and arterial vasa recta penetrate it. In the small intestine, these gaps are reinforced by the external longitudinal layer of the muscularis propria. In contrast, the outer muscle layer of the colon is gathered into three bands, the taenia coli, resulting in unreinforced sites where nerves and arteries cross the inner layer of the muscularis propria. Increased intraluminal pressure can cause the mucosa to herniate through the muscularis propria at these unreinforced sites. This is most prominent in the sigmoid colon, where exaggerated peristaltic contractions occur, particularly in association with low-fiber diets that lead to reduced stool bulk. As a result, diverticula are typically most common in the sigmoid colon in Western populations, whereas they tend to occur in the right colon in Asian and African populations. Diverticular inflammation, which may be caused by obstruction of the orifice, defines diverticulitis.

MORPHOLOGY

Colonic diverticula are 0.5 to 1 cm diameter flask-like outpouchings that are uniformly spaced, reflecting the regular distribution of penetrating nerves and arteries (Fig. 17.43A). The diverticular wall is formed by the mucosa, including the muscularis mucosae, and separated from the serosal surface by a thin layer of subserosal adipose tissue, sometimes in association with compressed submucosa or attenuated muscularis propria. Inflammation within this narrow region can therefore extend to cause serositis, and mucosal injury can result in perforation, development of pericolic abscesses, sinus tracts, and peritonitis. Hypertrophy of the inner circular layer of the muscularis propria is commonly associated with diverticulosis.

Clinical Features

Most individuals with diverticular disease remain asymptomatic throughout their lives. However, about 20% develop manifestations such as intermittent cramping, continuous lower abdominal discomfort, constipation, distention, or a sensation of never being able to completely empty the rectum. Patients sometimes experience alternating constipation and diarrhea that mimics IBS. Whether a high-fiber diet prevents or protects against diverticulitis is unclear, but diets supplemented with fiber may provide symptomatic improvement. Even when diverticulitis occurs, it most often resolves spontaneously. Surgical intervention is generally reserved for those with severe or recurrent diverticulitis.

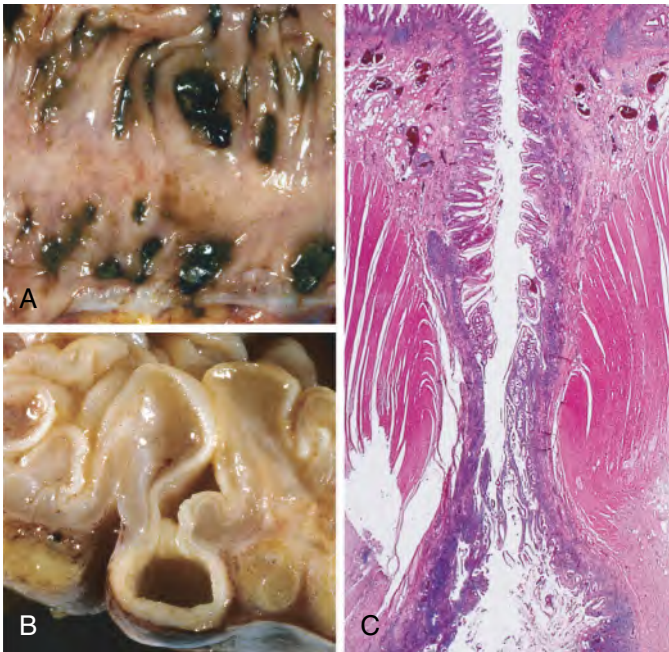


Fig. 17.43 Sigmoid diverticular disease. (A) Stool-filled diverticula are regularly arranged. (B) Cross section showing the outpouching of mucosa through the muscularis propria. (C) Low-power photomicrograph of a sigmoid diverticulum showing protrusion of the mucosa through the muscularis propria.

KEY CONCEPTS

Enterocolitis and Related Disorders

- Inflammatory bowel disease (IBD) is an umbrella term for ulcerative colitis and Crohn disease. Indeterminate colitis is reserved for the few cases of IBD without definitive features of either ulcerative colitis or Crohn disease.
- Crohn disease most commonly affects the terminal ileum and cecum but can involve any site within the GI tract. Disease can be discontinuous, resulting in skip lesions, but pancolitis also occurs. In affected areas, disease usually affects the full thickness of the entire bowel wall. Noncaseating granulomas are common.
- Ulcerative colitis is limited to the colonic mucosa and submucosa, is continuous from the rectum, and ranges from only rectal disease to pancolitis; neither skip lesions nor granulomas are present.
- IBD is thought to arise from a synergy between genetic risk, mucosal and immune dysfunction, and dysbiosis (i.e., disease-associated changes in microbial composition).
- The risk of colonic epithelial dysplasia and adenocarcinoma is increased in IBD patients who have had colonic disease for more than 8 to 10 years.
- The two forms of microscopic colitis, collagenous colitis and lymphocytic colitis, both cause chronic watery diarrhea. The intestines are endoscopically normal, and the diseases are identified by their characteristic histologic features.
- Epithelial apoptosis is prominent in both graft-versus-host disease and checkpoint inhibitor-induced colitis.
- Irritable bowel syndrome is characterized by chronic, relapsing abdominal pain, bloating, and changes in bowel habits without gross or histologic pathology. The pathogenesis is not defined, but likely includes contributions by psychologic stressors, diet, the gut microbiome, abnormal GI motility, and increased enteric sensory responses to GI stimuli.

- Diverticular disease of the sigmoid colon is common in Western populations older than age 60 years. The causes include low-fiber diets, colonic spasm, and the unique anatomy of the colon. Inflammation of diverticula, diverticulitis, affects a minority of those with diverticulosis, but can cause perforation in its most severe form.

INTESTINAL POLYPS

Polyps are most common in the colon and rectum but, as discussed earlier, may occur in the esophagus, stomach, or small intestine. Most, if not all, polyps begin as small elevations of the mucosa. These are referred to as *sessile*, a term borrowed from botanists who use it to describe flowers and leaves that grow directly from the stem without a stalk. As sessile polyps enlarge, proliferation of cells adjacent to the mass and the effects of traction on the luminal protrusion may combine to create a stalk. Polyps with stalks are termed *pedunculated*. In general, intestinal polyps can be classified as nonneoplastic or neoplastic. The most common neoplastic polyp is the adenoma, which has the potential to progress to cancer. Nonneoplastic polyps can be classified as inflammatory, hamartomatous, or hyperplastic.

Hyperplastic Polyps

Colonic hyperplastic polyps are benign epithelial proliferations that are typically discovered in the sixth and seventh decades of life. The pathogenesis of hyperplastic polyps is incompletely understood, but they are thought to result from decreased epithelial cell turnover and delayed shedding of surface epithelial cells, leading to a “piling up” of goblet and absorptive cells. These lesions have no malignant potential. Their chief significance is that they must be distinguished from sessile serrated polyps, which are histologically similar but have malignant potential, as described later. It is also important to remember that epithelial hyperplasia can occur as a nonspecific reaction adjacent to or overlying any mass or inflammatory lesion and therefore can be a clue to the presence of an adjacent, clinically important lesion.

MORPHOLOGY

Hyperplastic polyps are most commonly found in the left colon and are typically less than 5 mm in diameter. They are smooth, nodular protrusions of the mucosa, often on the crests of mucosal folds. They may occur singly but are frequently multiple, particularly in the sigmoid colon and rectum. Histologically, hyperplastic polyps are composed of **mature goblet and absorptive cells with a serrated surface architecture** that is the morphologic hallmark of these lesions (Fig. 17.44). Serration is typically restricted to the upper third of the crypt, and crypt base epithelium may have a regenerative appearance.

Inflammatory Polyps

Inflammatory polyps may form because of repeated cycles of injury and healing. Polyps that are part of the *solitary rectal ulcer syndrome* are an example of a purely inflammatory lesion. These patients present with a clinical triad of rectal bleeding, mucus discharge, and an inflammatory lesion of the anterior rectal wall. The underlying cause is impaired relaxation of the anorectal sphincter that creates a sharp angle at the anterior rectal shelf and leads to recurrent abrasion and ulceration of the overlying rectal mucosa. Entrapment of this polyp in the fecal stream leads to mucosal prolapse. The distinctive histologic features include mixed inflammatory

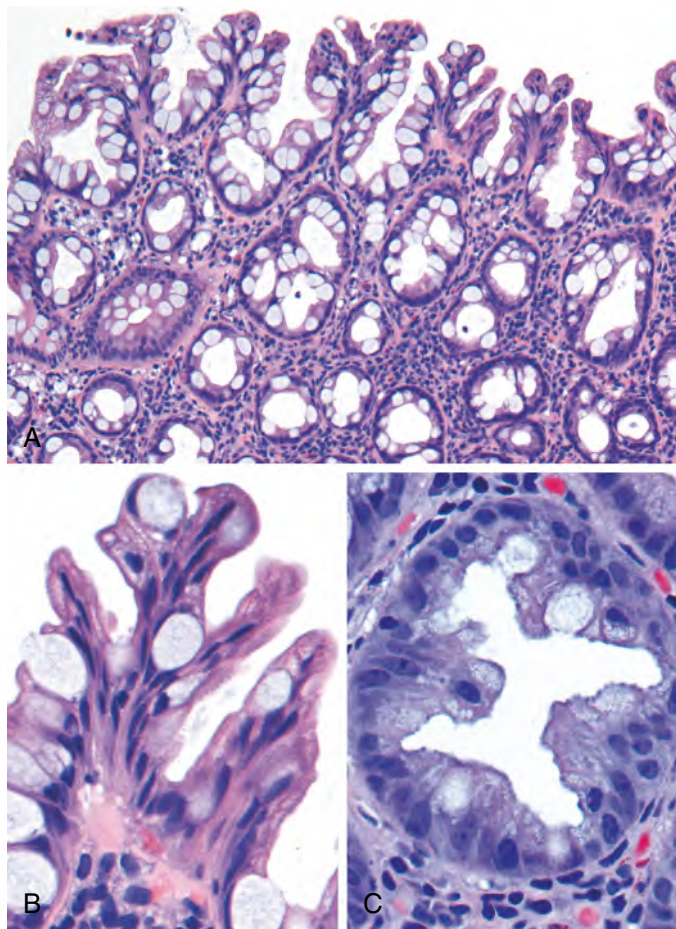


Fig. 17.44 Hyperplastic polyp. (A) Polyp surface with irregular tufting of epithelial cells. (B) Tufting results from epithelial overcrowding. (C) Epithelial crowding produces a serrated architecture when crypts are cut in cross section.

infiltrates, erosion, and epithelial hyperplasia together with prolapse-induced lamina propria fibromuscular hyperplasia (Fig. 17.45).

Hamartomatous Polyps

Hamartomatous polyps occur sporadically or as components of genetically determined or acquired syndromes (Table 17.9). Although they were originally thought to be developmental abnormalities, many hamartomatous polyp syndromes are caused by germline mutations in tumor suppressor genes or proto-oncogenes. Some of these syndromes are associated with increased cancer risk, either within the polyps or at other intestinal or extraintestinal sites. Thus, a subset of hamartomatous polyps can be considered to be premalignant, neoplastic lesions, much like adenomas. It is also important to recognize these polyps because of associated extraintestinal manifestations and the possibility that other family members are affected. Several of these syndromes are discussed next and others are summarized in Table 17.9.

Juvenile Polyps

Juvenile polyps are focal malformations of the epithelium and lamina propria that may be sporadic or syndromic. The majority occur in children younger than 5 years, but both sporadic and syndromic polyps can present at older ages. Most patients with juvenile polyposis syndrome are symptomatic by 20 years of age. Rectal bleeding is the most common presenting symptom.

Sporadic juvenile polyps, also referred to as retention polyps, are usually solitary and are frequently located in the rectum. Intussusception, intestinal obstruction, or rectal polyp prolapse (through the anal sphincter) can occur.

Autosomal dominant juvenile polyposis syndrome is characterized by a few to hundreds of hamartomatous polyps. In these cases, colectomy may be necessary to limit the chronic and sometimes severe hemorrhage associated with polyp ulceration. A minority of patients also have polyps in the stomach and small bowel that can undergo malignant transformation. Pulmonary arteriovenous malformations and other congenital malformations are recognized extraintestinal manifestations of juvenile polyposis.

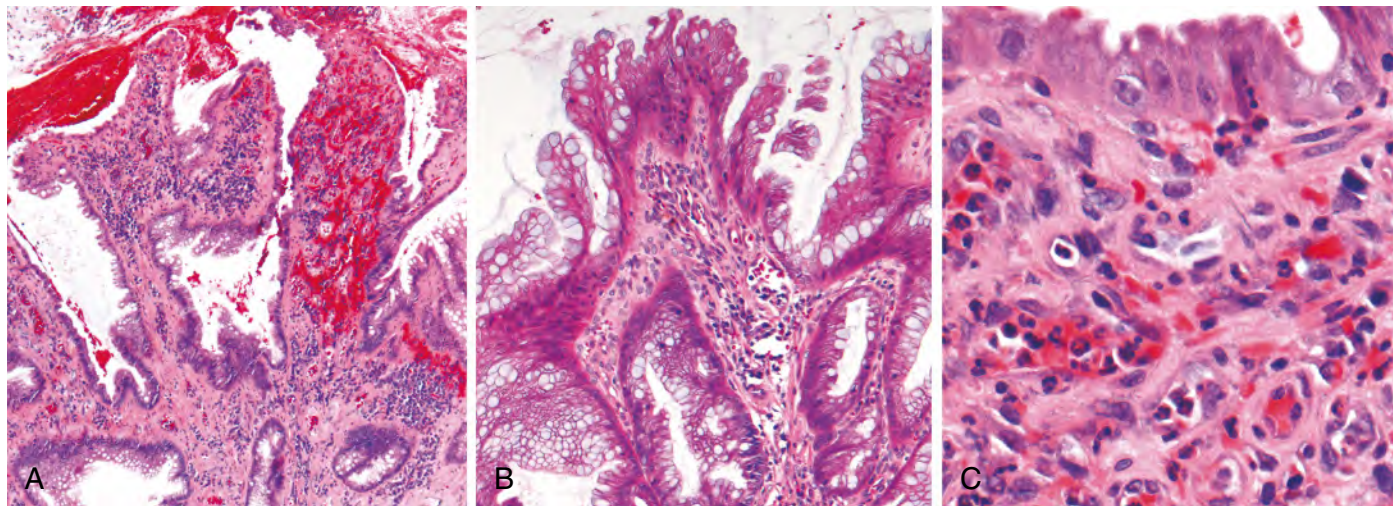


Fig. 17.45 Solitary rectal ulcer syndrome. (A) The dilated glands, proliferative epithelium, superficial erosions, and inflammatory infiltrate are typical of an inflammatory polyp. Accompanying smooth muscle hyperplasia within the lamina propria indicates that mucosal prolapse has also occurred. (B) Epithelial hyperplasia. (C) Granulation tissue-like capillary proliferation within the lamina propria caused by repeated erosion.

TABLE 17.9 Gastrointestinal Polyposis Syndromes

Syndrome	Mutated Genes; Pathway	GI Lesions	Selected Extra-GI Manifestations
Juvenile polyposis	<i>SMAD4</i> , <i>BMPR1A</i> ; TGF- β signaling pathway	Juvenile polyps; risk of gastric, small intestinal, colonic, and pancreatic adenocarcinoma	Congenital malformations, digital clubbing
Peutz-Jeghers syndrome	<i>STK11</i> ; AMP kinase-related pathways	Arborizing polyps; small intestine > colon > stomach; colonic adenocarcinoma	Pigmented macules; risk of colon, breast, lung, pancreatic, and thyroid cancer
Cowden syndrome, Bannayan-Riley-Ruvalcaba syndrome ^a	<i>PTEN</i> ; PI3K/AKT pathway	Hamartomatous/inflammatory intestinal polyps, lipomas, ganglioneuromas	Benign skin tumors, benign and malignant thyroid and breast lesions; no increase in GI cancers
Cronkhite-Canada syndrome	Nonhereditary, unknown cause	Hamartomatous polyps of stomach, small intestine, colon; abnormalities in nonpolypoid mucosa	Nail atrophy, hair loss, abnormal skin pigmentation, cachexia, and anemia. Fatal in up to 50%
Tuberous sclerosis	<i>TSC1</i> (hamartin), <i>TSC2</i> (tuberin); mTOR pathway	Hamartomatous polyps	Intellectual disability, epilepsy, facial angiofibroma, cortical (CNS) tubers, renal angiomyolipoma
FAP			
Classic FAP	<i>APC</i> ; <i>Wnt</i> signaling pathway	Multiple adenomas	Congenital RPE hypertrophy
Attenuated FAP	<i>APC</i>	Multiple adenomas	
Gardner syndrome	<i>APC</i>	Multiple adenomas	Osteomas, thyroid and desmoid tumors, skin cysts
Turcot syndrome	<i>APC</i>	Multiple adenomas	Medulloblastoma, glioblastoma
<i>MUTYH</i> -associated polyposis	<i>MUTYH</i> ; base excision repair	Multiple adenomas	Gastric and duodenal polyps

^aAlso called PTEN hamartoma-tumor syndrome.

CNS, Central nervous system; FAP, familial adenomatous polyposis; GI, gastrointestinal; mTOR, mammalian (also called molecular) target of rapamycin; RPE, retinal pigmented epithelium; TGF β , transforming growth factor beta.

MORPHOLOGY

Most juvenile polyps are less than 3 cm in diameter. They are typically pedunculated, smooth-surfaced, reddish lesions with characteristic cystic spaces. Microscopic examination shows these cysts to be dilated glands filled with mucin and inflammatory debris (Fig. 17.46). The remainder of the polyp is composed of lamina propria expanded by mixed inflammatory infiltrates. The muscularis mucosae may be normal or attenuated.

Although the morphogenesis of juvenile polyps is incompletely understood, it has been proposed that mucosal hyperplasia is the initiating event. This hypothesis is consistent with the discovery that mutations in pathways that regulate cellular growth cause autosomal dominant juvenile polyposis. The most common mutation identified is in *SMAD4*, which encodes a signaling intermediate in the TGF- β pathway. These patients often have both juvenile polyposis and hereditary hemorrhagic telangiectasia. In other cases, mutations in *BMPR1A*, which encodes a kinase that is a member of the TGF- β superfamily, cause disease (see Table 17.9). Together these mutations account for fewer than half of patients. Thus, other genes responsible for autosomal dominant juvenile polyposis remain to be discovered.

Dysplasia is rare in sporadic juvenile polyps. In contrast, juvenile polyposis syndrome is associated with dysplasia, both within the juvenile polyps and in separate adenomas. As a result, 30% to 50% of patients with juvenile polyposis develop colonic adenocarcinoma by age 45 years.

Peutz-Jeghers Syndrome

This rare autosomal dominant syndrome presents at a median age of 11 years with multiple GI hamartomatous polyps and mucocutaneous hyperpigmentation. The latter takes the form of dark blue to brown macules on the lips, nostrils, buccal mucosa, palmar surfaces of

the hands, genitalia, and perianal region. These lesions are similar to freckles but are distinguished by their presence in the buccal mucosa. Peutz-Jeghers polyps can initiate intussusception, which is occasionally fatal. Of greater importance, **Peutz-Jeghers syndrome is associated with a markedly increased risk of several malignancies.** Lifetime risk is approximately 40%, and regular surveillance is recommended, beginning at birth for sex cord tumors of the testes; in late childhood for gastric and small intestinal cancers; and in the second and third decades of life for colon, pancreatic, breast, lung, ovarian, and uterine cancers.

Pathogenesis

Germline heterozygous loss-of-function mutations in the gene *STK11* are present in approximately half of individuals with familial Peutz-Jeghers syndrome as well as a subset of patients with sporadic Peutz-Jeghers syndrome. *STK11* is a tumor suppressor gene that regulates AMP-activated protein kinase (AMPK) pathways, which control cell polarization and act as a brake on growth and anabolic metabolism (Chapter 7). As is common with other tumor suppressor genes, the function of the second “normal” copy of *STK11* is often lost through somatic mutation in Peutz-Jeghers syndrome-associated cancers.

MORPHOLOGY

The polyps of Peutz-Jeghers syndrome are most common in the small intestine, although they may occur in the stomach and colon and, with much lower frequency, in the bladder and lungs. Grossly, the polyps are large and pedunculated with a lobulated contour. Histologic examination demonstrates a characteristic **arborizing network** of connective tissue, smooth muscle, lamina propria, and glands lined by normal-appearing intestinal epithelium (Fig. 17.47). The arborization and presence of smooth muscle intermixed with lamina propria are helpful in distinguishing polyps of Peutz-Jeghers syndrome from juvenile polyps.

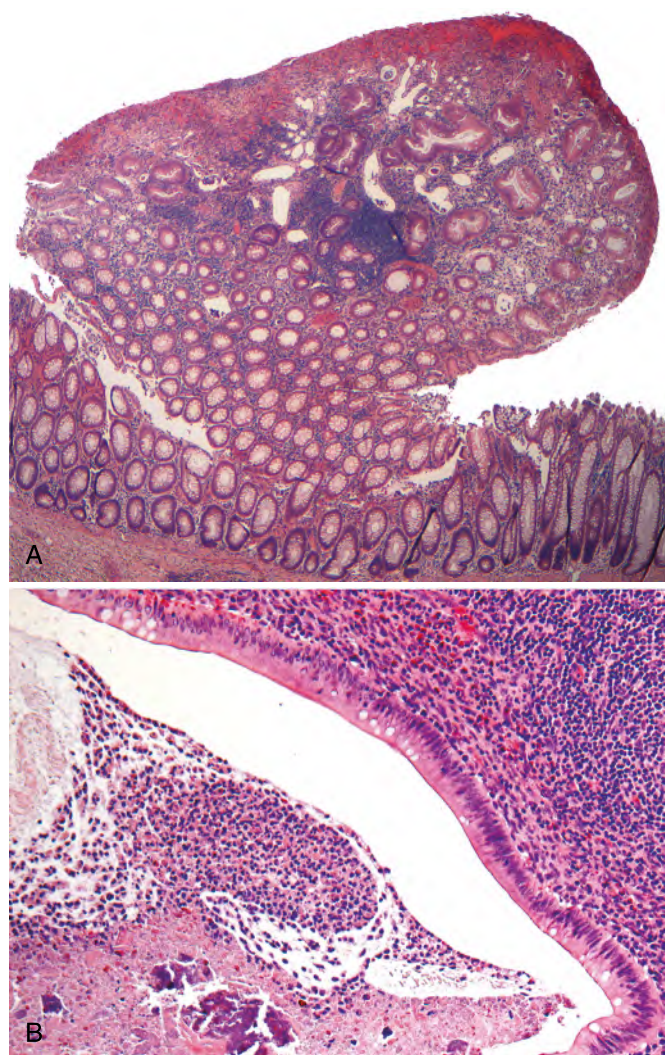


Fig. 1746 Juvenile polyposis. (A) Juvenile polyp. Note the surface erosion and cystically dilated crypts. (B) Inspissated mucus, neutrophils, and inflammatory debris can accumulate within dilated crypts.

Clinical Features

Because the morphology of Peutz-Jeghers polyps overlaps with that of sporadic hamartomatous polyps, the presence of multiple polyps in the small intestine, mucocutaneous hyperpigmentation, and a positive family history are critical to the diagnosis. Detection of *STK11* mutations can be helpful diagnostically in patients with polyps who lack mucocutaneous hyperpigmentation. However, the absence of *STK11* mutations does not exclude the diagnosis, as these mutations are not present in all patients.

PREMALIGNANT AND MALIGNANT LESIONS

The most common neoplastic intestinal lesions are colonic adenomas, which are polyps that are precursors to most colorectal adenocarcinomas. It is important, however, to realize that any mass lesion within the gastrointestinal tract, including adenocarcinomas, neuroendocrine tumors, stromal tumors, lymphomas, and even metastases from distant sites may form polyps. Although it accounts for 75% of the overall length of the GI tract, tumors are relatively uncommon in the small intestine. Among malignant small intestinal tumors, adenocarcinomas and well-differentiated neuroendocrine tumors have roughly equal incidence, followed by lymphomas and sarcomas.

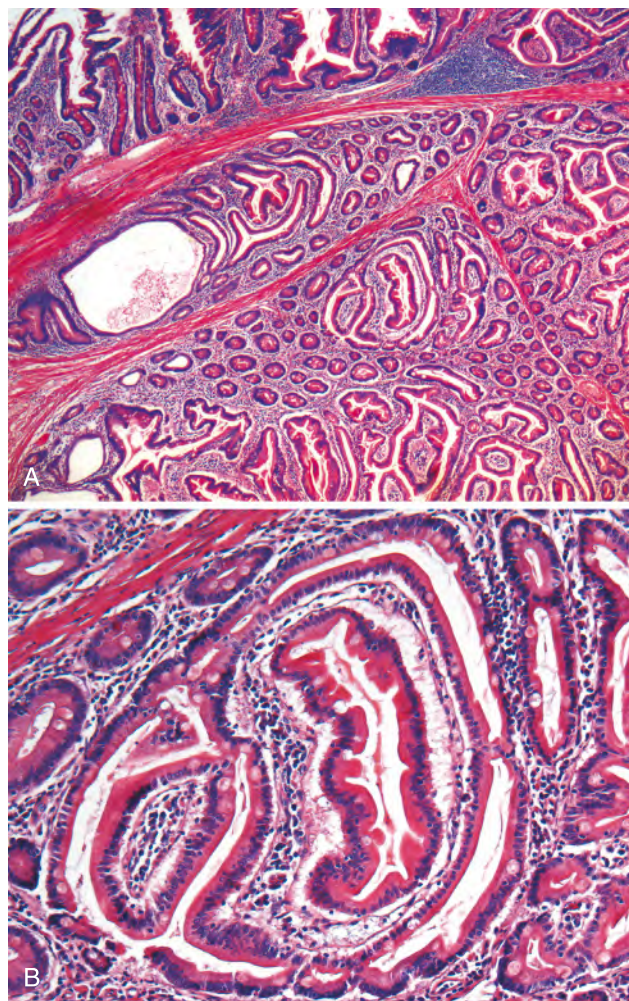


Fig. 1747 Peutz-Jeghers polyp. (A) Polyp surface (top) overlies stroma composed of smooth muscle bundles cutting through the lamina propria. (B) Complex glandular architecture and the presence of smooth muscle are features that distinguish Peutz-Jeghers polyps from juvenile polyps.

Adenomas

Adenomas are epithelial neoplasms that range from small pedunculated polyps to large sessile lesions. They develop in approximately 30% of adults living in the Western world by age 60 years. Colon adenomas are less common in Asia, but their incidence has risen (in parallel with increasing incidence of colorectal adenocarcinoma) as Western diets and lifestyles become more prevalent. Because adenomas are precursors to colorectal adenocarcinoma, surveillance colonoscopy is recommended beginning at age 45 years. Individuals at increased risk, including those with a family history of colorectal adenocarcinoma, typically begin screening at least 10 years before the youngest age at which a relative was diagnosed. Repeated surveillance colonoscopy is both diagnostic and therapeutic, as polyps that have been removed cannot become adenocarcinoma.

The prevalence of colorectal adenomas within populations correlates with that of colorectal adenocarcinoma, and the distributions of adenomas and adenocarcinoma within the colon are similar. Nevertheless, it must be emphasized that most adenomas do not progress to become adenocarcinomas. There are no tools presently available to identify adenomas that are destined to undergo malignant transformation. Most adenomas are clinically silent, with the exception of large polyps that can produce occult bleeding and anemia.

MORPHOLOGY

Colorectal adenomas are characterized by the presence of epithelial dysplasia. Typical adenomas range from 0.3 to 10 cm in diameter and can be pedunculated (Fig. 17.48A) or sessile, with the surface of both types having a texture resembling velvet or a raspberry (Fig. 17.48B). Histologically, **the hallmarks of epithelial dysplasia are nuclear hyperchromasia, elongation, and stratification** (Fig. 17.49C). These changes are most easily appreciated at the surface of the adenoma and are often accompanied by prominent nucleoli, eosinophilic cytoplasm, and a reduction in the number of goblet cells. Pedunculated adenomas have slender fibromuscular stalks (Fig. 17.48C) containing prominent blood vessels derived from the submucosa. The stalk is usually covered by nonneoplastic epithelium, but dysplasia can be present.

Adenomas can be classified as **tubular, tubulovillous, or villous** based on their architecture. These categories, however, have little clinical significance in isolation. Tubular adenomas tend to be small, pedunculated polyps composed of rounded, or tubular, glands (Fig. 17.49A). In contrast, villous adenomas are covered by slender villi (Fig. 17.49B). Villous adenomas harbor cancers more frequently than tubular adenomas, but this also correlates with larger size. As the name implies, tubulovillous adenomas have a mixed architecture.

Sessile serrated polyps, also referred to as sessile serrated adenomas, overlap histologically with hyperplastic polyps but are more commonly found in the right colon. Despite their malignant potential, these polyps **lack typical cytologic features of dysplasia** that are present in other adenomas. Histologically, these lesions can be differentiated from hyperplastic polyps by the presence of serrated architecture throughout the full length of the glands including the crypt base, crypt dilation, and lateral crypt extension (Fig. 17.49C).

Intramucosal carcinoma occurs when dysplastic epithelial cells breach the basement membrane to invade the lamina propria. These may extend into, but not through, the muscularis mucosae. Because functional lymphatic channels are absent in the colonic mucosa, intramucosal carcinomas have little or no metastatic potential and complete polypectomy is generally curative (Fig. 17.50A). Invasion beyond the muscularis mucosae (Fig. 17.50B), including into the submucosal stalk of a pedunculated polyp, constitutes invasive adenocarcinoma and carries a risk of spread to other sites. In such cases, several factors, including histologic grade of the invasive component, presence of vascular or lymphatic invasion, and distance from the invasive component to the margin of resection are considered in planning further therapy.

Although most colorectal adenomas are benign lesions, a small proportion harbor invasive cancer at the time of detection. **Size is the most important characteristic that correlates with risk of malignancy.** For example, while cancer is rare in adenomas less than 1 cm in diameter, nearly 40% of lesions larger than 4 cm in diameter contain foci of invasive cancer. High-grade dysplasia is also a risk factor for cancer to an individual polyp but does not confer an increased risk of cancer to other polyps within the same patient.

Adenocarcinoma of the colon is the most common malignancy of the GI tract and is a major cause of morbidity and mortality worldwide. Before describing the pathogenesis and morphology of colon cancer, we discuss two familial syndromes that have provided insight into mechanisms of these tumors.

Familial Adenomatous Polyposis

Familial adenomatous polyposis (FAP) is an autosomal dominant disorder in which patients develop numerous colorectal adenomas as teenagers. It is caused by germline mutation of the adenomatous polyposis coli, or *APC*, gene, which is a key negative regulator of the Wnt signaling pathway (Chapter 7). Approximately 75% of cases are inherited, while the remainder appear to be caused by de novo mutations.

At least 100 polyps are necessary for a diagnosis of classic FAP, but thousands may be present (Fig. 17.51). Except for their numbers, FAP-associated polyps are morphologically indistinguishable from sporadic adenomas. However, flat or depressed adenomas are also prevalent in FAP, and microscopic adenomas, consisting of only one or two dysplastic crypts, occur in otherwise normal-appearing mucosa.

Colorectal adenocarcinoma develops in 100% of untreated FAP patients, often before age 30 years and nearly always by age 50 years. As a result, prophylactic colectomy is the standard of care. Colectomy prevents colorectal cancer, but patients remain at risk for neoplasia at other sites. The ampulla of Vater and the stomach are common extracolonic sites of adenomas, and, if untreated, cancer, in FAP patients.

FAP is associated with a variety of extraintestinal manifestations including congenital hypertrophy of the retinal pigment epithelium, which can be detected at birth and may be an adjunct to early screening. Specific *APC* mutations have been associated with other manifestations of FAP and partly explain variants such as Gardner syndrome and Turcot syndrome (Table 17.10).

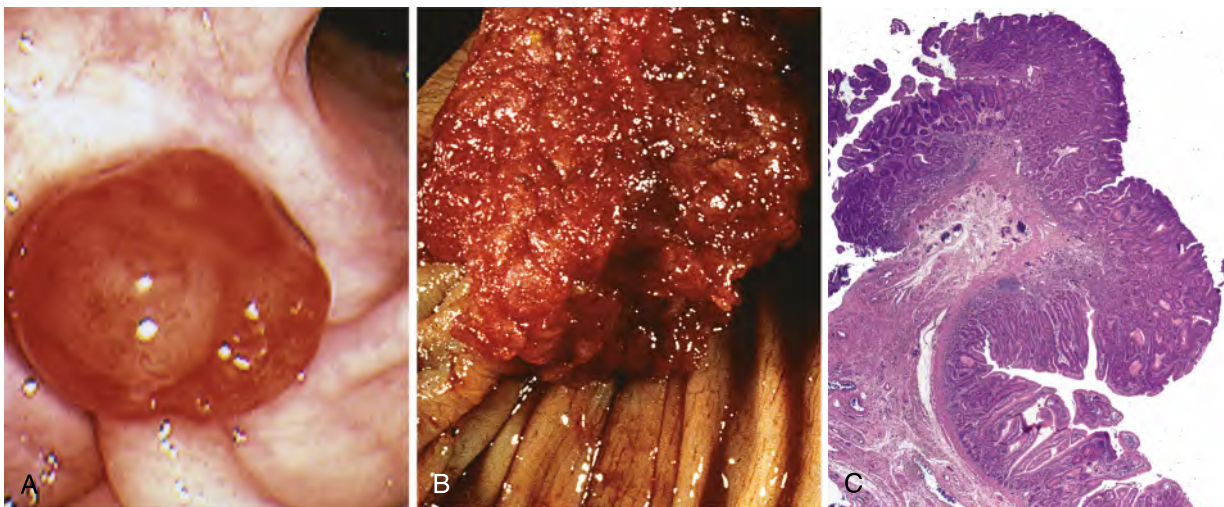


Fig. 17.48 Colonic adenomas. (A) Pedunculated adenoma (endoscopic view). (B) Adenoma with a velvety surface. (C) Low-magnification photomicrograph of a pedunculated tubular adenoma.

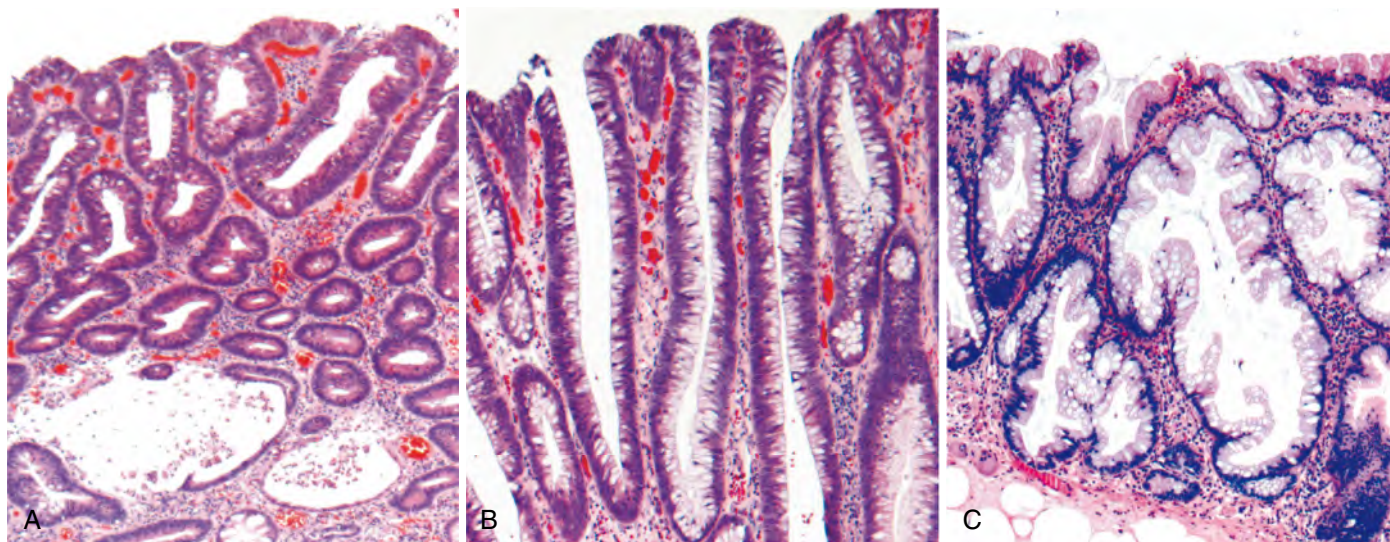


Fig. 17.49 Histologic appearance of colonic adenomas. (A) Tubular adenoma with a smooth surface and rounded glands. Active inflammation is occasionally present in adenomas; in this case, crypt dilation and rupture can be seen at the bottom of the field. (B) Villous adenoma with long, slender projections that are reminiscent of small intestinal villi. (C) Sessile serrated polyp lined by goblet cells without cytologic features of dysplasia. This lesion is distinguished from a hyperplastic polyp by extension of the neoplastic process to the crypts, resulting in lateral growth. Compare to the hyperplastic polyp in [Fig. 17.44](#).

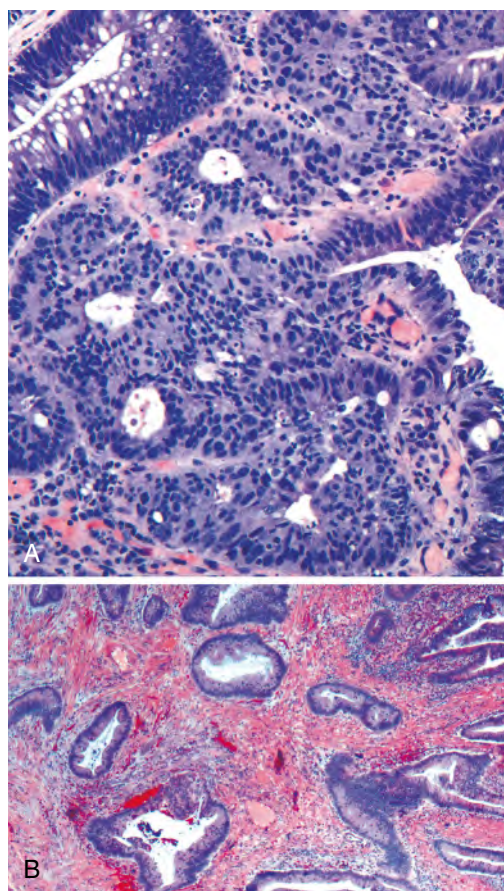


Fig. 17.50 Adenoma with carcinoma. (A) Cribriform glands interface directly with the lamina propria without an intervening basement membrane in this intramucosal carcinoma. (B) Invasive adenocarcinoma (*left*) beneath a villous adenoma (*right*). Note the desmoplastic response to the invasive components.

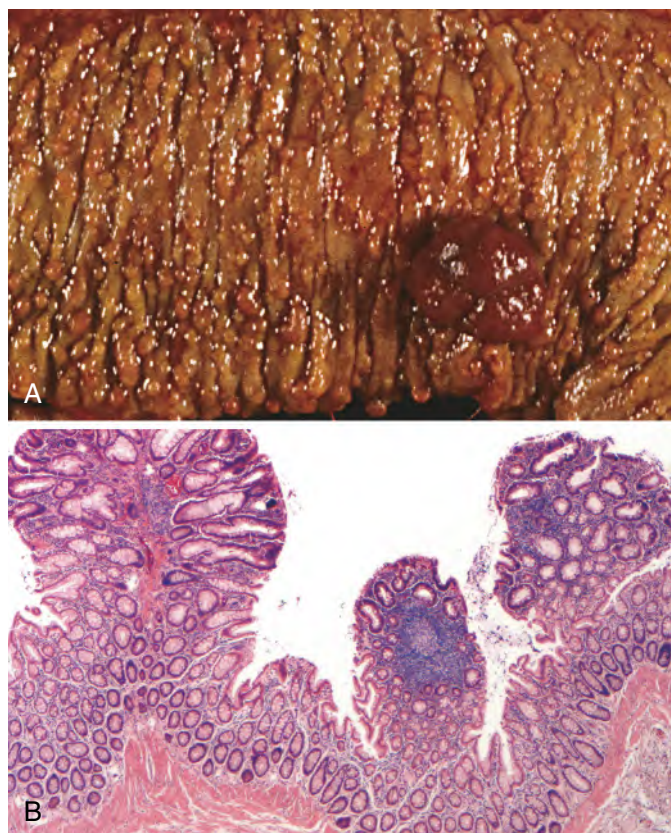


Fig. 17.51 Familial adenomatous polyposis. (A) Hundreds of small polyps are present throughout this colon with a dominant polyp (*right*). (B) Three tubular adenomas are present in this single microscopic field.

TABLE 17.10 Common Patterns of Sporadic and Familial Colorectal Neoplasia

Etiology	Molecular Defect	Target Genes	Transmission	Predominant Site	Histology
Familial adenomatous polyposis	APC/Wnt pathway	<i>APC</i>	Autosomal dominant	None	Tubular, villous; typical adenocarcinoma
<i>MYH</i> -associated polyposis	DNA mismatch repair	<i>MYH</i>	Autosomal recessive	None	Sessile serrated polyp; mucinous adenocarcinoma
Hereditary nonpolyposis colorectal cancer	DNA mismatch repair	<i>MSH2</i> , <i>MLH1</i>	Autosomal dominant	Right side	Sessile serrated polyp; mucinous adenocarcinoma
Sporadic colon cancer (70%–80%)	APC/Wnt pathway	<i>APC</i>	None	Left side	Tubular, villous; typical adenocarcinoma
Sporadic colon cancer (10%–15%)	DNA mismatch repair	<i>MSH2</i> , <i>MLH1</i>	None	Right side	Sessile serrated polyp; mucinous adenocarcinoma
Sporadic colon cancer (5%–10%)	Hypermethylation	<i>MLH1</i> , <i>BRAF</i>	None	Right side	Sessile serrated polyp; mucinous adenocarcinoma

Some polyposis patients without *APC* loss have biallelic mutations of the base-excision repair gene *MUTYH* (also referred to as *MYH*). This autosomal recessive disorder is termed *MUTYH-associated polyposis*, or *MAP*. In contrast to FAP, MAP is characterized by fewer than 100 polyps that appear at later ages. Colon cancer development is also delayed relative to FAP. Sessile serrated polyps and hyperplastic polyps, often with *KRAS* mutations, are also frequently present in MAP.

Hereditary Nonpolyposis Colorectal Cancer

Hereditary nonpolyposis colorectal cancer (HNPCC) is caused by inherited mutations in mismatch repair genes that encode proteins responsible for the detection, excision, and repair of errors that occur during DNA replication (Chapter 7). HNPCC, also known as *Lynch syndrome*, was originally described based on familial clustering of cancers at several sites, including the colorectum, endometrium, stomach, ovary, ureters, brain, small bowel, hepatobiliary tract, pancreas, and skin. HNPCC is thought to account for 2% to 4% of all colorectal cancers, making it the most common syndromic form of colon cancer. Colon cancers in HNPCC patients tend to occur at younger ages than sporadic colon cancers and are often located in the right colon (see Table 17.10).

Just as identification of *APC* mutations in FAP provided molecular insight into the pathogenesis of most sporadic colon cancers, unraveling the defects in HNPCC has shed light on the mechanisms responsible for most of the remaining sporadic cases. There are at least five mismatch repair genes, but most patients with HNPCC have mutations in *MSH2* or *MLH1*. Patients with HNPCC inherit one mutant gene and one normal allele. When the second copy is lost through mutation or epigenetic silencing, defects in mismatch repair lead to the accumulation of mutations at rates up to 1000 times higher than normal, mostly in regions containing short repeating sequences referred to as microsatellites. The human genome contains approximately 50,000 to 100,000 microsatellites, which are prone to undergo expansion during DNA replication and represent the most frequent sites of mutations in HNPCC. The consequences of mismatch repair deficiency and the resulting MSI are discussed next in the context of colonic adenocarcinoma.

Adenocarcinoma

Adenocarcinoma of the colon is the most common malignancy of the GI tract and is a major cause of morbidity and mortality worldwide. In contrast, the small intestine, which accounts for 75% of the overall length of the GI tract, is an uncommon site for benign and malignant tumors. Among malignant small intestinal tumors, adenocarcinomas

and well-differentiated neuroendocrine tumors have roughly equal incidence, followed by lymphomas and sarcomas.

Colorectal adenocarcinoma is responsible for nearly 10% of all cancer deaths worldwide. Approximately 1.2 million new cases of colorectal adenocarcinoma and 600,000 associated deaths occur each year worldwide. The incidence of these tumors is highest in North America, with the United States accounting for approximately 10% of worldwide cases and cancer deaths. This represents nearly 15% of all cancer-related deaths in the United States, second only to lung cancer. Australia, New Zealand, Europe, and (with changes in lifestyle and diet) Japan also have high incidences of colorectal adenocarcinoma. In contrast, rates are lower in South America, India, Africa, and South-Central Asia. Colorectal cancer incidence peaks at 60 to 70 years of age. Fewer than 20% of cases occur before age 50 years, but the incidence of disease before age 40 is increasing, for unknown reasons.

The dietary factors most closely associated with increased risk of colorectal cancer are low intake of unabsorbable vegetable fiber and high intake of refined carbohydrates and fat. Although these associations are clear, the mechanistic relationship between diet and risk is poorly understood. It is theorized that reduced fiber content leads to decreased stool bulk and altered composition of the intestinal microbiota. This change may increase synthesis of potentially toxic oxidative byproducts of bacterial metabolism, which would be expected to remain in contact with the colonic mucosa for longer periods of time because of reduced stool bulk. High fat intake also enhances hepatic synthesis of cholesterol and bile acids, which can be converted into carcinogens by intestinal bacteria.

In addition to dietary modification, pharmacologic chemoprevention is possible. Aspirin and other NSAIDs have a protective effect. This is consistent with studies showing that some NSAIDs cause polyp regression in FAP patients in whom the rectum was left in place after colectomy. It is suspected that this effect is mediated by inhibition of the enzyme COX-2, which is highly expressed in 90% of colorectal carcinomas and 40% to 90% of adenomas. COX-2 is necessary for production of prostaglandin E₂, which promotes epithelial proliferation, particularly after injury. TLR4, which recognizes lipopolysaccharide and is also overexpressed in adenomas and carcinomas, upregulates COX-2 expression.

Pathogenesis

Studies of colorectal carcinogenesis have provided fundamental insight into the general mechanisms of cancer evolution. These were discussed in Chapter 7; concepts that pertain specifically to colorectal carcinogenesis will be reviewed here.

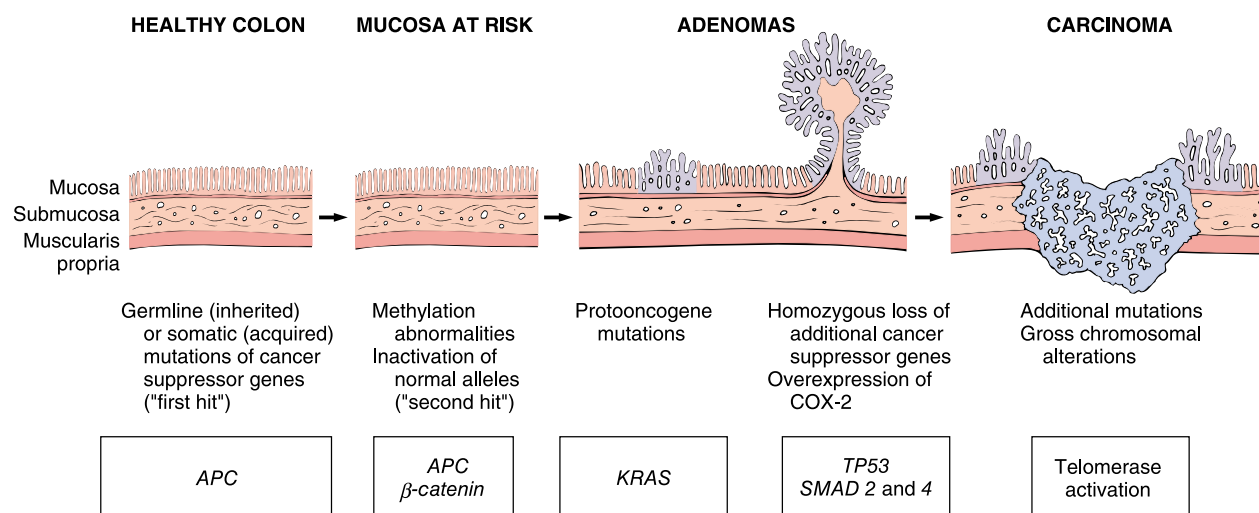


Fig. 17.52 Morphologic and molecular changes in the adenoma-carcinoma sequence. Loss of one normal copy of the tumor suppressor gene *APC* occurs early. Individuals born with one mutant allele are therefore at increased risk of developing colon cancer. Alternatively, inactivation of *APC* in colonic epithelium may occur later in life. This is the “first hit” according to the Knudson hypothesis (Chapter 7). The loss of the intact second copy of *APC* follows (“second hit”). Other changes, including mutation of *KRAS*, losses at 18q21 involving *SMAD2* and *SMAD4*, and inactivation of the tumor suppressor gene *TP53* lead to the emergence of carcinoma.

The combination of molecular events that lead to colonic adenocarcinoma is heterogeneous and includes genetic and epigenetic abnormalities.

Three pathways are responsible for most colorectal adenocarcinoma (see Table 17.10):

- The *APC*/β-catenin pathway, which is activated in the classic adenoma-carcinoma sequence
- The *MSI* pathway, which is associated with defects in mismatch repair and accumulation of mutations mainly in microsatellite repeat regions of the genome
- The *CpG island hypermethylation phenotype (CIMP)*, marked by silencing of genes such as *MLH1*

All these pathways involve the stepwise accumulation of multiple mutations but differ in the genes involved and the mechanisms by which mutations occur. Epigenetic events, the most common of which is methylation-induced gene silencing, may enhance progression along each pathway and predominate in the CIMP pathway.

- The classic adenoma-carcinoma sequence accounts for up to 80% of sporadic cancers and typically includes mutation of *APC* early in the neoplastic process (Fig. 17.52). Both copies of the *APC* gene must be functionally inactivated, either by mutation or by epigenetic silencing, for adenomas to develop. The *APC* protein normally binds to and promotes degradation of β-catenin, a component of the Wnt signaling pathway (Chapter 7). With loss of *APC* function, β-catenin accumulates and translocates to the nucleus, where it forms a complex with the DNA-binding factor TCF and activates the transcription of genes, including *MYC* and cyclin D1, that promote proliferation. The critical role of β-catenin in this pathway is demonstrated by the fact that many colon cancers without *APC* mutations harbor β-catenin mutations that prevent *APC*-dependent degradation, thereby allowing β-catenin to accumulate and activate Wnt signaling. Additional mutations accumulate including activating mutations in *KRAS* that promote growth and prevent apoptosis. *KRAS* mutations are present in fewer than 10% of adenomas less than 1 cm in diameter but are found in 50% of adenomas greater than 1 cm in diameter and in 50% of invasive adenocarcinomas, indicating this is a late event in

neoplastic progression. Carcinogenesis is also associated with mutations in tumor suppressor genes such as those encoding *SMAD2* and *SMAD4*, which are effectors of TGF-β signaling. Because TGF-β signaling normally inhibits the cell cycle, loss of these genes may allow unrestrained cell growth. The tumor suppressor gene *TP53* is mutated in 70% to 80% of colon cancers but is uncommonly affected in adenomas, suggesting that *TP53* mutations also occur at later stages of tumor progression. Loss of function mutations of *TP53* and other tumor suppressor genes are often caused by chromosomal deletions, demonstrating that chromosomal instability is a hallmark of the *APC*/β-catenin pathway. Alternatively, tumor suppressor genes may be silenced by methylation of CpG-rich zones, or CpG islands, within the 5' region that frequently includes the promoter and transcriptional start site of some genes. Telomerase expression also increases as lesions become more advanced.

- In patients with DNA mismatch repair deficiency, mutations accumulate in microsatellite repeats, a condition referred to as microsatellite instability (*MSI*). These are referred to as *MSI high*, or *MSI-H*, tumors. Some microsatellite sequences are located in the coding or promoter regions of genes involved in regulation of cell growth, such as the type II TGF-β receptor and the pro-apoptotic protein *BAX* (Fig. 17.53). Because TGF-β inhibits colonic epithelial cell proliferation, mutation of *TGFBR2*, which encodes the type II TGF-β receptor, can contribute to uncontrolled cell growth, while loss of *BAX* may enhance the survival of genetically abnormal clones.
- A subset of microsatellite unstable colon cancers without mutations in DNA mismatch repair enzymes demonstrate the CpG island hypermethylation phenotype (*CIMP*). In these tumors, the *MLH1* promoter is typically hypermethylated, thereby reducing *MLH1* expression and repair function. Activating mutations in the oncogene *BRAF* are common in these cancers. In contrast, *KRAS* and *TP53* are typically intact. The combination of *MSI*, *BRAF* mutation, and methylation of specific targets, such as *MLH1*, is the signature of this pathway of carcinogenesis.
- A small group of colon cancers display increased CpG island methylation in the absence of *MSI*. Many of these tumors harbor *KRAS* mutations, but *TP53* and *BRAF* mutations are uncommon. In contrast, *TP53* mutations are common in colon cancers that do not display a CIMP.

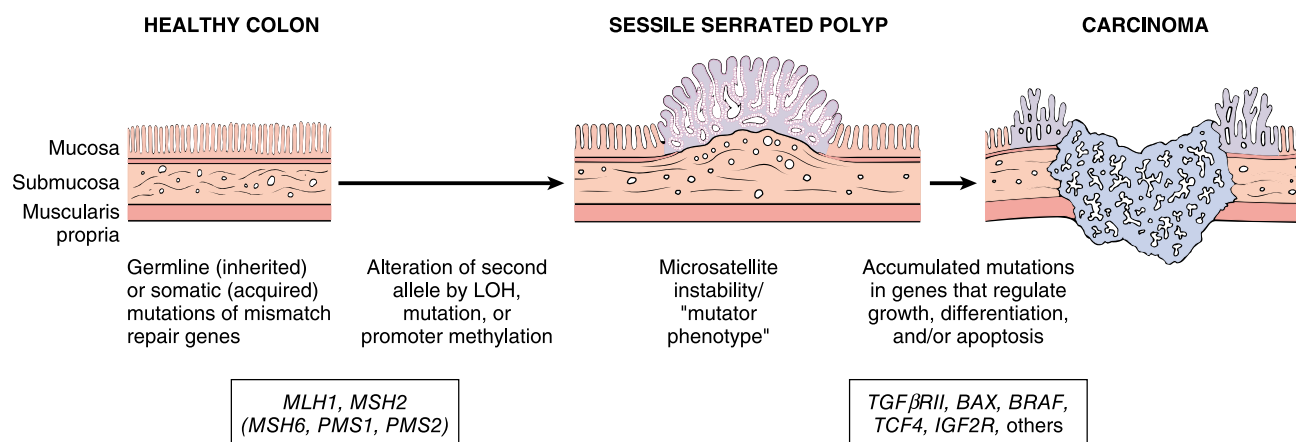


Fig. 17.53 Morphologic and molecular changes in the mismatch repair pathway of colon carcinogenesis. Defects in mismatch repair genes result in microsatellite instability and permit accumulation of mutations in numerous genes. If these mutations affect genes involved in cell survival and proliferation, cancer may develop.

While morphology cannot reliably define the underlying molecular events that lead to carcinogenesis, certain correlations have been associated with mismatch repair deficiency and MSI. These molecular alterations are common in sessile serrated lesions and cancers that arise from them. In addition, prominent mucinous differentiation and peritumoral lymphocytic infiltrates are more common in MSI invasive carcinomas. MSI tumors as well as those with a CIMP are frequently located in the right colon. MSI can be identified by the absence of immunohistochemical staining for mismatch repair proteins or by molecular genetic analysis of microsatellite sequences. These patients tend to be more responsive to checkpoint inhibitor therapy. Moreover, it is important to identify patients with germline mismatch repair protein mutations (i.e., HNPCC) because of the implications for genetic counseling, the elevated risk of a second malignancy of the colon or other organs, and, in some settings, differences in prognosis and therapy.

MORPHOLOGY

Overall, **adenocarcinomas are distributed approximately equally over the entire length of the colon.** Tumors in the **proximal colon often grow as polypoid, exophytic masses** that extend along one wall of the large-caliber cecum and ascending colon; these tumors rarely cause obstruction. In contrast, **carcinomas in the distal colon tend to be annular lesions that produce "napkin-ring" constrictions and luminal narrowing** (Fig. 17.54), sometimes to the point of obstruction. Both forms grow into the bowel wall over time. The general microscopic characteristics of right- and left-sided colonic adenocarcinomas are similar, and most tumors are composed of tall columnar cells that resemble the dysplastic epithelium of adenomas (Fig. 17.55A). The invasive component of these tumors elicits a strong stromal desmoplastic response, which is responsible for characteristic firm consistency of invasive colon cancers. Some poorly differentiated tumors form few glands (Fig. 17.55B). Others may produce abundant mucin that accumulates within the intestinal wall; these are associated with poor prognosis and may lead to pseudomyxoma peritonei. Uncommonly, colorectal tumors may be composed of signet-ring cells similar to those in gastric cancer (Fig. 17.55C).

Clinical Features

The availability of endoscopic screening combined with the knowledge that most carcinomas arise within adenomas presents a unique opportunity for cancer prevention. Unfortunately, colorectal cancers develop insidiously and may go undetected for long periods. Cecal and other right-sided colon cancers are most often called to clinical attention by the appearance of fatigue and weakness due to iron deficiency anemia.

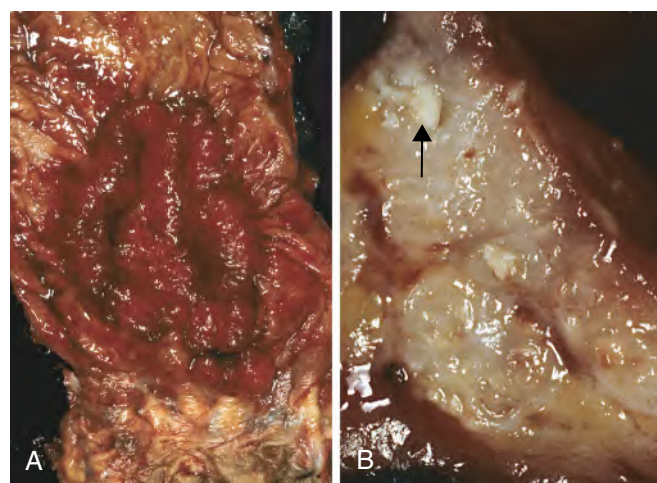


Fig. 17.54 Colorectal carcinoma. (A) Circumferential, ulcerated rectal cancer. Note the anal mucosa at the bottom of the image. (B) Cancer of the sigmoid colon that has invaded through the muscularis propria and is present within subserosal adipose tissue (left). Areas of chalky necrosis are present within the colon wall (arrow).

Thus, it is a clinical maxim that the underlying cause of iron deficiency anemia in an older male or postmenopausal female is GI cancer until proven otherwise. Left-sided colorectal adenocarcinomas may produce visible bleeding, changes in bowel habits, or cramping and left lower quadrant discomfort.

Although poorly differentiated and mucinous histologies are associated with worse prognosis, **the two most important prognostic factors are depth of invasion and the presence of lymph node metastases.** Invasion into the muscularis propria significantly reduces the probability of survival, which is decreased further by the presence of lymph node metastases (Fig. 17.56A). Metastases may involve regional lymph nodes, lungs (Fig. 17.56B), and bones, but because of portal drainage of the colon, the liver is the most common site of distant metastatic lesions (Fig. 17.56C). The rectum does not drain via the portal circulation; hence anal carcinomas that metastasize often circumvent the liver.

The prognostic significance of depth of invasion and lymph node involvement was originally recognized by Dukes and Kirklin and form the core of the TNM (tumor-nodes-metastasis) system (Table 17.11),

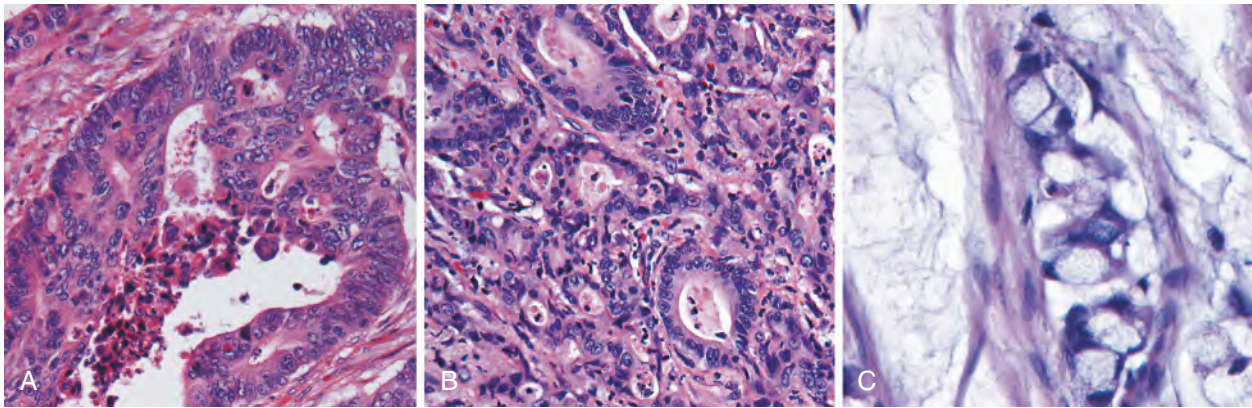


Fig. 17.55 Histologic appearance of colorectal carcinoma. (A) Well-differentiated adenocarcinoma. Note the elongated, hyperchromatic nuclei. Necrotic debris, present in the gland lumen, is typical. (B) Poorly differentiated adenocarcinoma forms a few glands but is largely composed of infiltrating nests of tumor cells. (C) Mucinous adenocarcinoma with signet-ring cells and extracellular mucin pools.

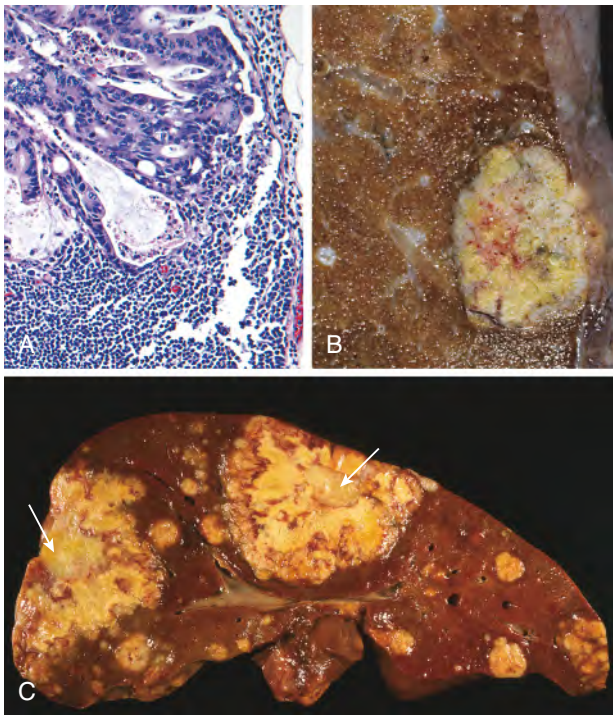


Fig. 17.56 Metastatic colorectal carcinoma. (A) Lymph node metastasis. Note the glandular structures within the subcapsular sinus. (B) Solitary subpleural nodule of colorectal carcinoma metastatic to the lung. (C) Liver containing two large and many smaller metastases. Note the central necrosis within the larger metastases (arrows).

which is used to define tumor stage (Table 17.12). Regardless of stage, it must be remembered that some patients with small numbers of metastases do well for years following resection of distant tumor nodules.

Five-year survival rates vary widely worldwide. The overall 5-year survival rate in the United States is 65% and ranges from 90% to 40% depending on stage. Survival rates in Europe, Japan, and Australia are similar, ranging from 60% (Switzerland, Japan) to 40% (Poland). Overall survival rates are somewhat lower in other countries, such as China, India, the Philippines, and Thailand (30% to 42%). In contrast, the 5-year survival rate in Gambia is only 4%, a measure of the challenges of delivering health care to under-resourced parts of the world.

TABLE 17.11 American Joint Committee on Cancer (AJCC) TNM Classification of Colorectal Carcinoma

TNM	
Tumor	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in situ, intramucosal carcinoma
T1	Tumor invades the submucosa
T2	Tumor invades the muscularis propria
T3	Tumor invades through the muscularis propria into subserosa or nonperitonealized pericolic/rectal tissues
T4	Tumor invades visceral peritoneum or invades or adheres to adjacent organ or structure
T4a	Tumor invades through the visceral peritoneum
T4b	Tumor extends through the serosa to directly invades or adheres to adjacent organs or structures
Regional Lymph Nodes	
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	One to three regional lymph nodes are positive (for metastasis)
N1a	One regional lymph node is positive
N1b	Two to three regional lymph nodes are positive
N1c	No regional lymph nodes are positive, but there are tumor deposits in the subserosa/mesentery or nonperitonealized pericolic, or perirectal/mesorectal tissues
N2	Four or more regional lymph nodes are positive
N2a	Four to six regional lymph nodes are positive
N2b	Seven or more regional lymph nodes are positive
Distant Metastasis	
M0	No distant metastasis by imaging, etc.
M1	Distant metastasis to one or more distant sites or organs or peritoneal metastasis
M1a	Metastasis to one site or organ without peritoneal metastasis
M1b	Metastases to two or more sites or organs without peritoneal metastasis
M1c	Metastasis to the peritoneal surface, alone or with other site or organ metastases

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TABLE 17.12 Colorectal Cancer Staging System

	Stage		
	T	N	M
0	Tis	N0	M0
I	T1–T2	N0	M0
IIA	T3	N0	M0
IIB	T4a	N0	M0
IIC	T4b	N0	M0
IIIA	T1–T2	N1/N1c	M0
	T1	N2a	M0
IIIB	T3–T4a	N1/N1c	M0
	T2–T3	N2a	M0
	T1–T2	N2b	M0
IIIC	T4a	N2a	M0
	T3–T4a	N2b	M0
	T4b	N1/N2	M0
IVA	Any T	Any N	M1a
IVB	Any T	Any N	M1b
IVC	Any T	Any N	M1c

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KEY CONCEPTS

Benign and Malignant Proliferative Lesions of the Colon

- Intestinal polyps can be classified as nonneoplastic or neoplastic. The non-neoplastic polyps can be further subclassified as hyperplastic, inflammatory, or hamartomatous.
- Hyperplastic polyps are benign epithelial proliferations most commonly found in the left colon and rectum. They have no malignant potential and must be distinguished from sessile serrated polyps.
- Inflammatory polyps form because of chronic cycles of injury and healing.
- Hamartomatous polyps occur sporadically or as a part of genetic diseases. The latter include juvenile polyposis and Peutz-Jeghers syndrome, which are associated with increased risk of malignancy.
- Benign epithelial neoplastic polyps of the intestines are termed adenomas. The hallmark of these lesions, which are the precursors of colonic adenocarcinomas, is cytologic dysplasia.
- In contrast to typical adenomas, sessile serrated polyps lack cytologic dysplasia and share morphologic features with hyperplastic polyps.
- Familial adenomatous polyposis (FAP) and hereditary nonpolyposis colon rectal cancer (HNPCC) are the most common forms of familial colon cancer.
- FAP is caused by *APC* mutations. Patients who typically have more than 100 adenomas develop colon cancer before 30 years of age if left untreated.
- HNPCC is caused by mutations in DNA mismatch repair enzymes. HNPCC patients have far fewer polyps and develop cancer at older ages than FAP patients but younger ages than those with sporadic colon cancer.
- FAP and HNPCC typify distinct pathways of neoplastic transformation and progression that are also involved in most sporadic colon cancers. A third pathway characterized by CIMP underlies most remaining colon cancers.
- Nearly all colonic cancers are adenocarcinomas. The two most important prognostic factors are depth of invasion and the presence or absence of lymph node metastases.

ANAL CANAL

Tumors

The anal canal can be divided into thirds. The upper zone is lined by columnar rectal epithelium; the middle third by transitional epithelium; and the lower third by stratified squamous epithelium. Carcinomas of the anal canal may have typical glandular or squamous patterns of differentiation, recapitulating the normal epithelium of the upper and lower thirds, respectively (Fig. 17.57A). An additional differentiation pattern of squamous cell carcinomas, termed basaloid, is present in tumors populated by immature cells derived from the basal layer of transitional epithelium (Fig. 17.57B). These tumors are classified as squamous, but the archaic term cloacogenic carcinoma is often applied when the entire tumor displays a basaloid pattern. Alternatively, basaloid differentiation may be mixed with mucinous differentiation. Squamous cell carcinoma of the anal canal is frequently associated with infection by high-risk strains of human papilloma virus (HPV), whereas infections with low-risk HPV strains are associated with *condyloma acuminatum* (Fig. 17.57C). The major risk factor for these HPV-associated lesions is anal sex.

Hemorrhoids

Hemorrhoids affect about 5% of the general population and develop secondary to persistently elevated venous pressure within the hemorrhoidal plexus. The most frequent predisposing influences are straining at defecation (e.g., in constipation) and venous stasis of pregnancy. Hemorrhoids may also develop secondary to portal hypertension. The pathogenesis of hemorrhoids (anal varices) in portal hypertension is similar to that of esophageal varices; anal varices are both more common and much less serious. Anal and perianal varices connect portal and caval venous systems, thereby relieving the venous hypertension.

Hemorrhoids often present with pain and rectal bleeding, particularly bright red blood seen on toilet tissue. In this case, colon cancer, which can also be present in those with hemorrhoids, must be excluded. Except for pregnant females, hemorrhoids are rarely encountered in persons younger than age 30 years. Hemorrhoidal bleeding is not generally a medical emergency and can be treated by sclerotherapy, rubber band ligation, or infrared coagulation. Extensive or severe internal or external hemorrhoids may be removed surgically by hemorrhoidectomy.

APPENDIX

Acute Appendicitis

The appendix is a normal true diverticulum of the cecum that is prone to acute and chronic inflammation. Acute appendicitis is most common in adolescents and young adults, with a lifetime risk of 7%; males are affected slightly more often than females. In adults, cancer of the appendix must be considered. Despite the prevalence of acute appendicitis, the diagnosis can be difficult to confirm preoperatively and may be confused with mesenteric lymphadenitis (often secondary to unrecognized *Yersinia* infection or viral enterocolitis), acute salpingitis, ectopic pregnancy, mittelschmerz pain (caused by minor pelvic bleeding at the time of ovulation), and Meckel diverticulitis.

Pathogenesis

Acute appendicitis is thought to be initiated by progressive increases in intraluminal pressure that compromise venous outflow. In 50% to 80% of cases, acute appendicitis is associated with overt luminal obstruction, usually caused by a small stone-like mass of stool, or fecalith, or, less commonly, a gallstone, tumor, or mass of worms (oxyuriasis vermicularis). Periappendiceal lymphoid hyperplasia (e.g., after viral infection) and the resulting appendiceal compression has also been proposed as a mechanism of venous and luminal obstruction. Stasis of luminal contents allows bacterial proliferation that leads to ischemia and inflammation.

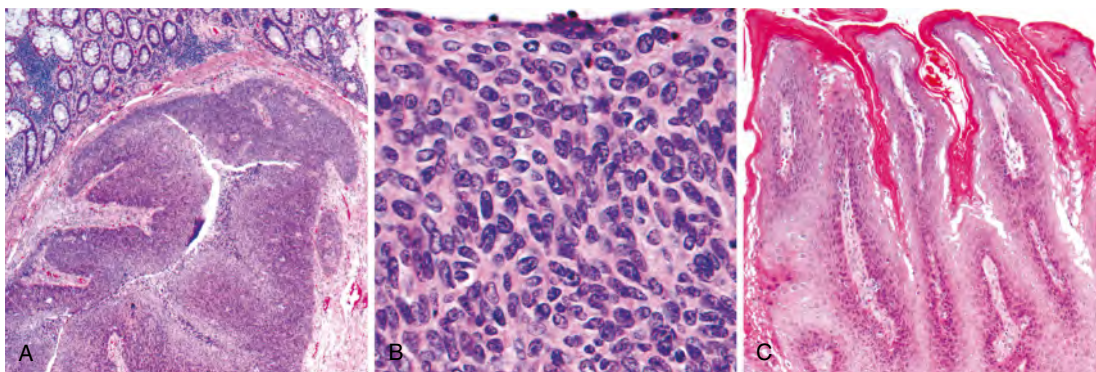


Fig. 17.57 Anal tumors. (A) This anal squamous cell carcinoma demonstrates a multilayered organization reminiscent of squamous mucosa. The adjacent rectal mucosa is normal. (B) This anal squamous cell carcinoma with basaloid features is composed of hyperchromatic cells that resemble the basal layer of normal squamous mucosa. (C) Condyloma acuminatum with verrucous architecture.

MORPHOLOGY

In early acute appendicitis, subserosal vessels are congested, and there is a modest perivascular neutrophilic infiltrate within all layers of the wall. The inflammatory reaction transforms the normal glistening serosa into a dull, granular, erythematous surface. Although mucosal neutrophils and focal superficial ulceration are often present, these are not specific markers of acute appendicitis. Diagnosis of acute appendicitis requires **neutrophilic infiltration of the muscularis propria**. In more severe cases a prominent neutrophilic exudate generates a serosal fibrinopurulent reaction. As the process continues, focal abscesses may form within the wall (acute suppurative appendicitis) and perforation can occur. Further compromise of appendiceal vessels can result in ischemia, hemorrhagic ulceration, and gangrenous necrosis that extends to the serosa, creating acute gangrenous appendicitis, which can be followed by rupture and suppurative peritonitis.

Clinical Features

Typically, early acute appendicitis produces periumbilical pain that ultimately localizes to the right lower quadrant, followed by nausea, vomiting, low-grade fever, and a mildly elevated peripheral white cell count. A classic physical finding is the *McBurney sign*, deep tenderness located two-thirds of the distance from the umbilicus to the right anterior superior iliac spine (McBurney point).

Regrettably, classic signs and symptoms of acute appendicitis are often absent. In some cases, a retrocecal appendix may generate right flank or pelvic pain, while a malrotated colon may give rise to appendiceal pain in the left upper quadrant. As with other causes of acute inflammation, there is neutrophilic leukocytosis. In some cases, the peripheral leukocytosis may be minimal or, alternatively, so great that other causes are considered. The diagnosis of acute appendicitis in young children and the very old is particularly problematic, since other causes of abdominal emergencies are prevalent in these populations, and the very young and old are also more likely to have atypical clinical presentations.

Given these diagnostic challenges, it should be no surprise that even highly skilled surgeons remove normal appendices. This is preferred to delayed resection of a diseased appendix, given the significant morbidity and mortality associated with appendiceal perforation. Other complications of appendicitis include pylephlebitis, portal venous thrombosis, liver abscess, and bacteremia.

Tumors

Several tumors occur in the appendix. The most common is the well-differentiated neuroendocrine (carcinoid) tumor. It is usually discovered incidentally at the time of surgery or pathologic examination of a

resected appendix. This neoplasm, which is almost always benign, most frequently forms a solid bulbous swelling at the distal tip of the appendix, where it can reach 2 to 3 cm in diameter. Although intramural and transmural extension may be evident, nodal metastases are very infrequent, and distant spread is exceptionally rare.

Conventional adenomas, mucin-producing and non-mucin-producing adenocarcinomas, and the uncommon adenocarcinoid tumor also occur in the appendix and may cause obstruction and enlargement that mimics acute appendicitis.

Mucinous neoplasms of the appendix occur in adults, most often in the sixth decade of life. They are characterized by mucinous epithelial proliferation, extracellular mucin, and pushing tumor margins; an infiltrating invasive pattern classifies the tumor as a mucin-producing adenocarcinoma.

Mucin can dissect through the wall to the peritoneal surface, causing appendiceal rupture. Tumors are classified as low-grade or high-grade appendiceal mucinous neoplasms (LAMN, HAMN) based on histologic features. *GNAS*, which encodes *Gαs*, is frequently mutated and thought to contribute to excessive mucin production. *KRAS* mutations are also common. Genes involved in colon cancer, such as *APC*, *TP53*, and *SMAD4*, are typically intact in LAMN but may be mutated in HAMN. When confined to the appendix, prognosis of LAMN is excellent. Outcomes are more variable in HAMN. Peritoneal spread of LAMN, HAMN, or mucin-producing adenocarcinomas (of the appendix or colon) can lead to *pseudomyxoma peritonei*, in which tenacious, semisolid mucin fills the abdomen. This disseminated intraperitoneal disease may be held in check for years after cytoreduction surgery and hyperthermic intraperitoneal chemotherapy. Heating allows better penetration of the chemotherapy. The long-term prognosis remains guarded because of potential recurrence of the tumor.

KEY CONCEPTS

Hemorrhoids and Appendiceal Disease

- Hemorrhoids are collateral vessels that develop secondary to persistently elevated venous pressure within the hemorrhoidal plexus. They also occur in portal hypertension.
- Acute appendicitis is most common in children and adolescents. It is thought to be initiated by increased intraluminal pressure and compromised venous outflow.
- The most common tumor of the appendix is the well-differentiated neuroendocrine (carcinoid) tumor, which is almost always benign.
- Peritoneal dissemination of mucinous neoplasms can cause pseudomyxoma peritonei.

PERITONEAL CAVITY

The peritoneal cavity houses the abdominal viscera and is lined by a single layer of mesothelial cells; these cover the visceral and parietal surfaces and are supported by a thin layer of connective tissue to form the peritoneum. Here we discuss inflammatory, infectious, and neoplastic disorders of the peritoneal cavity and retroperitoneal space.

PERITONITIS

Peritonitis may result from bacterial invasion or chemical irritation and is most often due to:

- *Perforation* of abdominal viscera, which results in leakage of luminal materials that can lead to infection and immune activation.
- Leakage of bile or pancreatic enzymes, which produces *sterile peritonitis*.
- *Perforation or rupture of the biliary system*, which evokes a highly irritating peritonitis, usually complicated by bacterial superinfection.
- *Acute hemorrhagic pancreatitis* (Chapter 19), which is associated with leakage of pancreatic enzymes and fat necrosis. Damage to the bowel wall may allow bacteria to spread to the peritoneal cavity.
- *Foreign material*, including that introduced surgically (e.g., talc and sutures), may induce foreign body–type granulomas and fibrous scarring.
- *Endometriosis*, which causes hemorrhage into the peritoneal cavity, acts as an irritant.
- *Ruptured dermoid cysts*, which release keratins and induce an intense granulomatous reaction.

Bacterial peritonitis occurs when bacteria from the GI lumen are released into the abdominal cavity, most commonly following intestinal perforation. *E. coli*, streptococci, *S. aureus*, enterococci, and *C. perfringens* are implicated most often.

Spontaneous bacterial peritonitis develops in the absence of an obvious source of contamination. It is seen most often in patients with cirrhosis and ascites and less frequently in children with nephrotic syndrome. Diagnosis is based on the presence of neutrophils in ascitic fluid and positive bacterial cultures; *E. coli*, streptococci, and *Klebsiella* species are the most frequently involved organisms.

MORPHOLOGY

The cellular inflammatory response is composed primarily of dense collections of neutrophils and fibrinopurulent debris that coat the viscera and abdominal wall. Serous or slightly turbid fluid begins to accumulate and becomes suppurative as infection progresses. Subhepatic and subdiaphragmatic abscesses may form. With the exception of tuberculous peritonitis, the reaction usually remains superficial.

Sclerosing Retroperitonitis

Sclerosing retroperitonitis, also known as idiopathic retroperitoneal fibrosis or *Ormond disease*, is characterized by dense fibrosis that may extend to involve the mesentery. Although the cause of sclerosing retroperitonitis is unknown, many cases are now thought to fall within the spectrum of IgG4-related sclerosing disease, an immunoinflammatory disorder that can lead to fibrosis in a wide variety of tissues. This entity is described in more detail in Chapter 6.

TUMORS

Primary malignant tumors arising from the peritoneal lining, *mesotheliomas*, are similar to mesotheliomas of the pleura and

pericardium. Peritoneal mesothelioma is almost always associated with high levels of asbestos exposure. Rarely, primary benign and malignant soft tissue tumors may develop within the peritoneum and retroperitoneum. The most common of these is desmoplastic small round cell tumor. This is an aggressive tumor that occurs in children and young adults. It bears a strong morphologic resemblance to Ewing sarcoma (Chapter 26) and like Ewing sarcoma is characterized by a reciprocal translocation, t(11;22) (p13;q12), that produces a *EWS* and *WT1* fusion gene.

Secondary tumors may involve the peritoneum by direct spread or metastatic seeding, resulting in peritoneal carcinomatosis. As discussed earlier, mucinous carcinomas, particularly those of the appendix, may cause pseudomyxoma peritonei.

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Liver and Gallbladder

Ryan M. Gill and Sanjay Kakar

CHAPTER CONTENTS

Liver and Bile Ducts, 749

General Features of Liver Disease, 749

- Mechanisms of Injury and Repair, 749
 - Hepatocyte and Parenchymal Responses, 749*
 - Scar Formation and Regression, 750*
- Liver Failure, 751
 - Acute Liver Failure, 751*
 - Chronic Liver Failure and Cirrhosis, 753*
 - Portal Hypertension, 754*
 - Acute-on-Chronic Liver Failure, 755*

Infectious Disorders, 755

- Viral Hepatitis, 755
 - Hepatitis A Virus, 756*
 - Hepatitis B Virus, 756*
 - Hepatitis C Virus, 758*
 - Hepatitis D Virus, 760*
 - Hepatitis E Virus, 760*
 - Clinicopathologic Syndromes of Viral Hepatitis, 760*
- Bacterial, Fungal, and Parasitic Infections, 762

Autoimmune Hepatitis, 763

Drug- and Toxin-Induced Liver Injury, 764

Steatotic Liver Disease, 766

- Alcohol-Associated Liver Disease, 766
- Metabolic Dysfunction–Associated Steatotic Liver Disease, 769

Inherited Liver Disease, 771

- Hemochromatosis, 771
- Wilson Disease, 773
- α_1 -Antitrypsin Deficiency, 774

Cholestatic Disease, 775

- Bile Formation and Secretion, 775
- Pathophysiology of Hyperbilirubinemia, 776
- Physiologic Jaundice of the Newborn, 777
- Hereditary Hyperbilirubinemia, 777
- Large Bile Duct Obstruction, 777
- Cholestasis of Sepsis, 778
- Primary Hepatolithiasis, 778
- Neonatal Cholestasis, 779
 - Extrahepatic Biliary Atresia, 779*
 - Nonobstructive Neonatal Cholestasis, 779*
- Autoimmune Cholangiopathies, 779

Primary Biliary Cholangitis, 780

Primary Sclerosing Cholangitis, 781

Vanishing Bile Duct Syndrome, 782

Structural Anomalies of the Biliary Tree, 782

- Choledochal Cyst, 782*
- Fibropolycystic Disease, 782*

Circulatory Disorders, 783

- Impaired Blood Flow Into the Liver, 783
 - Hepatic Artery Compromise, 783*
 - Portal Vein Obstruction and Thrombosis, 784*
- Impaired Blood Flow Through the Liver, 784
- Hepatic Venous Outflow Obstruction, 785
 - Hepatic Vein Thrombosis, 785*
 - Sinusoidal Obstruction Syndrome, 785*
- Passive Congestion and Centrilobular Necrosis, 785

Hepatic Disease Associated With Pregnancy, 786

- Preeclampsia and Eclampsia, 786
- Acute Fatty Liver of Pregnancy, 787
- Intrahepatic Cholestasis of Pregnancy, 787

Nodules and Tumors, 787

- Nonneoplastic Mass Lesions, 787
 - Focal Nodular Hyperplasia, 787*
 - Other Nonneoplastic Mass Lesions, 787*
- Benign Neoplasms, 788
 - Cavernous Hemangioma, 788*
 - Hepatocellular Adenoma, 788*
- Primary Malignant Neoplasms, 790
 - Hepatoblastoma, 790*
 - Hepatocellular Carcinoma, 790*
 - Malignant Biliary Tumors, 792*
 - Other Primary Hepatic Malignant Tumors, 793*
 - Metastasis, 793*

Gallbladder, 794

Congenital Anomalies, 794

Cholelithiasis (Gallstones), 794

Epidemiology, 794

Cholecystitis, 796

- Acute Cholecystitis, 796*
- Chronic Cholecystitis, 796*

Gallbladder Carcinoma, 797